

Euclidean Condensate Theory: A Narrative Companion

Conditional Routes from the Φ -Medium to Lorentzian Kinematics,
Gravity, Quantum Structures, and Observational Programmes

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Abstract

Euclidean Condensate Theory (ECT) asks whether familiar Lorentzian spacetime, gravity and quantum structures can arise as effective descriptions of a fundamentally Euclidean scalar medium. This companion presents that research programme in reader-oriented form, separating exact mathematics in declared models from conditional completions, phenomenological diagnostics and Open physical questions.

ECT takes as an input—not yet as a dynamically derived physical state—an ordered branch in which the scalar field has a nonzero gradient. Its direction has an exact $O(3)$ stabiliser. Independently, a supplied scalar principal tensor is Lorentzian exactly when $\beta(\beta - \alpha) < 0$; on the retained healthy branch $\beta > 0$, this becomes $\alpha > \beta > 0$. This classification supplies one conditional scalar characteristic cone. It does not by itself derive physical time, a universal observable metric, tensor gravity, or common matter, photon and gravitational cones.

The macroscopic programme studies separately supplied scalar–tensor completions, one conditional cosmological background, and galactic response diagnostics. Their calculations are model-internal or fitted: the common action, physical metric, matter-to-gravity bridge, full likelihoods and fundamental particle content remain Open. The quantum programme develops conditional Hilbert-space and open-system tools, exchange topology, decoherence and record diagnostics. It does not yet derive a physical Born measure, a unique outcome, a Bell forward law or a gravitational record channel; imported collapse-model calculations are external comparisons, not ECT rates. For colour, supplying a positive-definite Hermitian inner product and a normalised complex volume form for three internal complex directions reduces their frame symmetry to $SU(3)$. This is exact conditional bundle mathematics; its physical connection, action, matter map, confinement and mass gap remain Open.

Across these sectors, the concrete output is a set of exact conditional theorems, controlled proofs of class, reproducible stress tests and scoped no-go results. Future cone-matching, Bell, galaxy and cosmology completions must state their observable predictions and rejection criteria in advance; failure then rejects the named completion, not automatically the whole ECT programme.

The aim of the companion is therefore not to present a completed unification, but to explain one common condensate architecture and show precisely which steps are established, which depend on supplied closures or matching, which are phenomenological, and which remain research programmes.

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Reader’s orientation

ECT begins with a change in explanatory order. Instead of taking Lorentzian spacetime, particles and quantum states as primitive, it asks whether they can be reconstructed as effective structures of a deeper medium. The primitive model object is a real scalar field Φ on a four-dimensional Euclidean manifold. At that level there is no physical time coordinate, no causal cone, no particle spectrum and no observer-independent split into space and time.

The first additional input is an ordered branch in which the scalar field has a nonzero gradient. In the technical preprint this adopted branch datum is called P4. Its direction has an exact $O(3)$ stabiliser, but that group-theory fact does not make the distinguished coordinate physical time. A separately supplied scalar principal tensor can be classified as hyperbolic in a stated coefficient domain; it then supports conditional relativistic-form scalar wave equations. A physical clock, a universal metric and the matter, photon and tensor cones still require their own actions and matching maps.

This distinction sets the structure of the whole programme. ECT asks which stationary action and state could realise the ordered branch; how gravity and matter could acquire a common metric and source law; how coherent amplitudes, Hilbert-space structure, environmental records and probabilities could be reconstructed; and how cosmological and galactic observables could follow from one owner-complete action-to-data chain. The present work closes only selected mathematical steps inside supplied models. The gravity, quantum-measure, physical-colour and global cosmology programmes are not already derived from one scalar action.

The companion follows those questions in layers. It first introduces the Euclidean medium and the conditional scalar characteristic structure, then turns to gravity and cosmology, quantum and open-system reconstruction, and a separately adopted internal $SU(3)$ bundle used to organise the colour completion problem. Detailed derivations and full evidence remain in the technical preprint; the purpose here is to make the dependency structure and physical meaning readable.

Claim levels. Level A means an exact result in the declared model and assumptions with an explicit reproducible derivation; checks verify but do not replace that derivation, and it is not automatically a physical ECT result. Level B marks an adopted structural input or a conditional closure with declared missing physical steps. Level C marks a fit, benchmark, stress test or candidate mechanism. Open marks a missing physical owner or unresolved programme. A reduced action coefficient becomes a physical action only after its action-valued normalisation is supplied; the operational action scale S_0 is then separately calibrated by $S_0 = \hbar$, not derived topologically.

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Glossary of abbreviations and symbols

This companion article uses a compact set of abbreviations and symbols; the fully technical glossary is in the preprint (Tables of Symbols and Abbreviations). The items below cover the companion itself.

Abbreviations.

ECT	Euclidean Condensate Theory (this work)
SSB	Spontaneous symmetry breaking
EFT	Effective field theory
GR	General Relativity
QFT	Quantum field theory
QM	Quantum mechanics
QED	Quantum electrodynamics
QCD	Quantum chromodynamics
SM	Standard Model of particle physics
EW	Electroweak (sector)
RG	Renormalisation group
GEE	Generalised Einstein equations
PES	Principle of Euclidean Stationarity
PES-R	Response–persistence/record taxonomy; not a universal persistent-sector selection theorem
CDE-Φ	Cohesive defect elasticity: post-order branch-EFT acceptance criterion, not a pre-physical postulate or action owner
BR1	Branch rule 1 (supplied coherent-sector closure; not a dynamical branch-selection rule)
OP	Open problem (labelled OP- <i>name</i> in the preprint)
DP	Dynamical principle
LIV	Lorentz-invariance violation
CPT	Charge–Parity–Time (symmetry theorem)
WKB	Wentzel–Kramers–Brillouin (semi-classical approximation)
OS	Osterwalder–Schrader (axioms / reconstruction theorem)
RP	Reflection positivity
KG	Klein–Gordon (equation)
CP	Charge–Parity (symmetry)
CKM	Cabibbo–Kobayashi–Maskawa (quark-mixing matrix)
PMNS	Pontecorvo–Maki–Nakagawa–Sakata (lepton-mixing matrix)
MSW	Mikheyev–Smirnov–Wolfenstein (matter effect in neutrino oscillations)
VEV	Vacuum expectation value
UV	Ultraviolet (high-energy regime)
NLO	Next-to-leading order
PPN	Parametrised post-Newtonian (formalism)
WEP	Weak equivalence principle
BH	Black hole
GW	Gravitational wave
CMB	Cosmic microwave background
BAO	Baryon acoustic oscillations

Abbreviations (continued).

CDM	Cold dark matter (as in Λ CDM)
DM	Dark matter
DE	Dark energy
JWST	James Webb Space Telescope
ISW	Integrated Sachs–Wolfe (effect / cross-correlation)
CC	Cosmic chronometers (differential-ages $H(z)$ probe)
RSD	Redshift-space distortions
SH0ES	Supernovae and H_0 for the Equation of State (local H_0 programme)
SPARC	Spitzer Photometry and Accurate Rotation Curves (galaxy sample)
BTFR	Baryonic Tully–Fisher relation
RAR	Radial acceleration relation
MOND	Modified Newtonian dynamics
NFW	Navarro–Frenk–White (dark-matter halo profile)
FLRW	Friedmann–Lemaître–Robertson–Walker (metric)
DESI	Dark Energy Spectroscopic Instrument
LISA	Laser Interferometer Space Antenna
LIGO	Laser Interferometer Gravitational-Wave Observatory
LSST	Legacy Survey of Space and Time (Vera C. Rubin Observatory)
LHC	Large Hadron Collider
GUT	Grand unified theory
BMV	Bose–Mazumdar–Morley / Marletto–Vedral (gravity-induced entanglement proposals) [1, 2]
BMS	Bondi–Metzner–Sachs (asymptotic symmetry group at null infinity)
GIE	Gravity-Induced Entanglement
HPS	Hawking–Perry–Strominger (soft-hair proposal)
LOCC	Local Operations and Classical Communication
PCCG	Post-quantum Classical Gravity (Oppenheim hybrid framework)
QES	Quantum Extremal Surface

Principal symbols.

Φ	Fundamental scalar field on the 4D Euclidean manifold (the Φ -medium).
X^A, n_A	Euclidean coordinates / composite unit director $n_A = \partial_A \Phi / \partial \Phi $ where the gradient is nonzero; P4 adopts its aligned ordered-branch value, while the stationary action, state and selection history remain Open.
μ^2, λ	Bare-potential mass and self-coupling parameters ($V = -\frac{\mu^2}{2} \Phi^2 + \frac{\lambda}{4} \Phi^4$).
ϕ_0	Bare-potential minimum $\sqrt{\mu^2/\lambda}$; independently matched through the Level-B ansatz $\phi_0 = \zeta_\phi \bar{M}_{\text{Pl}}$, $\zeta_\phi > 0$, $\zeta_\phi = O(1)$, whose microscopic/tensor explanation remains Open.
u_0	Supplied P4 nonzero-gradient amplitude, GeV^2 ; the brackets in $\langle \partial_A \Phi \rangle$ denote the declared coarse-grained datum, not a vacuum expectation value derived from an owned state.

Principal symbols (continued).

α, β	Independently supplied dimensionless anisotropic/isotropic stiffness couplings; $\hat{c}_* = 1$, equivalently $\alpha = 2\beta$, is the Level-B coefficient-ratio benchmark, while $\beta = 1$ is a separate overall normalisation convention. The pair $(\alpha, \beta) = (2, 1)$ combines those two choices; it is not selected by common field normalisation or by dynamics. Their realised physical values and domain remain Open.
$\hat{c}_*$	Dimensionless scalar principal slope in common-length coordinates, $\hat{c}_*^2 = \beta/(\alpha - \beta)$; $\hat{c}_* = 1$ is a Level-B unit-slope coefficient benchmark, not a field-rescaling identity.
c_u	Independent conversion speed between the Euclidean length coordinate and a physical clock.
c_{char}	Physical scalar characteristic speed after the clock map, $c_{\text{char}} = c_u \hat{c}_* / \mathcal{N}_\Phi$; $c_{\text{char}} = c$ is the adopted Level-B scalar-clock calibration, while equality to photon and tensor cones remains Open.
v_2	Electroweak matching scale ≈ 246 GeV (second energy-dimension scale).
$\mathcal{I}_{\text{EW}}$	Logarithmic bookkeeping ratio $\mathcal{I}_{\text{EW}} = \ln(\phi_0/v_2) \approx 36.8$ for the displayed matched scales; it organises OP-EW-scale but is not a derived ECT hierarchy mechanism.
$M_G, \bar{M}_{\text{Pl}}$	Conditional tensor-EFT scale / reduced Planck mass $\approx 2.435 \times 10^{18}$ GeV; $\kappa_n \rightarrow M_G^2$ Open.
S_0	Generic physical action normalisation used only after an owner map; the equality $S_0 = \hbar$ is the adopted Level-B physical calibration, while the microscopic owner and cross-sector universality remain Open.
$\iota_0^{\text{EFT}}, \iota_{0,\text{bal}}^{\text{eff}}$	Dimensionless fixed-core and balanced phase-route coefficients; not actions and not interchangeable.
$K_\vartheta^{\text{1D}}, \mathcal{T}_{\text{loop}}$	Reduced compact-modulus metric and line functional entering the conditional T1 reduction; one common 4D owner is required.
$\iota_{0,\text{bal}}^{(\vartheta)}$	Dimensionless modulus-route coefficient $\sqrt{2K_\vartheta^{\text{1D}}\mathcal{T}_{\text{loop}}}$; distinct until an owner proves a map.
$\Sigma_\theta, \Sigma_\vartheta$	Independent finite positive owner-specific action-dimension normalisations; only $\Sigma_\theta \iota_0$ or $\Sigma_\vartheta \iota_{0,\text{bal}}^{(\vartheta)}$ can map to operational S_0 .
κ_b	Reduced conditional carrier-bending coefficient after extracting Σ_ϑ ; the physical action-valued coefficient is $\Sigma_\vartheta \kappa_b$. When present it changes the reduced minimum and splitting comparison.
$G_N, \hbar, h_P, c$	Observed gravitational, quantum, and luminal constants used as distinct matching targets, with $h_P = 2\pi\hbar$; no common derivation from P1–P6 is established.
G_*	Natural-unit scalar-action normalisation $(8\pi\bar{M}_{\text{Pl}}^2)^{-1}$ used with natural-unit $T_{\mu\nu}$; $G_*^{\text{SI}} = \hbar c^5/[8\pi(\bar{M}_{\text{Pl}}[\text{J}])^2]$ is a separate conversion target. Its match to observed G_N requires a local response calculation.
$G_{\text{div}}(z)$	Natural-unit divided-equation coefficient $G_*/F(z)$; $G_{\text{div}}^{\text{SI}} = G_*^{\text{SI}}/F$ only after the separate conversion, and neither is by itself a locally measured or cosmological response.
G_{Cav}	Natural-unit weak-body response after scalar canonical coupling, range and body charges are fixed; $G_{\text{Cav}} = G_{\text{div}}(1 + \alpha_0^2)$ only in the unscreened light-scalar limit.

Principal symbols (continued).

$\Lambda_{\text{div}}(z)$	Divided-equation potential coefficient $U/[\bar{M}_{\text{Pl}}^2 F]$; the observed expansion response requires the full background, branch and physical metric/clock map.
$\mathcal{X}_{\mu\nu}^{\text{cand}}[n]$	Candidate anisotropic tensor form used only as an external proxy; not an ECT stress tensor.
$T_{\mu\nu}^{(n)}$	Orientation stress defined only by $T_{\mu\nu}^{(n)} = -2(\sqrt{-g})^{-1}\delta S_n/\delta g^{\mu\nu}$ for a specified covariant $S_n[g, n, \lambda_n]$; presently Open.
$\Theta_{\mu\nu}$	Reserved shorthand only after such a variation; absent from the displayed scalar-only metric equation, before and after division by F .
ϕ_u	Amplitude coordinate $\beta_\phi^{-1} \ln(u/u_\infty)$.
ψ_χ	Anisotropy coordinate $\ln(\chi/\chi_{\text{vac}})$; no $u \leftrightarrow \chi$ bridge is derived.
q	Generic scalar coordinate in the coordinate-invariant scalar audit; it does not identify u with χ .
$\mathcal{A}(q)$	Scalar kinetic combination $K + 3f_{,q}^2/(2f)$.
$\mathcal{E}(q, \rho_{\text{m}})$	Scalar local-response function $V_{,q} - 2(f_{,q}/f)V - (f_{,q}/2f)\rho_{\text{m}}$.
m_{loc}	Conditional local homogeneous response mass $m_{\text{loc}}^2 = \mathcal{E}_{,q}/\mathcal{A}$; not a finite-body screening mass.
$H_{\text{ECT}}(a)$	Physical ordered-branch expansion history; conditional two-slope curve computed, microscopic completion and physical map Open.
$f\sigma_8(z), r_s$	Restricted growth and imported-recombination sound-horizon diagnostics computed; full owners Open.
t_0^{ECT}	Matter-congruence duration from the branch-formation front; named-completion estimate exists after one dimensional scale calibration, while the physical depth/lapse map remains Open.
a_M	HRC gravity-bridge acceleration scale; $a_{M0} = cH_0/(2\pi)$ is a matched input, while its ECT origin and any environment/history law are Open.
$L_{\text{gal}} \equiv r_*$	Conditional transition length $\sqrt{G_N M_{\text{bar}}/a_M}$ inside the conditional HRC gravity functional; not an energy scale and not derived from the scalar ansatz.
ΛCDM	Standard cosmological model: cold dark matter + cosmological constant.
C_I	Global integration/state density in the separately supplied trace-free completion; its ECT action and observable map are Open.
$\rho_{\text{late},0}^{(\text{GR ref})}$	GR-reference measured-source convention $3\Omega_{\text{late},0}^{(\text{GR ref})} \bar{M}_{\text{Pl}}^2 H_0^2$ after the metric and Planck calibration are fixed; generic scalar-tensor constraints also contain $F_0, \dot{F}_0$ and any orientation stress.
n^A	Ordered unit direction in 4D Euclidean space; it gives an exact fixed-determinant orientation identity but not by itself a trace-free metric action.
$m_{\text{guide}}^{\text{DP}}, m_{\text{cross}}^{\text{coll-DP}}$	Distinct external quantities: the fixed-radius unit-coefficient guide $m_{\text{guide}}^{\text{DP}} \sim [\hbar R/(G_N \tau_{\text{exp}})]^{1/2}$ and the separately supplied unit-coefficient collisional comparison $m_{\text{cross}}^{\text{coll-DP}} \sim 3\hbar n_{\text{gas}} v_{\text{th}}/(4G_N \rho)$. Neither is an ECT prediction. They are distinct from the geometry-resolved hard-sphere point $C_{\text{DP}}(1) = 51/160$, $\tau_{\text{DP}}^{\text{ext}} = 70.6 \text{ ms}$, whose coefficient must not be inserted silently into the unit-coefficient formulas.
$D^R, N, N^{(S)}$	Retarded response kernel, action-normalised time-domain PES-R noise kernel, and its stationary spectral Fourier transform, respectively; D^R is physically distinct from the two noise representations, and all are state/channel dependent.

Principal symbols (continued).

$s, \hat{\Lambda}, P_s$	HRC dimensionless control parameter, constrained stability operator, and recruited spectral projector $P_s = \mathbf{1}_{(-\infty, s)}(\hat{\Lambda})$.
C, τ	Positive static HRC compliance and positive semifinite trace in the supplied response model.
$\mathcal{Q}, n_{\text{ch}}, r_{\text{ch}}$	HRC bounded channel-occupation operator, channel count, and selected-projector rank; all are distinct from the PES-R noise kernel $N^{(S)}$ and from global radial variables.
$\Xi_{ab}^{\text{full}}(T)$	Full log-modulus diagnostic $-\ln F[a, b; T] $ for a named preparation and protocol; it need not be nonnegative for a generic correlated initial preparation.
$\Phi_{ab}(T)$	Nonnegative dimensionless pairwise record/dephasing exponent for named alternatives, channel, state and observation protocol, set equal to Ξ_{ab}^{full} only after full contractivity is established; the positive Gaussian bulk contribution is Φ_{ab}^{bulk} . It is not an outcome, entropy or metric-signature variable.
$\Gamma_{\text{tr}}[q; \eta]$	Translation/deformation susceptibility along a declared direction η , finite only when the directional record derivative lies in the noise quadratic-form domain with finite $N^{1/2}$ -norm; otherwise only extended lower/upper diagnostics are available. It is not a mean force or lifetime.
$\gamma_a, \Gamma_{E,a}$	Decay/leakage rate of an admissible mode and its energy width $\Gamma_{E,a} = S_0 \gamma_a$ after the action-scale matching.
$\Sigma[\gamma]$	Protocol-specific stochastic entropy production $\ln(dP_F/dP_R^\Theta)(\gamma)$ on a common forward path space after transporting the reverse measure. One-way $P_F \ll P_R^\Theta$ defines it P_F -a.s.; reciprocal/unit-IFT identities require $P_F \sim P_R^\Theta$, and a pointwise path-probability ratio is only common-density shorthand.
Δn_A	Orientation shift; a map to physical gravitational memory has not been derived.

Part I

Foundations: From the Φ -Medium to Emergent Laws

1. Motivation: why question the foundations?

Modern physics rests on two extraordinarily successful pillars: General Relativity and quantum field theory. They coexist in quantum field theory on curved spacetime, semiclassical gravity and the controlled low-energy effective field theory of gravity. Einstein gravity is perturbatively nonrenormalisable as a fundamental quantum field theory, but this does not make its low-energy EFT uncontrollable. The unresolved questions concern a ultraviolet completion, a self-consistent dynamical quantum-spacetime description and the measurement/gravity interface.

Standard formulations of General Relativity and quantum field theory take a Lorentzian spacetime structure as part of their starting data, while several quantum-gravity programmes investigate whether spacetime or causal structure is emergent. The spacetime interval

$$ds^2 = -c^2 dt^2 + dx^2 + dy^2 + dz^2 \quad (1)$$

contains one coordinate with a minus sign and three with a plus. For a physical spacetime metric this Lorentzian signature $(-, +, +, +)$ supplies the time/space distinction, hyperbolic propagation and causal ordering. In ECT the derived statement is narrower: one supplied scalar principal tensor has that eigenvalue signature; the physical clock, metric and cross-sector causal structure remain Open. An irreversible thermodynamic arrow requires the separate retarded influence-functional/decoherence closure. But *why* does this minus sign exist?

ECT asks the narrower question of whether a physical Lorentzian observable structure can descend from its proposed Euclidean medium. This is not a literature-wide novelty claim and the displayed scalar characteristic tensor does not yet answer the physical question.

The group-theoretic route must be stated carefully. Both $O(4)$ and $O(1, 3)$ have six generators; they are distinct compact and non-compact real forms, and neither is a lower-symmetry subgroup of the other. For the vector datum supplied by P4, the exact group-theoretic statement is only that its stabiliser is $O(3)$; the present theory does not yet derive a stationary physical state or its spontaneous selection. Independently, when the supplied scalar principal tensor satisfies $\beta(\beta - \alpha) < 0$ (equivalently $\alpha > \beta > 0$ on the retained $\beta > 0$ component), its eigenvalue signs are hyperbolic. Thus ECT asks whether one Euclidean medium can dynamically own both a physical $O(4) \rightarrow O(3)$ ordered state and a common hyperbolic observable structure; neither ownership follows from the displayed group algebra alone, and Lorentzian signature is not derived by an $O(4) \rightarrow O(1, 3)$ symmetry breaking.

Furthermore, path integrals, instantons, lattice gauge theory and constructive reconstruction methods make extensive use of Euclidean formulations. Their computational and reconstruction utility under stated hypotheses does not imply that Euclidean ontology is fundamental. ECT adopts that ontological hypothesis independently and must test it through its own physical reconstruction.

Euclidean Condensate Theory (ECT) explores whether observed Lorentzian spacetime can arise as an effective, hyperbolic observable structure of a Euclidean medium. The present derivation

is narrower than that programme: P4 supplies a Euclidean direction, while an independently supplied scalar principal tensor is hyperbolic exactly when $\beta(\beta - \alpha) < 0$; on the retained $\beta > 0$ component this is $\alpha > \beta > 0$. A physical clock/lapse, universal observable metric, gravity action and common photon, matter, fermion and tensor cones remain Open completion targets. The wider architecture investigates how coherent quantum structures, irreversible reduced dynamics, constants and particle sectors might be matched without treating those targets as already derived.

2. The Φ -medium: what it is and what it is not

At the foundation of ECT lies a single real scalar field Φ , defined on a four-dimensional Euclidean manifold $\mathcal{M}^4$. This manifold has no built-in time: all four coordinates $X^A = (x, y, z, w)$ enter with the same positive-definite Euclidean metric δ_{AB} . There is no preferred direction, no causal structure, no light cone.

In the minimal P1–P6 sector, Φ is the sole field. Its dynamics is governed by an $O(4)$ -invariant action with a self-interaction potential — a double-well potential familiar from the Higgs mechanism and superfluidity. In many ways Φ is analogous to the order parameter of a superfluid or ferromagnet, but operating at a much deeper level: it describes the minimal medium sector from which spacetime is intended to emerge. The reverse-engineered P7 extension adds an internal colour module and is not part of this scalar-only basis.

It is essential to understand what Φ is *not*. It is not a quantum field defined on a pre-existing spacetime — that would presuppose the very thing ECT aims to derive. It is not a metric or gravitational field; physical metric and gravity remain separate completion programmes rather than automatic collective properties. It is not an inflaton, Higgs field, or dark matter particle, though candidate completions may assign some modes analogous roles at specific scales. And it is not a completed “hidden variable” model of quantum mechanics. Φ is a *pre-spacetime*, *pre-quantum*, *pre-gravitational* model entity.

A subtle but critical point: the “physics” of Φ at the fundamental Euclidean level cannot be described using energy, force, or potential in their usual sense. Those concepts presuppose Lorentzian time, which does not yet exist. The reduced Euclidean functional $\mathcal{I}_E^{\text{P3}}[\Phi]$ and its separately normalised physical action $S_E^{\text{P3,phys}} = A_{\text{P3}}\mathcal{I}_E^{\text{P3}}$ are functionals of field configurations on the four-dimensional manifold, not time integrals of a Lorentzian Lagrangian. Importing energy language into the pre-time Euclidean domain is a common source of confusion and must be avoided.

The Φ -medium has specific declared properties, explicitly stated in the technical preprint (S1–S10): it is continuous and differentiable (S1); it is scalar — no internal spin or index structure at the fundamental level (S2); it fills the entire four-dimensional Euclidean space (S3); its dynamics is purely relational — no external clock, no preferred reference frame (S4); it admits ordered and disordered configurations (S5); no single configuration is privileged a priori (S6); it supports both smooth and singular (defect) configurations (S7); it is deformable and its response is read through invariant relational data rather than passive relabelling (S8); ordering need not be globally uniform (S9); and candidate topological sectors compatible with a declared configuration target may be admitted (S10). S10 does not identify the complete physical vacuum/configuration space or prove that every abstract target-space class has a finite-action physical representative.

The proposed property S11 is split to avoid conflating kinematics with dynamics. **S11a** is

conditional infinitesimal local deformability: sufficiently small regular perturbations that preserve the declared boundary data and topology remain admissible. This is a tangent-space fact, not a new force law. **S11b** is the stronger candidate that some ordered branches possess finite-action transition paths; it does not assert that relaxation traverses them. Neither statement contains gravity. ERP- Φ below is a narrower constitutive specialisation and remains an extension rather than a consequence of P1–P6.

3. The six postulates: formalising the Φ -medium

The working postulates P1–P6 are adopted defining inputs of the minimal scalar realisation. A postulate is not demoted to Open merely because it is not derived from an earlier postulate: Open status instead attaches to a missing constructive owner, state/selection mechanism, microscopic descent or observable map. Exact consequences of named inputs are Level A conditional on those inputs; adopted structural or physical matchings are Level B; fits, toys and benchmarks are Level C.

P1 — Four-dimensional Euclidean arena (formalises S1, S3). The fundamental arena is an adopted connected four-dimensional flat Euclidean carrier $\mathcal{M}^4$ with metric δ_{AB} . The local theory is formulated in flat charts. Whenever global equivalence of points is used, P1 additionally adopts a homogeneous boundary/topology class, canonically $\mathcal{M}^4 \simeq \mathbb{R}^4$; a generic flat quotient or a carrier with a boundary need not have that global symmetry. No direction is distinguished and no time coordinate is fundamental.

P2 — $O(4)$ -invariant action (formalises the symmetry of S3, S4). The reduced P3 functional is $O(4)$ invariant, $\mathcal{I}_E^{\text{P3}}[\Phi] = \mathcal{I}_E^{\text{P3}}[\Lambda \cdot \Phi]$, with $(\Lambda \cdot \Phi)(X) = \Phi(\Lambda^{-1}X)$. On the homogeneous flat carrier, translation invariance follows separately from the absence of explicit X -dependence; together translations and rotations form $E(4) = \mathbb{R}^4 \rtimes O(4)$. Thus $O(4)$ is the linear isotropy group, not the full isometry group. The physically normalised counterpart $S_E^{\text{P3,phys}} = A_{\text{P3}} \mathcal{I}_E^{\text{P3}}$ has the same symmetry for constant $A_{\text{P3}} > 0$; symmetry alone fixes neither that prefactor nor a path-integral measure.

P3 — Minimal scalar microdynamics (formalises S1, S2). The fundamental field is a single real scalar $\Phi(X)$ with a minimal local action:

$$\mathcal{I}_E^{\text{P3}}[\Phi] := \int d^4X \left\{ \frac{1}{2} (\partial_A \Phi)^2 + V(\Phi) \right\}, \quad S_E^{\text{P3,phys}} := A_{\text{P3}} \mathcal{I}_E^{\text{P3}}, \quad A_{\text{P3}} > 0, \quad (2)$$

$$V(\Phi) = -\frac{\mu^2}{2} \Phi^2 + \frac{\lambda}{4} \Phi^4, \quad \mu^2 > 0.$$

The integral is dimensionless in the printed natural-unit convention. A separately supplied Euclidean measure uses $e^{-\kappa_{\text{P3}} \mathcal{I}_E^{\text{P3}}} = e^{-S_E^{\text{P3,phys}}/\Sigma_{\text{PI,P3}}}$, where $\Sigma_{\text{PI,P3}} > 0$ and $\kappa_{\text{P3}} = A_{\text{P3}}/\Sigma_{\text{PI,P3}} > 0$. The equality $\kappa_{\text{P3}} = 1$ is only a possible normalisation convention; neither action owner, their ratio, nor a map to S_0 or $\hbar$ is derived. The double-well potential has two minima at $\pm\phi_0$, where $\phi_0 = \sqrt{\mu^2/\lambda}$. The single real scalar is an adopted economical one-component field content, not a uniqueness theorem; isotropic multiplets or additional fields are possible in named completions. P3 contains no independent director. Where $|\partial\Phi| > 0$, the composite coordinate $n_A = \partial_A \Phi/|\partial\Phi|$ adds no second local degree of freedom, but it is singular at zero gradient and changes the operator and constraint bookkeeping. Treating n_A as an independent dynamical field would enlarge P3 and requires a separately supplied action.

P4 — Adopted physically relevant ordered-gradient branch (formalises the ordered-sector input of S5 and S9). ECT adopts, as a Level-B structural postulate, the physically relevant branch $\langle \partial_A \Phi \rangle = u_0 \delta_{Aw}$, $u_0 > 0$. Conditional on this nonzero vector, its stabiliser inside $O(4)$ is exactly $O(3)$ (Level A group mathematics). The displayed quartic P3 action does not make the affine representative stationary and therefore does not derive P4. The complete ordered-sector action, state, stability domain, formation and selection history remain Open.

P5 — Deformability and relational response of the ordered branch (formalises S8). The ordered medium can deform and reconfigure. Physical differences are read through invariant strain, gradient, defect, boundary-response and, once a bundle exists, holonomy data. Passive relabelling does not create a new state, but P5 does not declare arbitrary pointwise changes of the compact phase or physical director to be gauge transformations.

This is analogous to the response of familiar media: a crystal is probed through strain, a superfluid through phase differences and circulation, and a nematic through director distortions relative to boundaries. The common property is deformability with relational readout, not a common microscopic gauge structure.

A local phase redundancy, compensating connection and physical gauge field require a separately supplied bundle, quotient (or reduced Hessian), kinetic action, state and matter map. The constrained spacetime director $n_A = \partial_A \Phi / |\partial \Phi|$ remains physical relative to other fields; P5 does not make it a local gauge orbit (Section 11.1).

P6 — Multiplicity of configurations and non-uniform ordering (formalises S6, S9, S10). No single configuration is singled out a priori. Ordered configurations need not be globally uniform: spatially varying condensate backgrounds, domain structures, and transition regions are all admissible.

In addition to these six postulates, the theory uses a supplied effective branch rule:

BR1 — Smooth coherent sector (adopted Level-B effective closure). BR1 declares a smooth, non-vanishing complex collective field $\Phi_{\text{eff}} = \rho e^{i\theta}$ on the retained effective sector. Single-valuedness of Φ_{eff} on a closed contour that does not intersect a defect core then defines a map $\mathcal{C} \rightarrow S^1$; a locally lifted phase obeys $\oint_{\mathcal{C}} \partial_A \theta dX^A = 2\pi n$, $n \in \mathbb{Z}$. That winding theorem is exact inside the declared non-vanishing compact-phase closure. BR1 defines the calculation sector; it does not dynamically select a physical branch. Microscopic descent, formation and selection of the BR1 domain remain Open. The existence of the complex field and its phase stiffness are not consequences of the real-scalar action P3 plus P4. A fixed-core evaluation may define ι_0^{EFT} , and a separate optional closure may additionally supply a positive line tension; neither input is part of BR1 and both physical ECT owners remain Open (Section 9).

ECT does not take external time as primitive. P4 adopts the physically relevant ordered-gradient branch as a Level-B structural input and supplies its analysis chart; the conditional scalar characteristic signature requires an independently supplied principal tensor and coefficient domain. The remaining items often grouped under “emergence” have different status: Hilbert-space and Schrödinger routes hold only in declared supplied subsectors; Born weights, physical clocks, universal signal speed and gravity remain Open owners; the galactic HRC relation is a conditional static diagnostic; and dark-sector particle content is not fixed.

4. Supplied ordered-gradient structure and conditional scalar signature

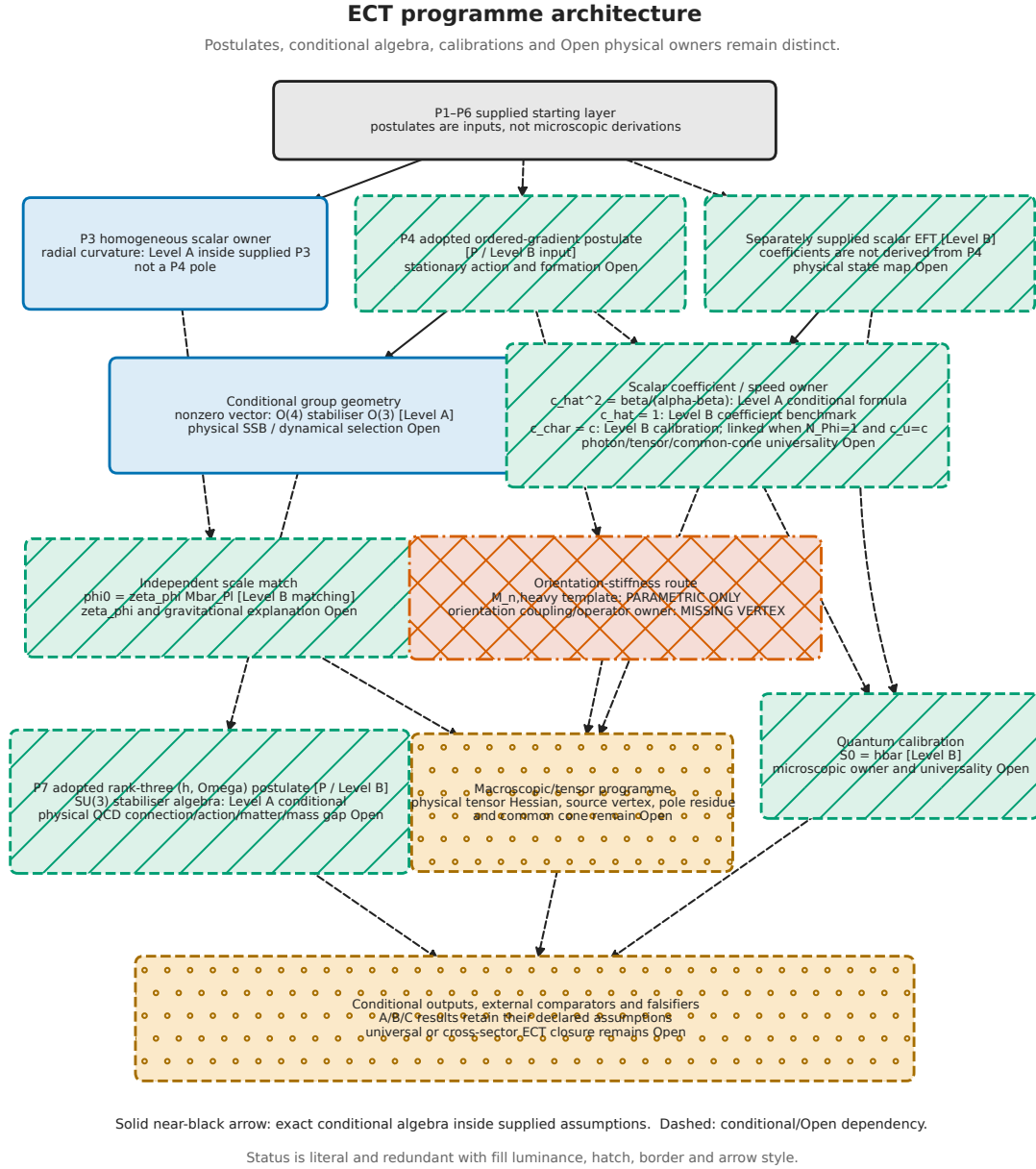


Figure 1: Architectural dependency map for the minimal P1–P6 scalar sector. P4 supplies a nonzero vector datum with stabiliser $O(3)$; within the declared local algebraic scalar-operator inventory, residual symmetry fixes the two-tensor basis but not its coefficients. The displayed downstream geometric and coherent branches are conditional programmes, not simultaneous Level-A consequences: physical SSB and branch formation, the realised coefficient sign, physical clock/metric/common cones, gauge/tensor sectors and record dynamics retain their individual B/Open owners.

Readability note. This portrait figure is the reader-scale map. Its colours are redundant with literal status text and border/edge style; no arrow upgrades a node.

Spontaneous symmetry breaking (SSB) is one of the most powerful organising principles in physics. A spherical magnet has no preferred direction, until it cools below the Curie temperature

and all spins align. A perfectly round ball on the tip of a Mexican hat has no preferred position, until it rolls down and selects one valley. In both cases, the underlying laws are symmetric, but the realised state is not. These are analogies for a possible completion of P4, not a derivation of such an event in ECT.

P4 adopts a physically relevant nonzero vector datum whose stabiliser in $O(4)$ is exactly $O(3)$ (Fig. 1). Three transverse directions remain equivalent; one aligned coordinate (w) is specified as a candidate time-length direction. The physical stationary state, spontaneous $O(4) \rightarrow O(3)$ selection and clock interpretation are separate Open maps.

The ordered-gradient chart is an organising input, not a derived transition or a derivation of all familiar physics. The bare Euclidean condensate distinguishes no time or causal cone. Once the P4 datum and independent scalar-EFT coefficients are supplied, one can classify the scalar principal tensor. A Lorentzian scalar cone requires $\beta(\beta - \alpha) < 0$; on the retained $\beta > 0$ component this is $\alpha > \beta > 0$. Formation of the chart, realised coefficient selection, the physical clock/metric, other-sector cones, excitation spectrum and decoherence hierarchy require separate owners and remain conditional/Open.

The candidate mechanism that would have to justify this chart is *gradient condensation*. Within the adopted one-scalar, first-derivative inventory, $\partial_A \Phi$ is the lowest-canonical-dimension local *polar-vector* order parameter. A nonzero scalar $\langle \Phi \rangle$ is $O(4)$ -invariant, whereas a supplied nonzero $\langle \partial_A \Phi \rangle = u_0 \delta_{Aw}$ has stabiliser $O(3)$. This is not a uniqueness theorem for all one-scalar order parameters: a nematic composite such as $\langle \partial_A \Phi \partial_B \Phi - \delta_{AB} (\partial \Phi)^2 / 4 \rangle$ could select an unoriented axis while $\langle \partial_A \Phi \rangle = 0$, with a different target and stabiliser. In the present P4 notation the brackets denote the adopted coarse-grained datum, not a state-owned vacuum expectation value. The group kinematics is exact conditional on that datum; it does not establish a transition, vacuum or spontaneous selection.

The gradient field can be decomposed into an amplitude and a direction:

$$\partial_A \Phi = u \cdot n_A, \quad u = |\partial \Phi| \geq 0, \quad n^A n_A = 1. \quad (3)$$

In the supplied ordered-sector chart, $\langle u \rangle = u_0$ and $\langle n^A \rangle = \delta^A_w$. This decomposition organises the conditional EFT; its physical state and selection remain Open. Each component plays a distinct role:

The **amplitude sector** contains the homogeneous P3 radial coordinate with $m_{\Phi, \text{rad}}^2 = 2\mu^2$ (Level A inside P3). A heavy, short-ranged ordered-branch pole is only an OP-grad/F7 benchmark closure. Its stationary P4 action, Hessian, pole, correlation length, gravitational coupling and matching to the P3 coordinate remain Open; no independent Planck-length theorem is used.

The **orientation sector** (δn^A): the abstract orbit of the supplied nonzero vector, $O(4)/O(3) \simeq S^3$, admits three director coordinates. Its identification with a physical vacuum manifold is Open. On a smooth patch of the strict scalar-only exact pullback, integrability locally parametrises the linear soft variation by one scalar coordinate χ (Level-A conditional kinematics). This coordinate count does not by itself prove that χ is a propagating physical mode or exclude scalar modes in other completion sectors. A one-mode physical interpretation additionally requires a supplied reduced action, complete constraint/boundary quotient and physical projector (Level B); the scalar count of the completed theory remains Open. At fixed α, β , about homogeneous $\bar{n} = w$, and to first order in a tangent director perturbation, the fixed image has no linear TT/static-density seed: $h_{ij}^{\text{TT}} = h_{ww} = 0$. This is not a physical graviton, while quadratic, nonlinear, composite,

nonlocal, additional-field and dynamical-parameter routes remain parametric/Open. A physical spin-2 Hessian, vertex, TT projector and canonical normalisation remain Open.

The **phase mode** (θ , in the coherent sector $\Phi_{\text{eff}} = \rho e^{i\theta}$): connected to gauge fields and quantum dynamics.

The dynamical origin of gradient condensation from the bare action P3 is an Open theoretical programme (OP-grad in the preprint). A general local first-derivative scalar density may depend on both Φ and $X = (\partial_A \Phi)^2$ through $F(\Phi, X)$ and need not split as $P(X) + V(\Phi)$. Inside the restricted separable inventory, a pure-gradient action $\int P(X)$ can admit an affine stationary family when $P_X(u_*^2) = 0$; this is a conditional toy route. Once the P3 potential $V(\Phi)$ is retained, an affine nonconstant profile also requires $V'(u \cdot X + \Phi_c) = 0$ pointwise and is not a stationary solution of the displayed quartic P3 model. Thus neither restricted route owns the physical P4 state. Given the adopted P4 chart, conditional EFT calculations can nevertheless be organised. Recent sharpening in the preprint decomposes this programme into three recorded subproblems (OP3.1a–c) and localises the remaining OP3.1a gap to a single constraint-relaxation/coarse-graining bridge: the candidate shortcut bridges — the EFT-level orientation route in the strict integrable regime and the potential-sector route — have been computed and excluded as completions.

An important restricted result (the “ $P(X)$ no-orientation theorem” in the preprint) concerns the strict homogeneous scalar-gradient seed: its two-derivative Hessian does not supply an independent transverse orientation stiffness. This is an A-math statement under those assumptions, not a theorem about every $P(X)$ completion, constrained reduction, inhomogeneous background, extra field, or higher-gradient operator. A physical orientation stiffness may arise only after such additional owners are specified. A covariant $S_n[g, n, \lambda_n]$, its constraints, and its metric variation are still required before a physical orientation stress or gravitational source is obtained (Open).

5. Conditional Lorentzian classification in the supplied scalar branch

Once the P4 vector datum is supplied, residual $O(3)$ symmetry fixes only the local algebraic two-tensor basis $\{\delta^{AB}, n^A n^B\}$ within the declared scalar-operator inventory. The coefficients α, β , their realised domain and the physical action owner are independent inputs. The object that owns the displayed principal polynomial is the dimensionless characteristic quadratic form, not yet a physical action:

$$\mathcal{Q}_{2,\varphi} = \frac{1}{2} \int d^4 X \left[K^{AB} \partial_A \varphi \partial_B \varphi + m_{\text{eff}}^2 \varphi^2 \right], \quad K^{AB} = \beta \delta^{AB} - \alpha n^A n^B, \quad (4)$$

$$\mathcal{I}_{L,\varphi}^{(2)} = \varsigma_L \mathcal{Q}_{2,\varphi}, \quad S_{L,\varphi}^{(2),\text{phys}} = A_{L,\varphi} \mathcal{I}_{L,\varphi}^{(2)}, \quad A_{L,\varphi} > 0. \quad (5)$$

Here $A_{L,\varphi}$ is a finite, positive, spacetime- and field-independent action scale on the displayed quadratic patch. It is independent of the cone and is not identified with S_0 , $\hbar$ or another sector’s normalisation. Writing $\Delta = \alpha - \beta$, the full Lorentzian domain is $\beta \Delta > 0$, equivalently $\beta(\beta - \alpha) < 0$. Positive quadratic kinetic and gradient energy fixes $\varsigma_L = -\text{sgn } \beta$. This companion retains $\beta > 0$, $\Delta > 0$ and hence $\varsigma_L = -1$ for its explicit mostly-plus formulas. This sign and Legendre algebra are Level-A mathematics inside the supplied quadratic model; derivation of the realised component, relative sector sign and $A_{L,\varphi}$ from P1–P6 remains Open.

This tensor has eigenvalues β (triply degenerate, perpendicular to n^A) and $\beta - \alpha$ (along n^A).

On the full real coefficient plane the exact classification is

$$\begin{aligned}\beta(\beta - \alpha) > 0 & : \text{nondegenerate elliptic,} \\ \beta(\beta - \alpha) < 0 & : \text{Lorentzian/hyperbolic,} \\ \beta = 0 \text{ or } \alpha = \beta & : \text{degenerate.}\end{aligned}$$

For the rest of this companion, shorthand “ $\alpha > \beta$ Lorentzian window” means only the retained component with $\beta > 0$ and hence $\alpha > \beta > 0$. In that component the principal tensor has signature $(-, +, +, +)$ and a scalar characteristic cone; identifying it with the physical metric and the cones of matter, photons and tensors remains Open. The pair $(\alpha, \beta) = (2, 1)$ is a canonical supplied benchmark, not a vacuum-selection theorem. P4/SSB does not determine the realised coefficient region; physical branch selection remains Open/matched.

The dimensionless common-length slope is:

$$\hat{c}_* = \sqrt{\frac{\beta}{\alpha - \beta}}, \quad (6)$$

The dimensional notation is fixed as follows. The ordered Euclidean coordinate is the *time-length* $x^0 \equiv w$, not an SI time. Along an ordered worldline the physical proper time is defined by

$$d\tau = \frac{\mathcal{N}_\Phi dx^0}{c_u}, \quad (7)$$

Only for an aligned planar branch with constant unit lapse $\mathcal{N}_\Phi = 1$ and $c_u = c$ may one use the shorthand $t = \tau = w/c$. The unit-slope coefficient benchmark $\hat{c}_* = 1$ is a Level-B unit-slope choice, not a field-rescaling identity. In the general relation it and the scalar-clock target have distinct owners, and a different clock/conversion map can calibrate $c_{\text{char}} = c$ at another supplied slope. On the particular slice $\mathcal{N}_\Phi = 1$, $c_u = c$, however, they are algebraically linked: $c_{\text{char}} = c$ if and only if $\hat{c}_* = 1$. In general $c_{\text{char}} = c_u \hat{c}_* / \mathcal{N}_\Phi$; a time variable is never equated directly to the length coordinate. After the clock/lapse convention is fixed, ECT adopts $c_{\text{char}} = c$ as the Level-B scalar-clock calibration. This does not identify the photon, matter or tensor cones, whose kinetic owners and common-cone map remain Open.

The strict result is a conditional signature classification, not dynamical sign selection. Within the stated local algebraic scalar-operator inventory, residual symmetry fixes the tensor basis, and the exact full-domain condition $\beta(\beta - \alpha) < 0$ gives one sign opposite to the other three. On the retained component this is $\alpha > \beta > 0$. P4 fixes neither the coefficients nor this region, the physical clock, metric, photon/matter/tensor cones or a realised vacuum. The classification is A-math conditional; physical branch selection remains Open.

Comparison with other theories. In Einstein–aether theory (Jacobson, 2004), the anisotropy is introduced by a vector field; Hořava–Lifshitz gravity breaks Lorentz invariance explicitly; CDT imposes Lorentzian causal structure on the lattice; and the Hartle–Hawking Euclidean arena is a construction for a wave function. ECT instead studies a P4-supplied ordered scalar-gradient branch whose quadratic scalar operator has Lorentzian signature in a declared coefficient window. That scalar statement is exact inside the supplied EFT; a physical observable metric and the photon, matter and tensor cones remain Open.

6. Conditional equation inventory: Euclidean scalar problem, KG form, and supplied Schrödinger sector

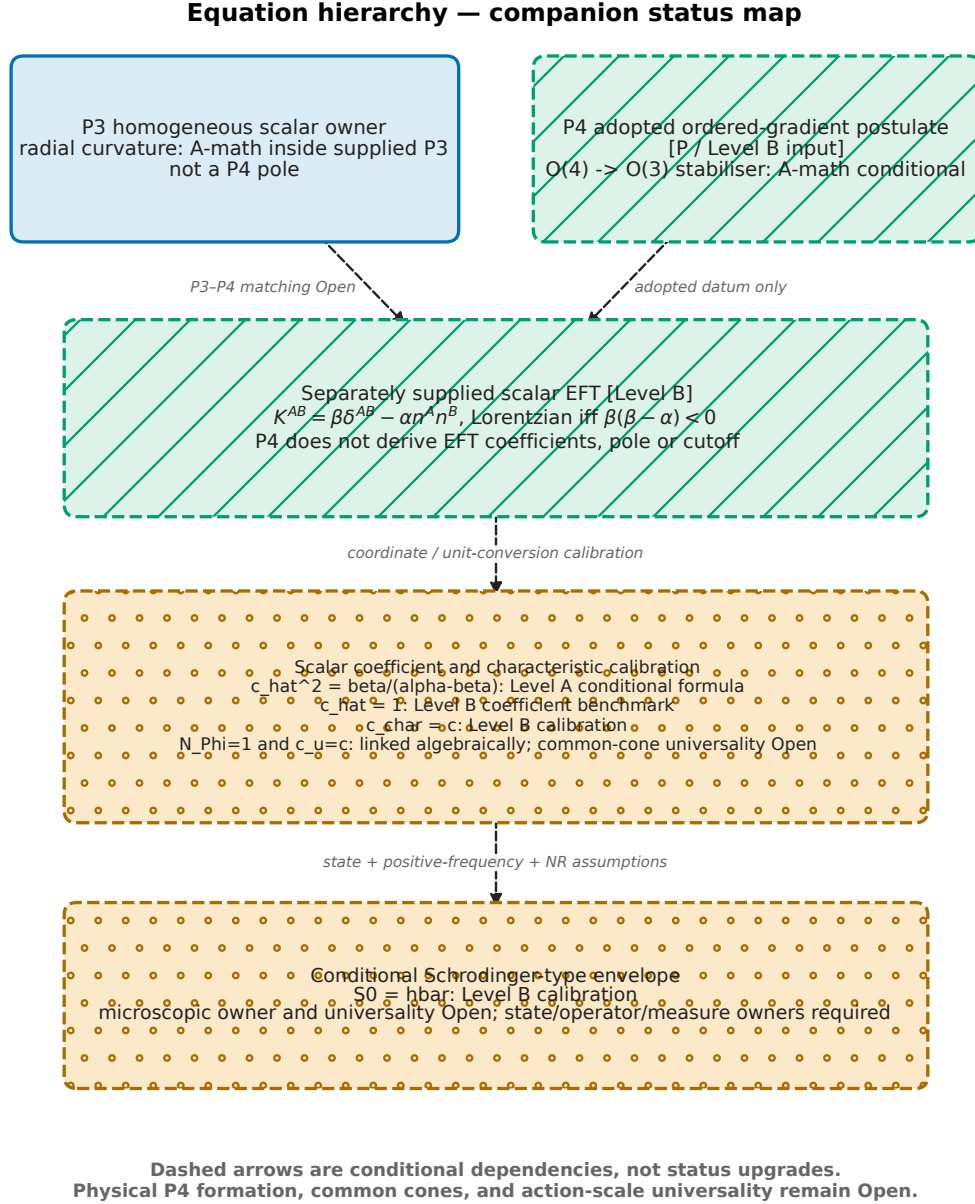


Figure 2: Status-separated equation inventory. Level 0 is the bare Euclidean scalar boundary problem; Level 1 is the supplied ordered-branch scalar equation; Level 2 is its local constant-lapse Klein–Gordon form after the clock map; and Level 3 is only a leading non-relativistic Schrödinger envelope after a norm-preserving complex law, Hamiltonian/domain, state and positive-frequency branch, weak interaction, slow-envelope regime, physical clock and $S_0 = \hbar$ calibration are separately supplied. The shorthand $t = \tau = w/c$ additionally requires an aligned planar patch with $\mathcal{N}_\Phi = 1$ and $c_u = c$. The separate $\hat{c}_*$ choice fixes only the scalar common-length slope. Arrows organise conditional reductions and never upgrade the literal status written in a node.

The conditional Lorentzian classification has a direct consequence for the supplied scalar equations (Fig. 2). The Euclidean field equation $\delta^{AB}\partial_A\partial_B\Phi - V'(\Phi) = 0$ is a four-dimensional *boundary-value problem*: boundary data specify that problem, but fix a unique solution only after existence, uniqueness and stability/well-posedness are proved for the stated nonlinear equation, domain and boundary class. Those global properties remain Open.

On an aligned locally homogeneous, frozen-coefficient constant-lapse patch, once the ordered branch and clock map are supplied, the scalar equation has the *Klein–Gordon form* $\partial_\tau^2\varphi - c_{\text{char}}^2\nabla^2\varphi + M_{\text{eff}}^2\varphi = 0$. The common-length and physical-clock gap coefficients are related by

$$\widehat{M}_{\text{eff}}^2 = \frac{m_{\text{eff}}^2}{\alpha - \beta}, \quad M_{\text{eff}}^2 = \frac{c_{\text{u}}^2}{\mathcal{N}_{\Phi}^2} \widehat{M}_{\text{eff}}^2 = \frac{m_{\text{eff}}^2 c_{\text{char}}^2}{\beta}.$$

Here M_{eff} is a physical-clock frequency/gap coefficient (with units of inverse time), not by itself a physical mass. Only after the standard relativistic matter interpretation, the same physical clock and an action unit $\hbar$ are supplied may one define a mass m by $M_{\text{eff}} = mc_{\text{char}}^2/\hbar$; neither the cone nor the clock map derives that matching. The symbol t may replace τ ; $t = w/c$ only in the canonical special case stated above. If the lapse or coefficients vary, additional derivative terms occur and the displayed constant-coefficient Klein–Gordon equation is only the local frozen-patch form.

After a norm-preserving complex evolution law, Hamiltonian and domain, state space, boundary/positive-frequency branch, the physical mass matching $M_{\text{eff}} = mc_{\text{char}}^2/\hbar$, and a weak interaction whose supplied leading envelope is V are separately admitted, write the complexified positive-frequency component as $\varphi_+ = e^{-imc_{\text{char}}^2 t/\hbar}\psi$ (with a real field reconstructed from the paired frequency components when applicable). In the slow-envelope regime

$$\epsilon_t = \frac{\hbar\|\partial_t\psi\|}{mc_{\text{char}}^2\|\psi\|} \ll 1, \quad \epsilon_p = \frac{\hbar\|\nabla\psi\|}{mc_{\text{char}}\|\psi\|} \ll 1, \quad \epsilon_V = \frac{\|V\|}{mc_{\text{char}}^2} \ll 1,$$

and after the second envelope-time derivative and interaction-specific relativistic corrections are consistently neglected, the leading equation is $i\hbar\partial_t\psi = -\frac{\hbar^2}{2m}\nabla^2\psi + V\psi$. This is Level-A asymptotic mathematics conditional on all stated inputs and a controlled residual, not an exact equality of the original equations; a quantitative error bound requires the supplied interaction owner and regularity/domain estimates. ECT ownership of those inputs remains Level-B/Open, while $S_0 = \hbar$ is a separate Level-B empirical calibration.

This is a status-separated inventory rather than a generic descent theorem. The real phase action or Euclidean boundary-value problem alone derives neither the norm-preserving complex law nor its state, Hamiltonian, interaction and positive-frequency sector. Identification with the textbook Schrödinger equation, its physical states, controlled approximation domain and Born/detector map therefore remains Level-B/Open rather than a completed derivation from P1–P6.

The Euclidean formulation is a boundary-value problem rather than a Lorentzian Cauchy problem. This mathematical distinction is structural; its mapping to entanglement, delayed-choice settings, and laboratory probabilities remains B/Open and requires a physical state, boundary map, and measurement kernel.

7. Causal cone inheritance and what GW170817 tells us

A critical question for any emergent-spacetime theory is whether its separately specified candidate physical sectors share the observed relativistic cone. The supplied scalar model conditionally defines one characteristic cone, but this fact alone cannot assign the principal symbols of photons or gravitational waves. Cone universality is therefore a cross-sector completion test, not a theorem obtained merely from the existence of one condensate.

In August 2017, the LIGO/Virgo gravitational-wave detectors observed the neutron-star merger **GW170817**; Fermi and INTEGRAL detected the associated gamma-ray burst about 1.7 seconds later after a journey of roughly 130 million light-years [3]. Under the standard source-delay assumptions, the multimessenger bound is conventionally quoted as $-3 \times 10^{-15} \lesssim (c_{\text{GW}} - c)/c \lesssim 7 \times 10^{-16}$. This is an external constraint on any future physical ECT tensor completion, not a prediction of the displayed scalar/director map.

In generic Lorentz-violating or derivative-coupled completions, matching independently owned matter and tensor cones imposes model-dependent constraints. The Standard Model Extension parametrises sector-dependent Lorentz-violating coefficients.

The P4 datum supplies only the aligned director. The scalar characteristic tensor K^{AB} and its coefficients are separately supplied, while the phase sector requires three distinct quadratic owners. On a locally homogeneous, frozen-coefficient aligned patch with $K_\theta^{\text{eff}} > 0$ and the retained $\beta > 0$ component, define

$$\begin{aligned}\mathcal{I}_{\theta,E} &= \frac{K_\theta^{\text{eff}}}{2} \int d^4X \delta^{AB} \partial_A \theta \partial_B \theta, \\ \widehat{K}^{AB} &\equiv K^{AB}/\beta, \quad \mathcal{Q}_{\theta,K} = \frac{K_\theta^{\text{eff}}}{2} \int d^4X \widehat{K}^{AB} \partial_A \theta \partial_B \theta, \\ \mathcal{I}_{\theta,L} &= -\mathcal{Q}_{\theta,K} = \frac{K_\theta^{\text{eff}}}{2} \int dx^0 d^3x \left[\frac{(\partial_0 \theta)^2}{\tilde{c}_*^2} - (\nabla \theta)^2 \right].\end{aligned}$$

The first is a non-negative Euclidean quadratic functional whose principal operator is elliptic. Its kernel contains functions constant on every connected component. Positive definiteness on the admissible space requires a declared domain and boundary conditions and absence or removal of the full kernel; quotienting one constant zero mode alone is insufficient on a disconnected domain. Coercivity in the declared norm additionally requires a positive spectral gap (a Poincaré inequality), which need not follow from kernel removal on a non-compact domain. The second owns the characteristic polynomial, and the third is the positive-energy Lorentzian reduced action. A physical phase action requires $S_{\theta,L}^{\text{phys}} = \Sigma_\theta \mathcal{I}_{\theta,L}$ with finite positive, spacetime- and field-independent constant Σ_θ ; using the same Σ_θ in a Euclidean weight is an additional same-sector matching input. Unspecified higher operators and variable-coefficient terms are not included in this displayed quadratic truncation. Consequently the status-separated cone inventory is

$$\begin{aligned}c_{\text{phase}} &= c_{\text{char}} && \text{only in this supplied phase truncation after the same clock/lapse map,} \\ c_\gamma &\stackrel{?}{=} c_{\text{char}}, \quad c_{\text{GW}} \text{ Open.}\end{aligned}\tag{8}$$

A shared K^{AB} fixes only the common-length slope; equality of physical speeds additionally requires the same supplied physical clock/lapse map. A photon cone can coincide with it only after the Abelian gauge owner and kinetic normalisation are established; a GW cone requires

a physical constrained tensor Hessian. The displayed phase algebra is Level-A mathematics conditional on the supplied quadratic owners, while the action magnitude, common physical clock, their derivation and cross-sector universality remain Level-B/Open inputs. GW170817 is an external kill gate for a future ECT tensor completion. The current scalar characteristic cone does not identify the gravitational-wave cone.

Higher-derivative dispersion and sector-dependent corrections can be assessed only after each physical kinetic owner is supplied. Any completed ECT sector must satisfy the corresponding laboratory and multimessenger cone bounds; the current scalar result neither forbids nor predicts an unsuppressed tensor-sector difference.

8. Two energy scales and a separate galactic length

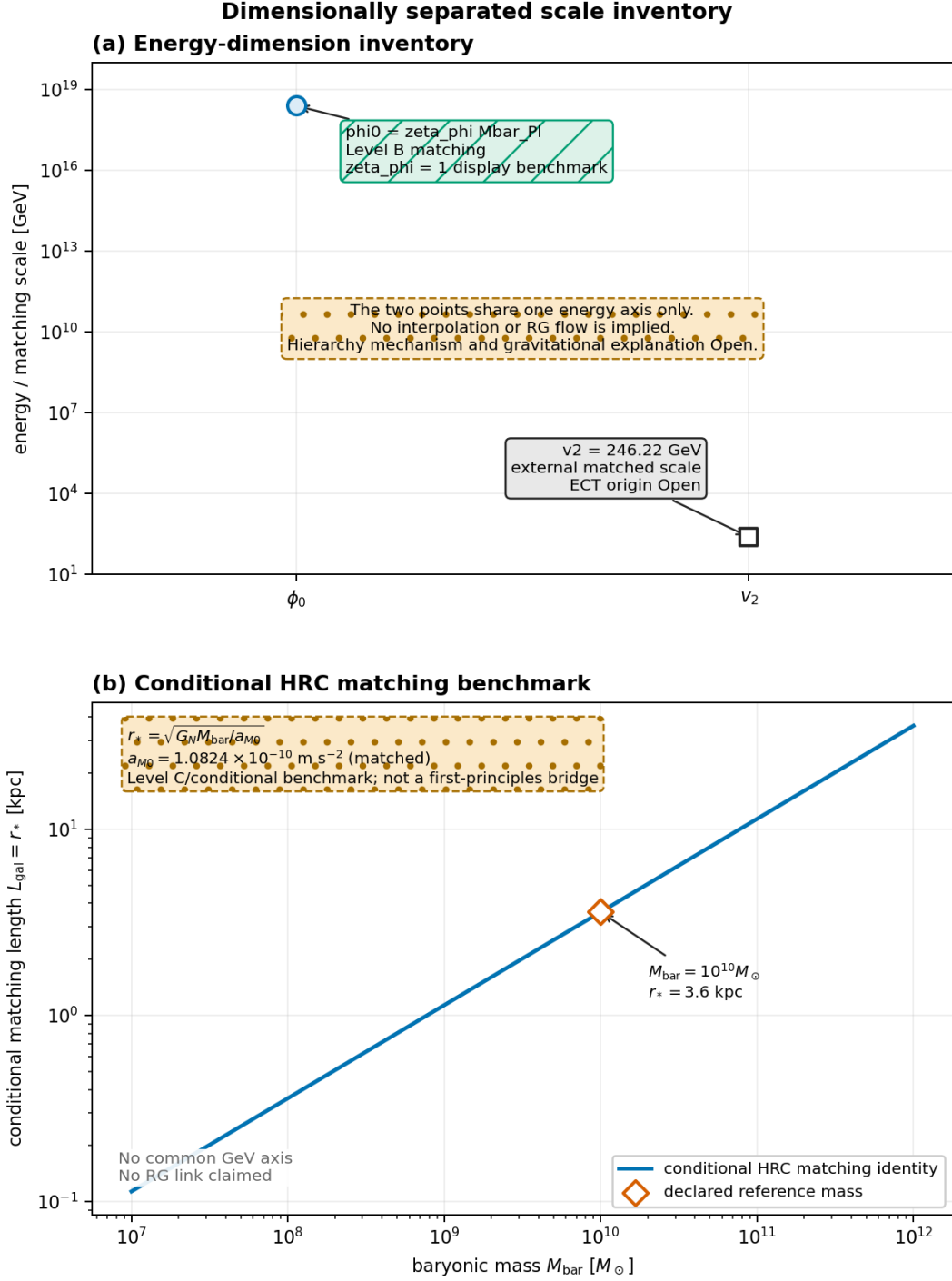


Figure 3: Dimensionally separated scale inventory. The left panel compares the two energy-dimension quantities ϕ_0 and v_2 ; they are independent inventory points, so no connecting curve or interpolation is implied. The gradient amplitude u_0 has units GeV^2 and is related only by the conditional dimensional estimate $u_0 \sim \phi_0 m_\sigma$. The right panel shows the distinct length $L_{\text{gal}} = r_* = \sqrt{G_N M_{\text{bar}} / a_M}$ inside the conditional HRC gravity functional. No common-axis RG flow or first-principles bridge among these objects is claimed.

The current inventory contains two quantities with energy dimension, ϕ_0 and v_2 , plus the dimension-two gradient amplitude u_0 and the separate Level-C galactic length L_{gal} (Fig. 3). They must not be plotted on one energy axis or described as three amplitudes generated by one RG mechanism.

The **Planck-like energy scale** is an independent matching input. ECT adopts

$$\phi_0 = \zeta_\phi \bar{M}_{\text{Pl}}, \quad \zeta_\phi > 0, \quad \zeta_\phi = O(1), \quad (9)$$

as a Level-B scale ansatz; the canonical benchmark $\zeta_\phi = 1$ gives $\phi_0 \simeq 2.435 \times 10^{18} \text{ GeV}$. The controlled matching inventory is

$$G_N^{\text{nat}} = \frac{1}{8\pi M_G^2}, \quad S_0 = \hbar \text{ (adopted calibration)}, \quad c_{\text{char}} = \frac{c_u \hat{c}_*}{\mathcal{N}_\Phi}, \quad \hat{c}_*^2 = \frac{\beta}{\alpha - \beta}. \quad (10)$$

The displayed Open bridge from ordered-sector stiffness to M_G contains no ϕ_0 and therefore does not derive the scale ansatz. The gradient amplitude instead has $[u_0] = \text{GeV}^2$ and only the dimensional estimate $u_0 \sim \phi_0 m_\sigma$; its value is not a third energy scale.

The **electroweak energy scale** $v_2 \simeq 246 \text{ GeV}$ is a matched secondary scale. The logarithm $\mathcal{I}_{\text{EW}} = \ln(\phi_0/v_2) \approx 36.8$ for the displayed matched scales and organises an Open non-polynomial scale-generation programme. This arithmetic ratio is not a derived naturalness mechanism; none of the candidate mechanisms has been derived from P1–P6.

The **galactic length** $L_{\text{gal}} = r_* = \sqrt{G_N M_{\text{bar}}/a_M}$ is conditional on baryonic mass and the fitted acceleration a_M inside the adopted Level-C functional. For $M_{\text{bar}} = 10^{10} M_\odot$ and $a_{M0} = 1.08 \times 10^{-10} \text{ m/s}^2$, it is about 3.4–3.7 kpc. Its scalar/ECT origin and environment law remain Open.

No additional energy scale is selected from (ϕ_0, v_2) by dimensional analysis alone: infinitely many composite functions have the required dimension. ECT adopts the Level-B programme hypothesis that the P3/P4, electroweak, infrared and HRC scales should admit a common-medium matching in a completed theory, but no present relation among ϕ_0 , u_0 , v_2 , a_M and m_χ follows without a separate named closure. In particular, the existence, action and dynamics of a proposed second ordering step remain Open. Higher-dimension Yukawa corrections are organized as powers of $v_2/M_{Y,\text{heavy}}$ for a separately supplied $M_{Y,\text{heavy}} > 0$; only the additional named match $M_{Y,\text{heavy}} = c_Y \phi_0$, $c_Y > 0$, turns this into powers of $v_2/(c_Y \phi_0)$. No equality to the orientation or neutrino heavy scale is implied. A neutrino or other application therefore requires its own operator, coefficient, flavour/state content and matching map.

9. Compact-phase winding and the conditional cohesive-defect loop programme

The compact phase gives a Level-A existence result for integer winding sectors inside its declared closure. Turning those sectors into a localised carrier, a distinguished physical action scale, quantum interference, or a complete discrete spectrum requires additional dynamics and matching.

In the coherent sector, the effective order parameter has the polar form $\Phi_{\text{eff}} = \rho e^{i\theta}$, with $\rho > 0$ along the contour under discussion. Single-valuedness applies to Φ_{eff} , not to a globally single-valued real lift of θ : on a closed contour $\mathcal{C}$ that does not intersect a defect core, Φ_{eff} defines a

map $\mathcal{C} \rightarrow S^1$. A locally lifted phase therefore obeys

$$\oint_{\mathcal{C}} \partial_A \theta dX^A = 2\pi n, \quad n \in \mathbb{Z}. \quad (11)$$

The integer is the degree of that map. This is not a quantum axiom; it is an exact topological statement only inside the supplied non-vanishing compact-phase closure. A contour crossing $\rho = 0$, or a homotopy allowed to pass through such a zero, lies outside that protected sector and can permit unwinding.

Inside the older phase-only fixed-core closure, the corresponding reduced elementary loop coefficient is

$$\iota_{\text{elem}}^{(\text{fixed core})} = 2\pi \iota_0^{\text{EFT}}, \quad (12)$$

and the physical action would be $S_{\text{elem}}^{\text{phys}} = \Sigma_{\theta} \iota_{\text{elem}}^{(\text{fixed core})}$. The dimensionless ι_0^{EFT} is supplied only after a loop length, transverse profile and reduced normalisation are admitted; Σ_{θ} requires its own four-dimensional owner. Compactness supplies the integer, not a finite nonzero global minimum: the phase-only reduced contribution falls as $1/L$ when the length is allowed to grow. Neither the product's value nor cross-sector ownership is derived, and only $\Sigma_{\theta} \iota_0^{\text{EFT}}$ may be mapped to S_0 and then separately calibrated to $\hbar$.

Post-order candidate constitutive hypothesis CDE- Φ . “Cohesive defect elasticity” is a simple acceptance criterion for a completed ordered-branch EFT: at least one stationary localised disturbance must heal to the ambient state, have finite positive vacuum-subtracted action per proper carrier length, and have a positive projected metric on every retained physical normalisable modulus. In plain language, extending the disturbance and changing a genuine internal coordinate must both cost positive action. CDE- Φ is not a pre-physical property, a new frozen postulate, or the owner of the equations. It supplies neither gravity nor quantisation, and it does not manufacture a compact modulus.

The actual owner must be one explicit four-dimensional completion action together with its state and boundary data, a stationary profile, a fixed vacuum subtraction and regulator, and the constraint/gauge/frame/zero-mode quotient used to define physical fluctuations. Suppose that owner supplies completion fields Ψ^I , a localised profile $\Psi_*^I(y; \vartheta)$ around a one-dimensional Euclidean carrier $X^A(s)$, and a genuine compact normalisable modulus $\vartheta \sim \vartheta + 2\pi$. A controlled collective-coordinate reduction [4] can then take the schematic form

$$S_E^{\text{sub,phys}} = \Sigma_{\vartheta} \mathcal{I}_E^{\text{sub}}, \quad \mathcal{I}_E^{\text{sub}} = \int ds \left[\mathcal{T}_{\text{loop}} + \frac{K_{\vartheta}^{1D}}{2} (D_s \vartheta)^2 + \frac{\kappa_b}{2} \mathcal{K}^2 + \dots \right]. \quad (13)$$

Here $\mathcal{K}$ is carrier curvature and D_s includes any derived connection. Twist, torsion, mixed curvature–modulus, nonlocal and higher-derivative terms hidden in the ellipsis must be retained whenever the owner allows them. Before constrained profile relaxation,

$$\mathcal{T}_{\text{loop}} = \frac{1}{\Sigma_{\vartheta}} \int d^3 y [\mathcal{L}_E(\Psi_*) - \mathcal{L}_E(\Psi_{\text{vac}})], \quad K_{\vartheta}^{1D} = \frac{1}{\Sigma_{\vartheta}} \int d^3 y G_{IJ} \partial_{\vartheta} \Psi_*^I \partial_{\vartheta} \Psi_*^J. \quad (14)$$

with the state, boundary data, regulator and quotient understood. The reduced modulus metric is the corresponding constrained Schur complement, not an unconstrained positive-looking integral; the physical action-valued coefficient is $\Sigma_{\vartheta} K_{\vartheta}^{1D}$. These formulas are a derivation template, not evidence that the minimal real-scalar P3 action has the required carrier or compact modulus.

The best-matched ECT-native owner candidate is a stabilised three-dimensional orientation texture whose worldline carries an independently established compact normalisable modulus. This remains Open: a static two-derivative texture has the usual Derrick-collapse obstruction, the exact-gradient branch can restrict the orientation image to a contractible hemisphere, and no required stabilising higher-gradient or independent-orientation completion has been derived from P1–P6.

The transverse d^3y measure describes a codimension-three curve in four Euclidean dimensions. Thus $L = \int_{\Gamma} ds$ is worldline proper length or Euclidean period, not the spatial radius of a texture. A spatial vortex loop persisting along the selected direction would instead trace a codimension-two worldsheet and requires a different reduction. A reparametrisation-invariant worldline representative makes the ensemble condition precise. For $\tau \in [0, 1]$, einbein $e(\tau) > 0$, $L = \int_0^1 e d\tau$ and $\vartheta(1) - \vartheta(0) = 2\pi n$,

$$S_{\text{wl}}^{\text{phys}}[e, \vartheta] = \Sigma_{\vartheta} \mathcal{I}_{\text{wl}}[e, \vartheta], \quad \mathcal{I}_{\text{wl}}[e, \vartheta] = \int_0^1 d\tau \left[\mathcal{T}_{\text{loop}} e + \frac{K_{\vartheta}^{1\text{D}}}{2e} \dot{\vartheta}^2 \right]. \quad (15)$$

For the following bending-free minimum, assume $K_{\vartheta}^{1\text{D}} > 0$, $\mathcal{T}_{\text{loop}} > 0$, $L > 0$ and $n \neq 0$. The ϑ equation then gives $K_{\vartheta}^{1\text{D}} \dot{\vartheta}/e = \text{const}$ and hence $\dot{\vartheta} = 2\pi n e/L$. Substitution produces

$$\mathcal{I}_n^{(\vartheta)}(L) = \mathcal{T}_{\text{loop}} L + \frac{2\pi^2 K_{\vartheta}^{1\text{D}} n^2}{L}, \quad L_{*,n} = \pi |n| \sqrt{\frac{2K_{\vartheta}^{1\text{D}}}{\mathcal{T}_{\text{loop}}}}, \quad \mathcal{I}_{n,\min} = 2\pi |n| \sqrt{2K_{\vartheta}^{1\text{D}} \mathcal{T}_{\text{loop}}}. \quad (16)$$

The modulus route defines the distinct coefficient

$$\iota_{0,\text{bal}}^{(\vartheta)} = \sqrt{2K_{\vartheta}^{1\text{D}} \mathcal{T}_{\text{loop}}}, \quad \iota_{0,\text{bal}}^{(\vartheta)} = \iota_{0,\text{bal}}^{\text{eff}} \text{ only after that owner map.} \quad (17)$$

For the bending-free elementary branch, the physically motivated matching chain may then be stated without conflating matching with derivation:

$$\iota_{0,\text{bal}}^{(\vartheta)} = \frac{\mathcal{I}_{1,\min}}{2\pi}, \quad \Sigma_{\vartheta} \iota_{0,\text{bal}}^{(\vartheta)} \xrightarrow{\mathcal{M}_{\text{bal}}} S_0 \xleftarrow{\mathcal{M}_{\hbar}} \hbar \implies \Sigma_{\vartheta} \mathcal{I}_{1,\min} = 2\pi \hbar = h_{\text{P}}, \quad 2K_{\vartheta}^{1\text{D}} \mathcal{T}_{\text{loop}} = (\hbar/\Sigma_{\vartheta})^2. \quad (18)$$

The first equality is exact in the supplied T1 model. The arrow $\Sigma_{\vartheta} \iota_{0,\text{bal}}^{(\vartheta)} \rightarrow S_0$ is an Open candidate-to-operational owner map. The final equality $S_0 = \hbar$ is the separately adopted Level-B physical calibration. Conditional on both steps, h_{P} is the calibrated elementary-loop physical action and $\hbar$ is the calibrated product $\Sigma_{\vartheta} \iota_{0,\text{bal}}^{(\vartheta)}$, not a dimensionless reduced coefficient. Neither the candidate owner map, Σ_{ϑ} , the calibration's microscopic explanation nor its cross-sector universality is derived from P1–P6 or T1.

The algebra is Level A inside the supplied one-coordinate model. Its interpretation as a physical stationary length requires the proper-time zero mode to belong to a declared closed-worldline measure or variational ensemble. If the duration is externally fixed, or the einbein zero mode is pure gauge, L is not varied and the displayed minimum is not a size prediction. With that ensemble supplied, the optimal gradient is independent of $|n|$, while $L_{*,n}$ and $\mathcal{I}_{n,\min}$ are linear in $|n|$. This makes a single multiwinding carrier exactly marginal against separated unit carriers in the two-term truncation; the $n = 0$ branch has no finite two-term minimum. Neither statement is a binding or population theorem.

If the same owner gives $\kappa_b > 0$, the curvature bound selects a circle at fixed length and the

conditional minimum becomes

$$\mathcal{I}_{n,\min}^{\text{circ}} = 2\pi\sqrt{2\mathcal{T}_{\text{loop}}(K_{\vartheta}^{\text{1D}}n^2 + \kappa_b)}. \quad (19)$$

Bending therefore changes the reduced minimum and, for a fixed Σ_{ϑ} , the corresponding physical action scale; it makes the result nonlinear in $|n|$, and removes the bare two-term splitting marginality. It cannot be invoked while retaining the old numerical minimum unchanged. In the restricted space of nonempty smooth one-component closed curves, the $n = 0$ branch has $L_{*,0}^{\text{circ}} = \pi\sqrt{2\kappa_b/\mathcal{T}_{\text{loop}}}$ and $\mathcal{I}_{n=0,\min}^{\text{circ}} = 2\pi\sqrt{2\kappa_b\mathcal{T}_{\text{loop}}}$. The empty vacuum and core-fill-in paths are absent from that curve space, so this is not a proof of a stable or metastable physical loop. If one identical positive per-carrier κ_b is used and every other interaction is omitted, the connected-circle truncation obeys $\mathcal{I}_{n,\min}^{\text{circ}} < n\mathcal{I}_{1,\min}^{\text{circ}}$ for $n > 1$. This is exact same-core subadditivity in the supplied curve model, not a physical binding verdict: separated-core interactions, core relaxation, reconnection, splitting and the decay barrier remain Open. Stable Skyrme-type textures show only that an independent orientation field with a separately supplied positive higher-derivative operator can evade two-derivative Derrick collapse [5]; they are external proofs of class, not the ECT owner.

Nor does the asymptotic loop-gas count supply the missing physical normalisation. With coarse-graining length b , action denominator a_{measure} and connective factor μ , marginality requires $\mathcal{T}_{\text{loop}}b/a_{\text{measure}} = \ln\mu$. Only after independently imposing $\mathcal{I}_{1,\min}/a_{\text{measure}} = 2\pi$ does one obtain $L_*/b = \pi/\ln\mu = 1.51\text{--}1.64$ for representative $\mu = 6.774\text{--}8$. That circumference is shorter than two lattice steps and therefore lies outside both the large-loop counting asymptotics and a controlled thin-curve regime. This is a UV-control failure of the combined assumptions, not a derivation of $\hbar$.

A complete calculation must still derive the carrier and codimension, compact physical modulus, common coefficients and subtraction, projected Hessian, shape terms, topology and finite-action unwinding barrier, splitting interactions, Lorentzian continuation, measure and cross-sector normalisation. Only after those gates close could one compute a physical eigenmode spectrum, phase-slip lifetime, fragmentation diagram or universality test. PES-R enters only after the carrier, physical modes, state, noise and record protocol exist; it can test admissibility, persistence and records, but it supplies none of the missing owner data and does not turn any candidate scale into $\hbar$.

10. The arrow of time and the second-law programme

In the Euclidean picture, the coordinate w is one of four equivalent directions. It has no fundamental “forward” or “backward” orientation. P4 supplies a preferred Euclidean direction. After a supplied $\beta(\beta - \alpha) < 0$ scalar operator and a chosen initial-boundary orientation, the scalar problem admits retarded support. This does not identify physical time or a thermodynamic arrow.

ECT separates three statements. First, a supplied hyperbolic scalar operator admits retarded support after an initial-boundary orientation is chosen. Second, a positive pairwise noise kernel suppresses selected coherences in a declared reduced channel. Neither statement alone proves monotone von Neumann entropy.

An entropy theorem additionally requires the reduced map, initial state, entropy functional, and

hypotheses such as unitality, detailed balance, or contractivity relative to a stationary state. Under a declared contractive channel one may obtain

$$\frac{d}{dw}D(\rho_w||\rho_*) \leq 0, \quad (20)$$

but this is a conditional open-system result; the ECT-specific channel and stationary state remain Open. Ohmicity and Markovianity alone imply no universal scalar entropy-rate law.

Likewise, a Crooks relation requires named forward/reverse protocols, path measures, a measurable time-reversal map Θ , initial ensembles, and local detailed balance. The reverse measure must first be transported to the common forward path space, $P_R^\Theta(A) := P_R(\Theta A)$. One-way $P_F \ll P_R^\Theta$ defines the Radon–Nikodym entropy variable, whereas the reciprocal relation and a unit integral fluctuation identity require $P_F \sim P_R^\Theta$; singular or absolute irreversibility must be treated separately. A pairwise dephasing exponent is not this forward/reverse likelihood ratio. The ECT entropy-production/fluctuation-relation construction therefore remains an explicit research programme rather than a derived second law.

11. Spectrum, branching, gauge symmetries, and particles

Decoherence exponents are channel-, state-, and protocol-dependent, and no universal forward/reverse law or single object-placement axis has been derived.

The supplied ordered-chart bookkeeping separates three candidate sectors; their physical excitation content is owner-dependent:

Amplitude sector: The homogeneous P3 expansion has a radial coordinate with $m_{\Phi,\text{rad}}^2 = 2\mu^2$ (Level A inside P3). It does not identify a stationary P4 pole. A physical ordered-branch amplitude action, Hessian, pole, correlation length and matching to the P3 coordinate remain Open.

Orientation sector (δn^A): On a smooth strict scalar-only pullback, integrability reduces the independent linear soft parametrisation to one longitudinal coordinate χ (Level-A conditional kinematics). A physical propagating mode requires the separately supplied reduced action, constraint/boundary quotient and projector (Level B/Open). At fixed α, β , on the homogeneous $\bar{n}^A = \delta_w^A$ background and at tangent-director linear order, the fixed map has no TT/static-density seed; a massless spin-2 field is therefore an Open completion target rather than an induced mode of that scoped map. For a separately supplied linear scalar operator $(-\beta\nabla^2 + m_\sigma^2)\sigma = J$, $\beta > 0$, the exterior of a compact source solves the homogeneous modified-Helmholtz equation; every nonzero exterior multipole has Yukawa exponent $m_\sigma/\sqrt{\beta}$, but its amplitude may vanish. A nonzero exterior response over a distance L requires $m_\sigma/\sqrt{\beta} \lesssim L^{-1}$ (Level A inside that supplied operator). Conversely, an extended prescribed profile can be driven, so compact-source asymptotics are not a universal no-response theorem. At the separately declared point $\phi_\infty = \bar{M}_{\text{P1}} = 2.4353 \times 10^{18}$ GeV, $\lambda = 10^{-2}$ and $\beta = 1$, the proxy gives $m_\sigma = 3.444 \times 10^{17}$ GeV, $\xi_\sigma = 5.730 \times 10^{-34}$ m and $(m_\sigma/\sqrt{\beta})/[\hbar c/(1 \text{ kpc})] = 5.386 \times 10^{52}$. This is only a Level-C negative control for the named unsourced exterior pole, not a P4 identification, galactic fit or ECT prediction. Named C/Open soft-scalar completions may separately posit $g_\chi(k) = c_\chi k/f_\chi$ with independent c_χ, f_χ , or with $H_0^{(\text{freq})} > 0$ and $H_{0,E} := \hbar H_0^{(\text{freq})}$, the named branch sets $m_\chi c^2 = \zeta_H H_{0,E}$ for independent $\zeta_H > 0$, giving the reduced Compton proxy $\xi_\chi = c/(\zeta_H H_0^{(\text{freq})})$. The separate boundary choice $\zeta_H = 0$ is a massless branch with $\xi_\chi = \infty$ only as a limit; the quotient is not used there. At the explicit

orientation point $\zeta_H = \zeta_\phi = 1$, $\phi_0 = \bar{M}_{\text{P1}} = 2.4353 \times 10^{18} \text{ GeV}$ and $H_0^{(\text{freq})} = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$, the values are $m_{\chi,E} = 1.493 \times 10^{-33} \text{ eV}$ and $\xi_\chi = 1.322 \times 10^{26} \text{ m}$, while $m_{\chi,E}/\phi_0 = 6.131 \times 10^{-61}$. These are Level-C scale conversions, not a particle prediction. Neither branch is selected by P1–P6 or fixes abundance, dark-matter/dark-energy identity, HRC response or a laboratory signal. The physical P4 pole, action, state, source vertex and gravity/observable maps remain Open.

Phase mode (θ): In the coherent sector, the compact phase supplies winding sectors and a conditional Abelian-connection route. A physical, unbroken photon and its coupling to matter are not yet derived.

The supplied architecture admits conditional geometric and coherent effective descriptions, but they are not separated by a universal decoherence scalar. Macroscopic geometry is an ordered-chart long-wavelength closure; coherence and record formation depend on a declared subsystem/environment split, state, and protocol. A pairwise exponent can diagnose a reduced channel but does not define an ontological or metric-signature transition.

11.1. Gauge symmetries and the interaction sectors

ECT does not postulate all gauge structures as independent primitive symmetries. P5 supplies deformability and relational response, not a gauge bundle. A physical connection exists only after the bundle, quotient by genuinely redundant internal directions (or reduced Hessian), kinetic operator, state and matter representation have been specified. The supplied compact phase gives a *conditional* Abelian route; the orientation algebra gives only a candidate weak-sector analogy. For colour, the minimal $O(4)$ algebra admits no direct injective $\mathfrak{su}(3)$ Lie-subalgebra embedding; emergent/nonlinear, independently supplied bundle and defect-kernel routes are not excluded. P7 is an adopted Level-B frame-module postulate; its physical-colour identification and dynamics remain C/Open/Open, and it is not a necessary or unique completion.

Abelian gauge route (B/Open): The compact coherent phase $\theta \sim \theta + 2\pi$ supports a candidate principal- $U(1)$ connection once local bundle covariance is admitted. On the ordered background the leading parity-even quadratic basis contains both $F_{AB}F^{AB}$ and $(n^A F_{AB})(n_C F^{CB})$; Maxwell form is the special choice with zero coefficient for the second invariant, not a consequence of compactness, covariance or P5 alone. The physical gauge volume and reduced Hessian, kinetic normalisation, charge operator and matter representation, and the identification with $U(1)_Y$ or an unbroken $U(1)_{\text{em}}$ all remain Open. In particular, a charged condensate can Higgs rather than preserve the candidate photon, so masslessness must be demonstrated in the completed action.

Weak-sector programme (A algebra/ Open physical): The residual orientation structure after $O(4) \rightarrow O(3)$ has the exact Lie-algebra identity $\mathfrak{so}(3) \cong \mathfrak{su}(2)$, and its spinorial lift can be written in $SU(2)$ language. This algebra does not select a local internal $SU(2)_L$ bundle, physical chirality or parity violation. Hopf-fibered phase-orientation locking is a candidate global geometry only. Its dynamics, the maps $SU(2)_{\text{orient}} \rightarrow SU(2)_L$ and $U(1)_\theta \rightarrow U(1)_Y$, local gauge fields, matter representations, couplings, and protection of the secondary radial mode all remain Open.

Strong interaction (*restricted obstruction A; physical colour Open*): because $\dim \mathfrak{su}(3) = 8 > 6 = \dim \mathfrak{so}(4)$, $SU(3)$ colour cannot be realised as a direct injective Lie-subalgebra or residual-factor embedding in the displayed minimal $O(4)$ algebra. The theorem excludes only that route: emergent/nonlinear constructions, independently supplied bundles and defect-kernel routes re-

main possible. The rank-3 P7 module is an adopted reverse-engineered Level-B postulate whose positive-definite h and unit-norm nonvanishing Ω data give the fibrewise $SU(3)$ reduced structure group (Level-A conditional mathematics); global automorphisms form the associated gauge group of sections; they do not provide a physical colour connection, action or matter map. Anomaly cancellation alone does not select colour rank, and full QCD descent remains Open.

Gravity route (Level B/Open): an orientation-stiffness operator is a candidate route, while a covariant orientation action, its metric stress, exact TT projection, and physical normalisation remain Open (Part II).

Director coupling (Level B operator class / C/Open optional parametrisation; NOT IDENTIFIABLE as unrestricted species scaling): The ordered-branch EFT admits the candidate vector operator $\mu_5 \bar{\Psi} \gamma^A n_A \Psi$, but does not fix a nonzero coefficient. For $m_f \neq 0$, define $\beta_{5f} := \mu_{5f}/m_f$. A named optional additional hypothesis H_{5m} assumes that the coefficient is generated by the fermion-mass spurion, is analytic in m_f/ϕ_0 and has no constant term. Its leading truncation sets $\beta_{5f} = \zeta_{5f} m_f/\phi_0$ and $\mu_{5f} = \zeta_{5f} m_f^2/\phi_0$, where every dimensionless ζ_{5f} is an independent matching input and may vanish. Because the vector current is chirally even, the spurion owner, analyticity and zero constant term are extra assumptions rather than consequences of chiral symmetry or P1–P7; higher powers are allowed. Dimensional analysis does not select or universalise this ansatz. At $m_f = 0$ the quotient defining β_{5f} is undefined even if the analytic ansatz gives $\mu_{5f} = 0$; that endpoint must be stated directly in terms of μ_{5f} rather than identified with a 0/0 ratio. This parametrisation is only diagonal in a declared mass-eigenstate basis at a separately supplied fixed scale $\mu_{\text{match}} > 0$; Hermiticity restricts the diagonal ζ_{5f} to be real, while off-diagonal flavour coefficients require a separate Hermitian matrix owner. With an unconstrained independent ζ_{5f} for every nonzero-mass species, the formula fits any such diagonal coefficient set through $\zeta_{5f} = \mu_{5f} \phi_0/m_f^2$ and is therefore NOT IDENTIFIABLE as a cross-species scaling hypothesis. A testable branch would need a common coefficient, a representation-owned relation or another stated correlation. In a homogeneous timelike rest frame the vector operator is a number-density term proportional to the identity in spin space, so it produces no spin splitting; a fixed spurion supplies no source, propagator, body charges or screening. A nonzero coefficient is observable only if the resulting operator survives the relevant field redefinitions and relative-observable reduction. The parametrisation therefore implies no fifth force, spin precession or equivalence-principle violation without separate operators and observable owners.

The colour asymmetry. Colour is the only listed sector with the displayed direct-injective dimension obstruction inside the minimal $O(4) \rightarrow O(3)$ algebra. This does not exclude emergent/nonlinear, independent-bundle or defect-kernel routes; it motivates the dedicated comparison of the adopted P7 structural input with the remaining Open colour owners below (Section 54).

Common pre-geometric architecture. ECT places candidate interaction sectors within a common pre-geometric framework, while assigning different derivational status to each. The ordered condensate supplies a scalar characteristic geometry; a universal physical metric, tensor and gravitational source remain Open. The compact phase supplies the strongest conditional Abelian-connection route, but neither physical electromagnetism nor charges. The weak sector is only a candidate route at present: the orientation-sector $SU(2)$ lift and the Hopf-fibered locking geometry supply a possible pre-gauge structure, but the full chiral $SU(2)_L \times U(1)_Y$ Standard-Model realisation remains open. In colour, only a direct embedding in the displayed $O(4)$ algebra

is excluded; P7 supplies adopted Level-B internal frame data, while its physical-colour map, other completion classes and every physical gauge owner remain Open. Together with the candidate director-operator route, this yields a common condensate-based organisation without supplying derived interaction fields or observable fifth-force magnitudes.

11.2. Elementary particles

Particles as stable patterns. In this picture particles are sought as stable excitation patterns of the ordered condensate rather than as tiny hard balls placed inside spacetime. In a completed gauge/matter sector a photon would be an unbroken gauge excitation and an electron a charged fermionic excitation; neither identification is derived from the present P1–P6 action. A massive vector boson, in a completed electroweak closure, would be tied to a secondary ordering or locking scale. This is an organising picture, not yet a census of physical particles.

The bosonic content of ECT is best organised hierarchically, by structural level; it is not a unique census of physical particles. At Level 0 the supplied scalar EFT contains longitudinal soft and radial modes, while a physical graviton remains Open. Search channels remain conditional: an ultralight χ with a cosmologically populated state, non-negligible energy fraction, perturbation/response map and nonzero coupling would motivate background or structure tests. A composition-dependent weak-EP/Eötvös channel for ordinary test bodies would further require a non-universal or composition-dependent coupling of relevant range; strong-EP/Nordtvedt and PPN channels have separate self-gravity and metric owners. A heavy radial pole would require a separately specified portal, production history and lifetime before collider or early-Universe searches are selected. Mass alone chooses neither route. At Level 1 the compact phase supplies winding data and motivates a conditional Abelian completion; a physical unbroken photon, Maxwell normalisation, charges and matter map remain Open. At Level 2 the matched scale $v_2 \simeq 246$ GeV anchors a candidate secondary ordering/locking model, but its local chiral gauge theory and dynamical origin remain Open. At Level 3 the P7 internal structure supplies an adopted Level-B frame-module postulate, not a necessary or unique colour completion and not a derivation of a physical connection, action, matter map or gluons from P1–P6. A supplied reflection-positive free spinor kernel admits conditional OS/Dirac reconstruction. P1–P6 do not derive physical fermions, an interacting CAR sector, a local weak bundle or observed chiral representations; orientation alone cannot supply them. The relation $\pi_3(S^3) = \mathbb{Z}$ gives sector labels only, not a particle spectrum or mass law. With primary two-derivative stiffness the texture energy scales as $E_2(R) \propto R$ and collapses by Derrick scaling, so stabilisation needs additional dynamics whose owner remains Open.

Neutrino sector. The present condensate construction does not derive a physical chiral weak gauge field, a sterile state, a neutrino mass operator or a PMNS matrix. The canonical low-energy doublet is $H = \Phi_{\text{ew}}/\sqrt{2}$ with $H^0 = (v_2 + h)/\sqrt{2}$, so a supplied one-flavour seesaw with $M_R > 0$, $s_y = |y_\nu|$ and $m_D = s_y v_2/\sqrt{2}$ has the exact light Takagi singular value $m_\nu = [\sqrt{M_R^2 + 4|m_D|^2} - M_R]/2$ [6]. Only on the declared seesaw branch $\epsilon := |m_D|/M_R \ll 1$ does this reduce to $m_\nu = s_y^2 v_2^2/(2M_R) [1 + \mathcal{O}(\epsilon^2)]$, while a supplied Weinberg operator gives $m_\nu = |C_5|v_2^2/(2\Lambda)$ [7]. A named C/Open completion hypothesis may set only the characteristic heavy singular-value scale $M_H := \|M_R\|_2 = c_R \sqrt{\phi_0 v_2}$, $c_R > 0$, and another may set $\Lambda = c_\Lambda \phi_0$, $c_\Lambda > 0$. A full matrix branch instead requires a supplied dimensionless complex-symmetric matrix $\hat{C}_R$ in $M_R = \sqrt{\phi_0 v_2} \hat{C}_R$. The choices $c_R = c_\Lambda = 1$, $C_5 = 1$ and $\zeta_\phi = 1$ are unit-coefficient benchmarks, not preferred masses. They give $M_H = 2.4487 \times 10^{10}$ GeV and the independent unit-Wilson value

$m_\nu^{(5)} = 1.2447 \times 10^{-5}$ eV. In the same one-flavour normalisation the supplied 0.050 eV orientation scale would require $|C_{5,\text{eff}}|/(c_\Lambda \zeta_\phi) \simeq 4.02 \times 10^3$ in the relevant low-energy Takagi direction after RG and threshold matching. This is a required coefficient/scale combination, not a selected UV closure or prediction. On the one-flavour seesaw branch, inverting the first scale at the illustrative targets 0.050 eV and 0.009 eV gives $s_y = 6.355 \times 10^{-3}$ and 2.696×10^{-3} , respectively; these are fits, not predictions. Without a common UV owner, the relation between the Weinberg and type-I contributions is NOT IDENTIFIABLE, so they must not be added by default. If the same N_R fields induce C_5 , simultaneous inclusion double counts unless the matching $C_5/\Lambda = -Y_\nu M_R^{-1} Y_\nu^T$ and a subtraction prescription are supplied; an independently owned residual C_5 may instead add. Neither P1–P6 nor dimensional analysis selects the coefficients, Y_ν , flavour structure, RG convention or ultraviolet operator. Absolute mass, Dirac/Majorana character, ordering, mixing and leptogenesis remain Open.

Standard PMNS/MSW phenomenology may be supplied as the low-energy closure, but the displayed diagnostic is explicitly vacuum/free propagation with $V_{\text{MSW}} = 0$. On the declared constant-homogeneous perturbative branch, flavour-Hermitian a_L and c_L , a positive-nonsingular time-kinetic form and a self-adjoint spatial domain establish the Hermitian effective Hamiltonian and unitary propagation used here. The displayed $\|\delta\mathcal{K}\|_{\text{op}} < 1$ condition is sufficient, not necessary; for Hermitian $\mathcal{K}$, positivity is necessary and sufficient for the positive-congruence reduction, while the self-adjoint spatial domain remains a separate condition on the full generator [8, 9]. Hermiticity alone is insufficient; $c_{00} = -1$ is a singular one-flavour counterexample for the displayed minus-sign derivative operator. At leading order $C_{ab} = -(c_L)^{\mu\nu}_{ab} \hat{p}_\mu \hat{p}_\nu$ has the sign derived from that convention. The cited mostly-minus source convention instead has the plus-sign derivative term: with $\eta_H = -g_S$, $\gamma_H = \gamma_S$, $p_\mu^H = -p_\mu^S$, $D_H^\nu = -D_S^\nu$ and the covariant a_L, c_L arrays held fixed, it maps to the displayed house minus term while $A_H = A_S$ and $C_H = C_S$. The sources do not literally use the house display. These are imported SME operator classes, not ECT evidence. For antineutrinos the CPT-odd matrix changes sign with flavour conjugation, $A \mapsto -A^*$, whereas the CPT-even matrix retains its sign, $C \mapsto C^*$. Oscillations see only the traceless flavour projection. At a fixed direction, a supplied Hermitian $d = 4$ coefficient is a flavour matrix rather than a global scalar, and its identity part is invisible to oscillations. Comparing its separate eigenspan with the mass contribution is at most a coefficient-level diagnostic: unless the two matrices commute, equality of their separate spans is not an eigenvalue crossing of the full Hamiltonian. No coefficient, preferred direction, flavour embedding, equality scale or detector likelihood is derived from ECT. Above electroweak breaking the full lepton doublet, gauge-covariant derivative and charged-lepton partner operators are required; the neutrino-only tensor is not an independent gauge-invariant coefficient or a global SME fit. The ECT vertex, coefficients, flavour map and more general propagation domain remain Open. Consequently $0\nu\beta\beta$ discriminates Majorana completions in general but is not yet a distinctive ECT prediction. Relative to the minimal Standard-Model field content, the present ECT architecture contains an additional condensate scalar sector. The strict map provides the coordinate χ , while the radial response σ is treated as heavy only at the matched ordered point. The physical existence, mass, radiative protection and dark-sector role of a propagating χ excitation remain Open.

What remains open (*Level Open*): the origin of three fermion generations; the Yukawa coupling hierarchy; the fine-structure constant $\alpha = 1/137$. In one named C/Open Pöschl–Teller toy, if the attractive self-adjoint branch with $\kappa_{\text{PT}} > 0$, $\lambda_{\text{eff}} > 0$ and $\nu = \kappa_{\text{PT}}|y|/\sqrt{2\lambda_{\text{eff}}} > 0$ is supplied, the exact bound-state count gives three toy levels only for $2 < \nu \leq 3$, equivalently $2\sqrt{2\lambda_{\text{eff}}}/\kappa_{\text{PT}} < |y| \leq 3\sqrt{2\lambda_{\text{eff}}}/\kappa_{\text{PT}}$. This is a conditional wedge, not the withdrawn rectangular

window and not a derivation of three physical generations. A physical route still requires a Dirac operator, profile map, bundle, domain, boundary class and index/spectral owner; topology alone fixes neither chiral zero modes nor the generation count.

Three logical classes of excitations. In the Standard Model's elementary-field description, the field content and representations are supplied inputs. ECT instead organises its candidate sectors into different logical tiers. Category (i) contains the scalar/ordered-medium variables and supplied reduced excitations: the radial response σ at the matched point and the strict-map coordinate χ . Their physical spectra require their own actions, domains and projectors; in particular, neither a universally ultra-light scalar nor a physical graviton has been derived. Category (ii), the *externally supplied Standard-Model field targets* (photon, $W^\pm$, Z , Higgs, fermions, gluons), are analogous to impurity atoms in a lattice: compatible with the medium but not derived from the minimal scalar action. Category (iii) contains *conditional target-space topology*: for the supplied orientation orbit S^3 , $\pi_3(S^3) = \mathbb{Z}$ while $\pi_0 = \pi_1 = \pi_2 = 0$. This is exact target-space mathematics, but physical textures or the absence of walls, strings and monopoles require the complete physical vacuum/configuration space, action, boundary class and formation history, all of which remain Open. This tiering is an accounting choice of the present ECT programme, not a universal distinction from quantum field theory as a framework.

Multi-particle description and the status of Fock space. First-quantised N -body quantum mechanics fixes N ; variable-number descriptions use Fock-space or second-quantised formalisms, and relativistic creation/annihilation requires quantum field theory. ECT treats that structure as a reconstruction target, but has not yet derived it: a sector decomposition with variable excitation number, canonical creation/annihilation operators and physical states remains Open (OP-Q11). The condensate-reconfiguration analogy motivates the programme but is not its solution.

Multi-particle descriptions remain necessary for exchange sectors, coherent interactions, variable excitation number, and candidate common-medium correlations. A shared medium does not by itself establish nonfactorisation or entanglement; the physical state/channel is Open.

Figure 4 summarises the status and dependency logic of Part I.

12. Conclusions of Part I

12.1. Status-separated outputs and Open targets

From six postulates (P1–P6), supplied effective branches such as BR1, and the assumptions stated for each result, the following structures are obtained or targeted:

1. Scalar principal tensor is Lorentzian iff supplied $\beta(\beta - \alpha) < 0$; on the retained $\beta > 0$ component this is $\alpha > \beta > 0$ (A-math conditional); P4/SSB does not select that sign, physical branch, clock or universal metric (Open).
2. Scalar common-length slope $\hat{c}_*$ (A-math inside the supplied scalar model); $c_{\text{char}} = c_u \hat{c}_* / \mathcal{N}_\Phi$ additionally needs the clock map. After that convention is fixed, $c_{\text{char}} = c$ is the adopted Level-B scalar-clock calibration. Physical photon and tensor cone equality remains Open.
3. Conditional equation inventory: the Euclidean scalar boundary problem and supplied local KG-form scalar operator are distinct from a separately supplied complex Schrödinger

law. The leading NR envelope is A-level asymptotic mathematics only after the same physical clock, mass matching, stable positive-frequency component, Hamiltonian/domain, weak interaction, slow-envelope regime and residual control are supplied; a polar rewriting separately requires its nonzero-amplitude and regularity domain. ECT ownership remains B/Open.

4. Retarded causal response and positive pairwise noise in their declared sectors (*A under the stated kernels*).
5. ECT-specific monotone-entropy or fluctuation theorem (*Open*: reduced map, state, entropy functional, and detailed-balance/contractivity hypotheses required).
6. Candidate geometric/coherent sector architecture (*B/Open*); the scalar mode decomposition is Level A conditional only inside its supplied ansatz and does not establish a common physical owner.
7. The supplied compact phase has exact global algebra. A conditional Abelian connection additionally requires an independently admitted local bundle or reduced quotient; P5 supplies deformability, not local gauge redundancy (*A global/B-Open local*). Physical $U(1)_{\text{em}}$, an unbroken photon, gauge reduction, charges and the matter map remain Open.
8. Candidate $SO(3)/SU(2)$ bundle architecture (*B/Open*); the algebra isomorphism does not derive the physical chiral weak group $SU(2)_L$, its action or its matter representations. The dimension inequality excludes only a direct injective $SU(3)$ embedding in the displayed minimal algebra. P7 is the adopted Level-B rank-three bundle postulate with fibrewise $SU(3)$ reduced structure group; its physical connection, action and matter map, and alternative completion classes remain Open.
9. Scoped fixed-map result: at fixed α, β , about homogeneous $\bar{n} = w$, and at tangent-director linear order, $h_{ij}^{\text{TT}} = h_{ww} = 0$ (A inside that limit); quadratic/nonlinear/composite and physical spin-2 completions remain parametric/Open.
10. The phase-only closure defines $\iota_{\text{elem}}^{(\text{fixed core})} = 2\pi\iota_0^{\text{EFT}}$ after a core length/profile is supplied but has no nonzero global minimum as L varies. CDE- Φ is only a post-order acceptance criterion. A distinct texture-worldline modulus route has an exact conditional L -minimum after its proper-time ensemble is supplied and defines $\iota_{0,\text{bal}}^{(\vartheta)} = \sqrt{2K_{\vartheta}^{1D}\mathcal{T}_{\text{lopt}}}$. If its Open owner map and the separate Level B calibration $S_0 = \hbar$ hold in the bending-free $n = 1$ branch, then $\Sigma_{\vartheta}\mathcal{I}_{1,\text{min}} = 2\pi\hbar = h_P$. This is a calibrated conditional consequence, not an absolute derivation. Its 4D owner, profile, compact modulus, projected Hessian and bending/barrier data remain Open.
11. Existence of integer compact-phase winding sectors (*A*); physical action normalisation and spectrum reconstruction are B/Open.
12. Candidate director operator: A-math conditional in the restricted EFT basis. Its dimension-one coefficient μ_5 , sign, species dependence and observable bridge are Open; $\beta_5 \equiv \mu_5/m_f$ (when $m_f \neq 0$) is only a parametrisation, not a derived law or benchmark.
13. Discrete-symmetry, anomaly, and flavour architecture of the fermionic sector: P, T, C with conditional CPT via OS reconstruction (*A/B*); electroweak anomaly conditions verified with inherited content and the $\mathcal{L}_5$ transparency lemma (*A*). Supplied overlap matrices can

parametrise flavour data but derive neither CKM nor PMNS structure; the physical flavour operators, phases and ECT owner remain Open.

14. Neutrino sector: no physical chirality or mass mechanism follows from P1–P6. The standard one-flavour seesaw and Weinberg-operator normalisations hold only after their masses, coefficients and flavour content are supplied; PMNS/MSW, Dirac/Majorana character and $0\nu\beta\beta$ predictions require a supplied flavour completion (Open).

12.2. Key equations and principles of Part I

Conditional scalar principal tensor: $K^{AB} = \beta\delta^{AB} - \alpha n^A n^B$; the local algebraic basis is unique inside the supplied operator inventory, while α, β and the physical owner are independently supplied.

Scalar slope: $\hat{c}_*^2 = \beta/(\alpha - \beta)$; physical scalar speed: $c_{\text{char}} = c_u \hat{c}_*/\mathcal{N}_\Phi$; adopted scalar-clock calibration: $c_{\text{char}} = c$ after the clock/lapse convention, with photon and tensor cone equality Open.

Matching inventory: $G_N^{\text{nat}} = (8\pi M_G^2)^{-1}$, $S_0 \leftrightarrow \hbar$, and $M_G \sim \phi_0$ as a separate matching step; no direct amplitude-to-action-scale theorem is used.

Compact-phase winding: $\oint \partial_A \theta dX^A = 2\pi n$ (Level A after BR1 is supplied). The phase-only closure defines $\iota_{\text{elem}}^{(\text{fixed core})} = 2\pi\iota_0^{\text{EFT}}$. The post-order CDE- Φ criterion does not own a carrier; if one 4D action/state/profile/quotient supplies a compact worldline modulus, the reparametrisation-invariant route defines the distinct $\iota_{0,\text{bal}}^{(\vartheta)} = \sqrt{2K_\vartheta^{1D}\mathcal{T}_{\text{loop}}}$. Conditional owner and physical matching in the bending-free $n = 1$ branch then give $\Sigma_\vartheta \mathcal{I}_{1,\text{min}} = 2\pi\hbar = h_P$, without deriving the measured constant. Proper-time ensemble, Hessian, bending/barrier and universality remain Open.

Cone status: the supplied scalar and displayed phase truncations share K^{AB} and therefore only a common-length slope. Equality $c_{\text{phase}} = c_{\text{char}}$ additionally requires the same supplied physical clock/lapse map on the aligned frozen-coefficient patch; photon and tensor kinetic owners remain Open.

Entropy-production programme: ohmicity and Markovianity alone do not imply monotone von Neumann entropy; an ECT theorem remains Open pending the reduced channel, state, entropy functional, and detailed-balance/contractivity hypotheses.

12.3. Conditional tests, external gates, and Open programmes from Part I

GW170817 is an external kill gate for a future tensor completion; the current ECT action does not identify c_{GW} or satisfy that gate by derivation.

Director-matter programme: the admitted homogeneous timelike vector-current term is a number-density deformation with no spin splitting and no source-generated body force. Spin observables require an independent axial, derivative/Pauli or inhomogeneous operator and detector Hamiltonian; composition-dependent acceleration requires a separate source/body-force map. Neither completion, coefficient nor numerical ω_5 , Eötvös parameter or experimental reach is currently derived.

Propagation corrections: an Open parametrisation template for a future named photon completion may be written schematically as $\omega^2 = k^2[1 + \xi_n(k/\Lambda_\gamma)^n + \dots]$, with $n \in \mathbb{N}$, $\Lambda_\gamma > 0$, $k/\Lambda_\gamma \ll 1$ and

a domain that keeps $\omega^2 \geq 0$, only after its gauge-invariant operator, derivative order, CPT/parity and helicity/birefringence assignment are fixed. Until that operator is named this equation is a template, not a defined model. The coefficient ξ_n , scale Λ_γ , photon clock, source-lag model and detector likelihood are supplied inputs; no unit coefficient, numerical delay or ECT interpretation follows. Tensor delay likewise remains Open until a physical tensor operator, coefficient and scale are derived.

Casimir target: the supplied N -scalar Dirichlet BVP is A mathematics; the physical ECT boundary bundle, material operator, stress and force are Open.

Preferred-direction neutrino operators: the standard $d = 3$ CPT-odd and $d = 4$ CPT-even SME classes may be supplied, but the coefficient, direction, flavour operator and physical propagation map are not derived. No numerical PMNS/MSW correction or equality scale is therefore an ECT prediction.

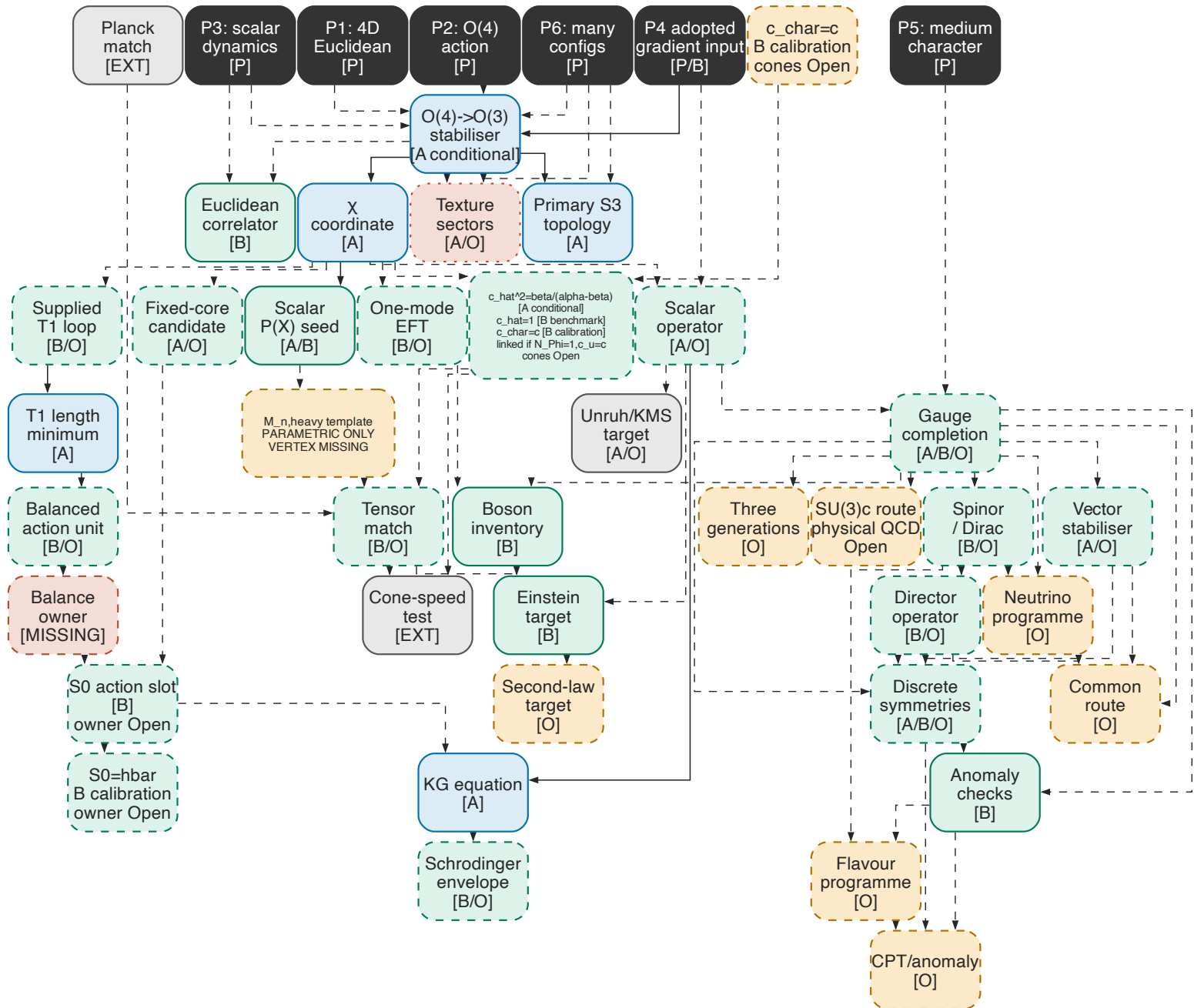
12.4. Part I status and dependency logic

Figure 4: Complete connected Part I status and dependency map. P4 and BR1 are supplied inputs; the scalar characteristic algebra is exact only inside its stated EFT, while local gauge, particle, tensor, action-scale and entropy owners remain Open. All canonical nodes and directed edges are shown in one colour-coded A4 reader map; arrows organise dependencies and never upgrade the status written in a node. The deterministic exact-key and machine-topology records preserve the canonical source labels and edge inventory.

Figure 4 restores the complete graph rather than splitting it into apparently independent routes. The PDF remains vector content for zooming; colour is redundant with border and edge style, and the literal node text is authoritative. Its allowed director-vector node denotes only a conditional vector-current operator: the homogeneous timelike term has no spin splitting or source-generated composition force, so those observables require separate completions.

Selected Part I status and dependency map

Status focus: P4 adopted P/B; O(4)->O(3) stabiliser A conditional; $c_{\text{hat}}^2 = \beta / (\alpha - \beta)$ A conditional; $c_{\text{hat}} = 1$ B coefficient benchmark; $c_{\text{char}} = c$ B calibration, linked when $N_{\text{Phi}} = 1$ and $c_u = c$; cone universality Open; M_n , heavy route PARAMETRIC ONLY / VERTEX MISSING.



P postulate/input | A exact in stated model | B conditional/calibration | C fit/toy | O Open | MISSING owner absent. Solid: exact declared step; dashed/dotted: conditional or Open dependency. Literal node status controls.

Part II

Macroscopic Physics: Gravity, Cosmology, and Observational Tests

13. Lorentzian kinematics in the homogeneous scalar/phase limit

In the homogeneous ordered scalar/phase model, where the condensate amplitude and direction are uniform, the characteristic metric reduces to the Minkowski form after the proper-time map, $ds^2 = -c_{\text{char}}^2 d\tau^2 + dx^2 + dy^2 + dz^2$. Lorentz transformations and the corresponding dispersion relation follow at Level A inside this displayed characteristic model. ECT separately adopts $c_{\text{char}} = c$ as a Level-B scalar-clock calibration after fixing the conversion/lapse convention. This is not yet a derivation of physical special relativity for clocks, rods, matter, photons or tensors: their kinetic operators, metric maps and cross-sector cone equality remain Open.

13.1. The scalar/phase characteristic-speed bound

The ordered scalar/phase equation has a characteristic-speed bound that is not inserted as Einstein's second postulate. Its physical interpretation as the universal speed bound requires separate completion of every propagating sector.

The argument rests on the *characteristic cone* of the ordered-branch scalar equation. Its principal symbol defines one cone with speed c_{char} . Under the stated hyperbolicity and regularity assumptions, the domain of dependence of that scalar problem is confined by this cone. The statement is Level A inside the declared scalar/phase model; it does not assign the principal symbols of photons, gravitons, or general matter.

For linear plane waves, a massive mode has $v_{\text{gr}} < c_{\text{char}}$ at finite wave number, with $v_{\text{gr}} \rightarrow c_{\text{char}}$ at high wave number; a massless mode has $v_{\text{gr}} = c_{\text{char}}$ for nonzero wave number. This is a statement about the centre of a narrow-band packet, not its causal front. Because the mass term is lower order, massive and massless modes have the same front speed c_{char} . In flat 3 + 1 dimensions the massive retarded kernel has both a cone-boundary term and an interior tail, while the massless retarded fundamental solution is supported on the cone.

The phase velocity of a massive mode formally exceeds c_{char} , but it transports neither information nor causal influence. The scalar causal bound is set by the characteristic/front speed, and its promotion to a universal physical speed bound remains Open.

13.2. Signal speed, not photon speed

Within the scalar/phase model, the limiting velocity is the signal speed of its characteristic cone, not yet a photon-specific constant.

A physical photon may realise the scalar boundary speed after the gauge matching is completed. The current action does not identify the gravitational-wave principal symbol, so GW170817 is an external tensor-completion gate rather than a structural ECT success.

Coordinate transformations preserving the homogeneous scalar/phase cone form the corresponding

Lorentz group, so c_{char} is invariant within that characteristic geometry. Identifying this invariant cone with the physical light cone remains conditional on the unbroken gauge owner, its reduced Hessian and kinetic normalisation, and the matter/photon metric map.

The scalar calibration $c_{\text{char}} = c$ is adopted only after the scalar clock/lapse convention is fixed. Identifying the photon cone with that calibrated scalar cone additionally requires the photon action, reduced Hessian, kinetic normalisation and observable metric; it does not follow from the scalar principal symbol.

14. Emergent gravity and the generalised Einstein equations

In ECT geometry is not the primitive ontology. The effective metric is a low-energy bookkeeping object for propagation on the ordered medium. ECT adopts the Level-B/Open programme hypothesis $H_{N \rightarrow G}$: a future macroscopic source tensor should arise by a declared metric-response and coarse-graining map from a stationary full ordered-sector action, with background subtraction, state and matching fixed. The P3 Noether tensor is not itself that universal matter-to-gravity vertex, and withdrawn off-shell P3 density/pressure proxies are not physical cosmological sources. The controlled result relevant here is narrower than the older orientation-stress narrative.

For the displayed scalar-only Jordan-frame ansatz, the consistently undivided metric equation can be written schematically as

$$F G_{\mu\nu} = 8\pi G_* T_{\mu\nu} + \mathcal{T}_{\mu\nu}^{(q)}, \quad (21)$$

where $\mathcal{T}_{\mu\nu}^{(q)}$ collects the scalar derivative and potential terms. This entire equation uses $\hbar = c = 1$: $G_* \equiv (8\pi \bar{M}_{\text{Pl}}^2)^{-1}$ and $T_{\mu\nu}$, U are natural-unit quantities. Dividing the *entire* equation defines the natural-unit coefficients $G_{\text{div}} = G_*/F$ and $\Lambda_{\text{div}} = U/(\bar{M}_{\text{Pl}}^2 F)$ as exact equation coefficients inside the declared ansatz. They are not by themselves Cavendish, finite-body, or cosmological responses. Those require scalar canonical normalisation and range, body charges, the complete background and the physical metric/clock map; matching G_* to the observed G_N is part of that transfer.

Within the Level B scalar ansatz, write $a(q) = f_{,q}/f$, $\mathcal{A} = K + 3f_{,q}^2/(2f)$, and $\mathcal{E} = V_{,q} - 2aV - (a/2)\rho_{\text{m}}$. On a fixed smooth root the implicit-function theorem gives $d \ln f / d\rho_{\text{m}} = a^2/(2\mathcal{E}_{,q})$ only when $\mathcal{E}_{,q} \neq 0$. The sign is non-negative on the separate stable-root branch $\mathcal{E}_{,q} > 0$; $\mathcal{A} > 0$ is an independent kinetic guard. Then $m_{\text{loc}}^2 = \mathcal{E}_{,q}/\mathcal{A}$ is a conditional local response, not a finite-body screening solution. Freezing a homogeneous reference value is an algebraic choice inside the ansatz, not a screening mechanism or a proof that the reference is dynamically reached.

Three objects must remain separate. M is the microscopic operator $\mu_5 \bar{\Psi} \gamma^A n_A \Psi$; S is this Level-B scalar ansatz; and G is the independently adopted Level-C galactic functional. $M \neq S$ by operator/variable content, while the physical-coordinate, source, force, and normalisation map $S \rightarrow G$ is Open. The compact-source Yukawa theorem constrains only a separately supplied scalar operator and source; without its pole, coefficient and physical vertex it yields no galactic-distance verdict, while an extended driven profile remains a distinct case.

The displayed action contains no independent orientation field n^A and therefore does not derive $T_{\mu\nu}^{(n)}$. The form $\mathcal{X}_{\mu\nu}^{\text{cand}}[n]$ is only an external proxy. A physical orientation stress requires a separately specified covariant $S_n[g, n, \lambda_n]$, its constraints, and the full metric variation (*Open*); the previously used projected-Riemann “identity” cannot replace those inputs. Dividing Eq. (21)

by F changes only the normalization of terms already present and does not create an orientation source.

The scalar cosmological equations are exact inside their declared action. A named two-slope action gives conditional background, age, distance and high-redshift clock outputs; P1–P6 do not select a unique physical completion, and full Hubble/JWST inference remains Open. No orientation contribution has been calculated. Practical galactic functional G is a separate Level-C fit. The cluster script is only a local-map order/argmax audit and not a morphology closure; neither follows from Eq. (21). A canonical finite-body scalar charge, thin shell, Solar-System profile, and PPN projection remain Open.

A fixed-metric scalar boundary-value problem for the same named near-GR two-slope state now supplies a narrower model-internal result. At the deliberately artificial scale $m_{\text{out}}R = 1$ and frozen $(r, \eta) = (0.99, 3333)$, its declared finite-window, asymptotic, and regular $r \rightarrow 1$ tail ratios are respectively 0.006399711753, 0.006393895852, and 0.006372442154. These are three different estimators or boundary limits, not measurements to be averaged. Their publication-zone producer, independent verifier, and payload manifest are respectively `scripts/cosmology/compute_r114_twoslope_finitebody_estimators.py`, `scripts/verification/r114/verify_r114_twoslope_finitebody_estimators.py`, and `data/cosmology_r114/MANIFEST_SHA256_v1.json`. This establishes nonlinear far-tail suppression inside the supplied fixed-metric BVP; the physical body charge, coupled metric, Cassini/WEP comparison, and full PPN profile remain Open.

At fixed α, β , about homogeneous $\bar{n} = w$, and to first order in a tangent director perturbation, the displayed map contains only mixed components h_{iw} and has no linear TT/static-density seed: $h_{ij}^{\text{TT}} = h_{ww} = 0$. This scoped Level-A negative result does not exclude quadratic, nonlinear, composite, nonlocal, additional-field or dynamical- α, β vertices. An independent physical tensor completion — constrained Hessian, matter source, positive residues, canonical normalisation, TT projector and nonlinear completion — and the physical matter/photon metric map remain Open.

15. Condensate evolution and cosmic history

In Λ CDM, the universe is modelled through a hot early phase, primordial expansion, nucleosynthesis, recombination, structure formation and late acceleration. ECT has not yet derived a complete alternative history; Fig. 5 records the owners still required.

The bare model uses a Euclidean arena without Lorentzian time; P4 then supplies an ordered-gradient analysis chart rather than a derived chronological transition. The dimension-two gradient amplitude u_0 and the energy-dimension matching scale ϕ_0 are distinct; no equation here sets $u_0 = \phi_0$. An ordering trajectory, physical branch selection, and any rapid-expansion phase require an explicit nonequilibrium completion and remain Open, so P4 is not a derived replacement for inflation.

Within a separately supplied post-order cosmological completion, familiar milestones are targets rather than derived outputs. A regular two-slope scalar completion demonstrates that a radiation–matter–late-de-Sitter history is possible, while the physical ECT action, state, ordering history, full scalar–metric response history and late source remain Open.

A new conditional consistency condition follows within the same charge-selective architecture: if a physical ordered branch and charged completion are supplied, a neutral relaxing amplitude mode cannot directly populate the gauge/radiation bath unless charged degrees of freedom are

Shape-agnostic external Euclidean view and internal cosmological history

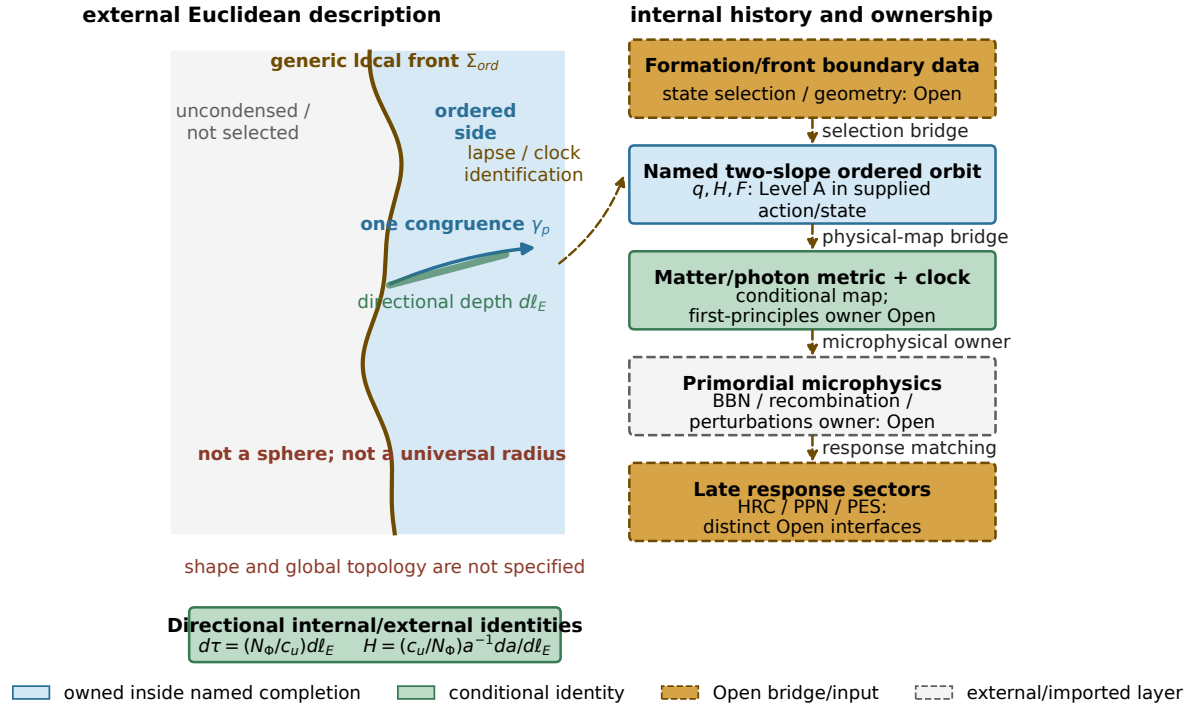


Figure 5: Shape-agnostic external/internal history map. The external panel depicts a generic local formation front and one action-owned congruence, not a spherical boundary or universal radius. P4 supplies an ordered direction and a candidate clock seed, but not the physical matter congruence, lapse or clock map. The displayed orbit is Level A only inside the named supplied action/state and coefficient assumptions; the metric/clock map is conditional Level B. Formation, the physical matter/photon metric, primordial and Boltzmann owners, HRC/PPN matching and PES kernels remain Open. Complementary background panels are reference-ordered in the technical preprint appendix *Visual atlas: conditional background and cosmology*; the companion does not duplicate that atlas. Colour is redundant with border style and literal status text.

available. The candidate sequence is therefore

gradient ordering $\rightarrow$ metric/ Φ -sector matching $\rightarrow$ charged-spectrum population $\rightarrow$ entropy production.

This is only a post-onset routing constraint; it does not specify an efficiency, timescale, or detailed thermalisation mechanism, and it makes no statement at the pre-emergent Euclidean level.

A notable conceptual distinction is the **age of the Lorentzian branch**. ECT defines it as the duration from a physical ordering event,

$$t_{0,\text{ECT}} = \int_{a_{\text{ord}}}^1 \frac{da}{aH_{\text{ECT}}(a)}.$$

The universal postulates need not select the integration constants or the dimensional unit of our realised condensate. A completed action must instead derive the matter-clock congruence and external depth/lapse map and then test one declared state. The supplied two-slope completion gives the conditional value $\tau_{2s} \equiv H_{2s}(0)\Delta t = 0.96368678$ and direct high-redshift clock budgets; these are ECT-candidate state outputs, not one universal law-level age. Every distance and structure calculation must use that same globally regular background and disclose the extra metric/perturbation inputs.

Profiling its single dimensional scale against the official 15-point BC03 cosmic-chronometer subset with the suggested covariance gives the conditional calibration $H_0 = 68.59 \pm 3.95 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and $t_0 = 13.74 \pm 0.79 \text{ Gyr}$ (linearised H_0 -only uncertainty). This Level-C result calibrates one realised orbit; it is neither a universal postulate-level age nor a resolution of the Hubble tension, and the matched flat control is observationally indistinguishable.

At late times, qualitative expanding or recollapsing scenarios remain possible, but their timescales require the missing cosmological completion.

16. The four classical cosmological problems

The standard hot-Big-Bang extrapolation motivates four well-known early-universe puzzles. Inflation is the conventional dynamical solution class. Its model space has separate questions about initial conditions, measures in eternal inflation, and ultraviolet sensitivity. In sufficiently long or high-scale inflationary histories, backward extrapolation can place observable modes below the Planck length; this conditional trans-Planckian concern is not a property of every inflationary history.

ECT organises structural or conditional non-inflationary candidate routes to the four puzzles; it does not yet supply a complete dynamical replacement for inflation or a physical primordial-spectrum prediction.

16.1. The horizon problem

The cosmic microwave background (CMB) — relic radiation from about 380 000 years after the Big Bang — is uniform to one part in 10^5 across the entire sky. In a decelerating hot-Big-Bang extrapolation with no earlier causal mechanism, sufficiently separated last-scattering regions have non-overlapping past light cones. Inflation is the standard solution class; other proposed early-universe mechanisms exist, so the observation does not logically select inflation alone.

Conditional post-ordering evolution of the supplied two-slope condensate

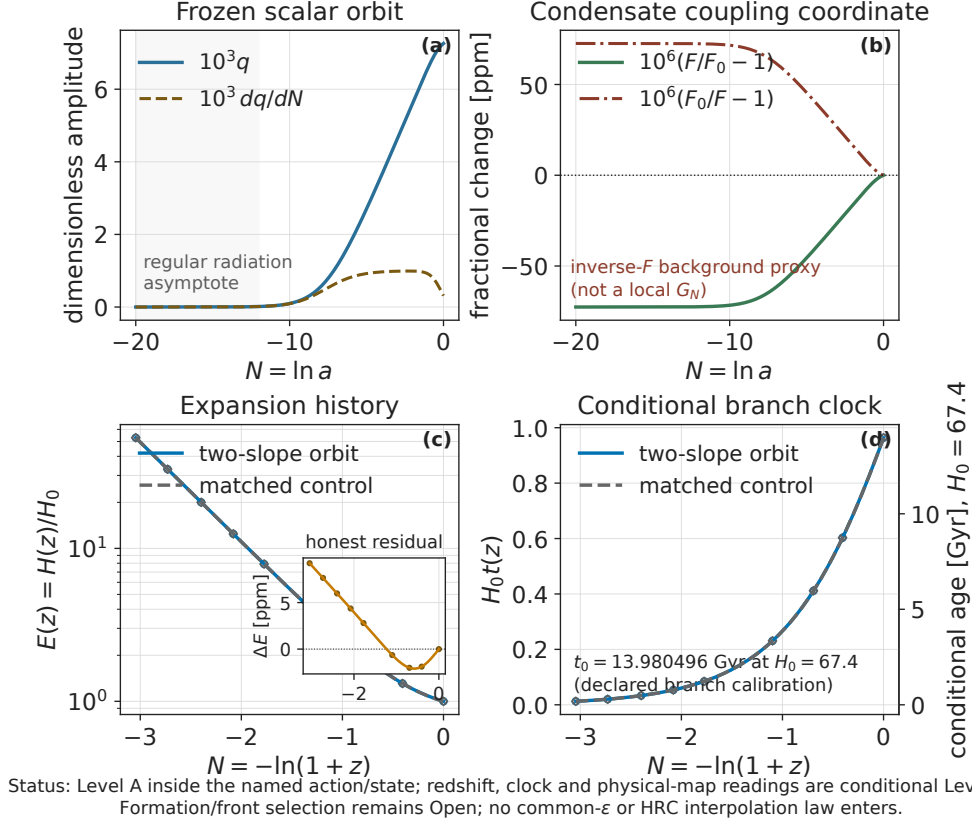
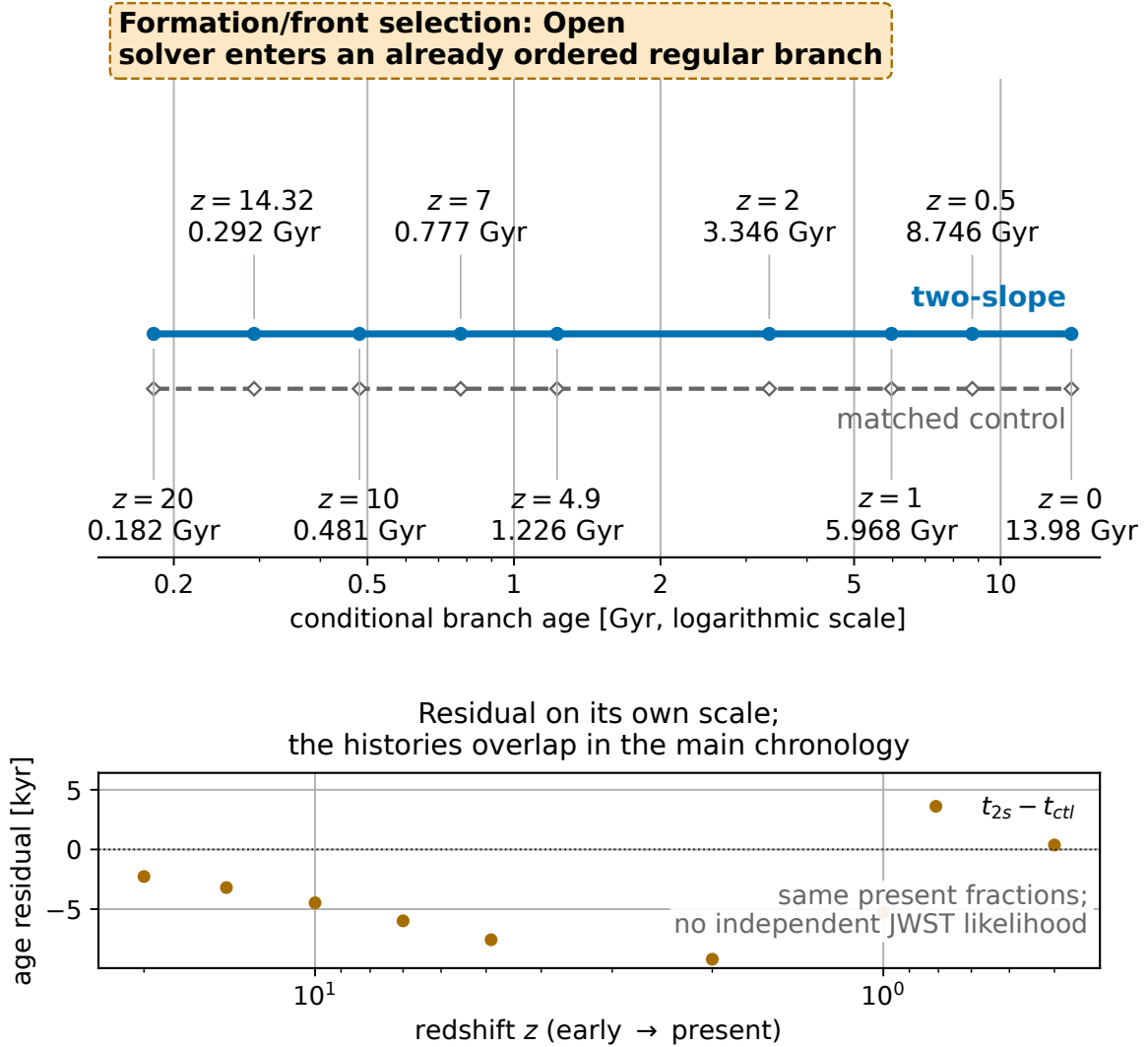


Figure 6: Conditional post-ordering orbit of the supplied two-slope completion. The symbol σ in this figure denotes the dimensionless coordinate of the named two-slope completion; it is not the microscopic radial fluctuation $\sigma = u - u_0$ used elsewhere. The inverse- F_s curve is a background proxy, not a locally measured G_N . Dynamical curves are Level A only inside the supplied action/state and coefficient assumptions; redshift, Gyr and physical-map readings are Level C conditional on the declared dimensional calibration. The integration starts on an already ordered regular radiation branch and does not derive the Open formation front. Panels (c), its inset and (d) show nine evaluated redshift nodes as unconnected markers; no intermediate interpolation is implied. *Display semantics:* Panels (a,b) are dense frozen solver histories without decorative markers; panels (c,d) use solver-owned dense background traces with the nine published diagnostic nodes shown separately.

Conditional chronology of the named two-slope ordered branch



Externally supplied calibration: $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $t_0 = 13.980496 \text{ Gyr}$.
 The separate BC03 clock fit ($H_0 = 68.586 \pm 3.949$) is not used here.
 Neither calibration is a universal P1-P6 prediction.

Figure 7: Conditional chronology of the named two-slope ordered branch. At the single externally supplied calibration $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$ the displayed state has $t(z = 10) = 0.4814276482 \text{ Gyr}$ and $t_0 = 13.9804960874 \text{ Gyr}$. These numbers share one external dimensional calibration and are neither universal P1–P6 ages, a Hubble-tension solution nor a JWST abundance calculation. The formation front, BBN/recombination microphysics and physical metric/clock map remain Open. The residual panel shows nine evaluated event redshifts as unconnected markers; no intermediate residual interpolation is implied.

The P1 Euclidean configuration class contains no supplied Lorentzian light-cone horizon; that is an exact statement about the declared configuration class. Interpreting it as a physically earlier epoch and demonstrating that the observed CMB uniformity is inherited by a later physical metric require ordering dynamics, a clock, a state and thermal transfer, and remain Level B/Open.

A useful condensed-matter analogy is a ferromagnet. Below the critical temperature, a sufficiently coherent domain orders as one region, and distant points inside that domain share the same orientation not because they exchanged late-time signals, but because they belong to one ordered phase. ECT motivates the corresponding candidate: a sufficiently coherent pre-Lorentzian ordering domain could seed correlated initial data. Whether the observed CMB uniformity is inherited requires a supplied state, nonequilibrium ordering history and transfer calculation and remains Level B/Open.

16.2. The flatness problem

Observations constrain the present spatial-curvature parameter tightly (for example, the cited Planck+BAO analysis gives $\Omega_K = 0.001 \pm 0.002$ at 68% confidence). A quoted early-time “fine-tuning” factor is meaningful only after an initial slice, background history and measure on initial data have been specified; no stand-alone 10^{-60} inference is used here.

P1 chooses a flat Euclidean substrate (δ_{AB}). This gives a candidate flatness route, but does not by itself prove the curvature of the emergent cosmological branch; the geometric/dynamical transfer is Level B/Open.

16.3. The relic and monopole problem

Grand Unified Theories can predict massive topological defects (magnetic monopoles). None has been observed. For the supplied P4 orientation target S^3 , $\pi_2(S^3) = 0$, so maps into that target carry no π_2 topological charge (A-math conditional). Identifying S^3 with the complete physical cosmological vacuum/configuration space and deriving a formation history are Open; therefore no physical ECT monopole anti-prediction follows yet. Later gauge-completion sectors have their own topology and abundance owners.

16.4. The primordial perturbation problem

The CMB contains tiny fluctuations ($\sim 10^{-5}$) with a nearly scale-invariant spectrum ($n_s = 0.9649 \pm 0.0042$, Planck 2018). At linear order on a smooth patch of the strict scalar-only pullback, integrability parametrises the soft ordering variation by one scalar coordinate χ (Level-A conditional kinematics). Whether it is a propagating physical mode, and whether it is the only scalar of the completed theory, requires a supplied reduced action, constraint/boundary quotient and physical projector (Level B/Open). An ordinary $O(4)$ -critical correlator does *not* by itself give the observed spectrum: it projects to a blue tilt. For the separately supplied translation-invariant pure-quartic strict-pullback kernel with constant $Z_u, \mathcal{C}_n > 0$, vanishing lower-order deformations, $k > 0$ and a controlled zero mode, the formal local equal-slice seed is exactly $P_\chi \propto k^{-3}$ and hence $n_\chi = 1$ (Level-A conditional mathematics). With lower-order terms, positivity alone is insufficient: the exact $(\delta_a, \delta_m, \rho)$ amplitude and bandwise tilt bounds

must hold throughout a finite quartic-dominance band above the infrared crossovers and below the EFT ultraviolet validity limit. The existence of this band, a physical reflection-positive covariance and state, the coefficient owner and running, the $\chi \rightarrow \zeta$ transfer, and the observed n_s, A_s, r remain Open.

A separate low-multipole branch may be retained only as an Open completion ansatz: a supplied low- k window, scale and signed amplitude would have to be propagated through one gauge-invariant perturbation, recombination/Boltzmann and likelihood chain. An ordering/coherence scale does not by itself select suppression, and P1–P7 predicts neither its sign nor a CMB quadrupole value.

16.5. Summary

The four classical puzzles are reformulated without introducing an inflaton field. The pre-Lorentzian Euclidean configuration class gives a structural route to horizon-scale homogeneity; the flat Euclidean substrate gives the baseline flat branch; and the supplied S^3 orientation target has $\pi_2(S^3) = 0$. The physical vacuum-manifold, transition and relic conclusions remain Open. In the perturbation route, a supplied pure-quartic ordering-sector kernel gives an exactly flat formal local seed, while the physical finite band, state, transfer and observed primordial spectrum remain Open. The first-principles probability of selecting the required coherent branch and the full quantitative derivation of (n_s, r, A_s) remain open.

17. The dark sector in ECT

The terms “dark matter” and “dark energy” organise observations that visible matter plus standard gravity does not account for by itself: rotation curves, gravitational lensing, large-scale structure and accelerated expansion. No dark-matter particle has been identified. A rigid Λ fits the background data well but is not uniquely established as a microscopic vacuum source. The present ECT applications are separate closure layers. The adopted Level-C galactic functional contains no particle-halo term, but its ECT matching and fundamental particle content remain Open; the late-time background closure has its own declared status.

17.1. Galactic no-halo fit and Open particle content

In Λ CDM, the flat rotation curves of galaxies, gravitational lensing in clusters, and the large-scale structure of the universe are attributed to an invisible component — dark matter — constituting roughly one quarter of the energy budget. Broad direct, indirect and collider searches have not established a dark-matter particle. Direct-detection experiments such as LZ strongly constrain standard WIMP parameter space, but a null search does not establish the absence of every particle or hidden-sector completion [10, 11].

The galactic mass-discrepancy results come from the HRC-only algebraic implementation of functional G (Level C), not from the corrected scalar trace equation or a derived orientation stress. Rotation-curve and BTFR/RAR statements retain their fit/conditional status. The separate cluster script is only an exact order/argmax-preservation audit of hand-set maps and supplies no morphology evidence. The physical $S \rightarrow G$ source/force map and any $T_{\mu\nu}^{(n)}$ remain Open.

17.2. Late-time acceleration: conditional scalar example, physical owner open

Observed late-time accelerated expansion is an empirical target. In Λ CDM it is represented by a cosmological constant. The current ECT action does not yet identify the physical late-time source, its state, or its equation-of-state history.

Inside the supplied positive-kinetic scalar action the dimensionally normalised bare pieces use $K_\phi^{\text{phys}} = \bar{M}_{\text{Pl}}^2 \omega \dot{\phi}^2 / 2$ and obey

$$\rho_\phi^{\text{bare}} = K_\phi^{\text{phys}} + U, \quad p_\phi^{\text{bare}} = K_\phi^{\text{phys}} - U, \quad w_\phi^{\text{bare}} \geq -1$$

when $\omega > 0$ and the bare denominator is positive. This is an exact bookkeeping identity, not an acceleration theorem: $\dot{F}$ and $\ddot{F}$ enter the full scalar–tensor expansion history. The local corrected result shows that the observable w_{eff} is not bounded below by -1 and can lie below -1 ; it does not construct or select a global crossing history. A physical ECT result requires one stable action-owned background and the complete distance, growth, photon and lensing likelihood.

The exact fixed determinant concerns only the fixed-coefficient orientation submanifold and does not prove vacuum decoupling. An explicitly supplied trace-free completion is classically shift covariant through $V + C_I$, while the distinct two-slope completion provides a regular late-acceleration proof of class but is not shift invariant. Their descent from the Φ medium, the physical metric and the source/state of the observed acceleration remain Open.

17.3. ECT reinterpretation of the dark sector

The adopted Level-C galactic functional fits selected mass-discrepancy benchmarks without a particle-halo term. This is a property of that fit model, not an ECT derivation or an inference that particle dark matter is absent. Its scalar matching, cosmological fraction budget, and fundamental particle content remain Open. A particle species independently shown to explain the galactic discrepancy would reject the minimal no-halo fit interpretation, not ECT as a whole; null direct-detection searches do not confirm the closure. The late-time background is a separate closure and must not be merged with this ontology statement.

18. The Hubble tension

The measured expansion history is a real target for ECT, but the current theory does not yet predict a numerical H_0 shift. A genuine calculation must derive one ordered-branch background, the physical photon metric, recombination and the sound horizon, and must analyse early- and late-time data in one likelihood. The scalar-only background equations are exact inside their declared model, yet their closure functions, orientation stress, state and initial data are not fixed by P1–P6. They therefore permit many different $H(z)$ histories.

The age of the Universe is correspondingly not a number that can be imported from a reference cosmology. ECT defines it along the physical matter congruence from the ordered-side branch-formation front,

$$t_{0,\text{ECT}} = \int_{a_{\text{ord}}}^1 \frac{da}{a H_{\text{ECT}}(a)}.$$

The outside description is the corresponding clock-lapse integral over Euclidean arc length. P1–P6

need not select the integration constants or unit scale of our realised state; the completion must derive the congruence, physical metric and depth/lapse map. The supplied-completion estimates below are therefore conditional ECT-candidate ages, not one universal law-level number.

A useful existence result survives this ceiling. A fully specified two-exponential scalar–tensor action with positive kinetic determinant has an exact frozen-radiation branch, an exact dust tracker, one stable late-time vacuum, and a two-solver trajectory connecting the three regimes without an attached high-redshift background. This is a proof that a regular scalar completion exists in the nearby model class. It is not yet the ECT background: the microscopic theory has not derived that action, physical metric and clock map. The dimensionless clock integral is a conditional age of that ECT-candidate state rather than a universal P1–P6-selected value. For a second near-GR member of that same supplied family the model-owned dimensionless interval is $\tau_{2s} \equiv H_{2s}(0)\Delta t = 0.9636867786$; a disclosed conditional dimensional scale calibration to $H_0 = 67.40$ or $73.04 \text{ km s}^{-1} \text{ Mpc}^{-1}$ gives respectively 13.9805 or 12.9010 Gyr. These are conditional ages of the named completion, not a unique ECT prediction.

The same orbit directly gives $E(z) = H(z)/H(0)$, distances and high-redshift clock budgets. For the 67.40 conditional dimensional scale calibration its ages at $z = (4.9, 7, 10, 14.32, 20)$ are (1.2265, 0.7769, 0.4814, 0.2924, 0.1818) Gyr. Under explicitly imported standard photon/recombination inputs it changes the sound horizon by only -0.00316% relative to the matched control. A restricted quasistatic growth diagnostic changes the normalised growth by less than 0.002% through $z = 20$. In a still more restricted same-metric, zero-slip, sub-horizon closure, the instantaneous Weyl-amplitude carrier changes by only 0.00267–0.00292% and its conformal-derivative carrier by 0.00308–0.00813% over $0 \leq z \leq 2$. These are not projected lensing or horizon-scale ISW predictions. These conditional numbers show that this near-GR member neither materially shifts Hubble inference nor resolves early maturity.

At the nine prespecified computed nodes, every change in the high-redshift clock is negative (down to -10.491167% at $z = 10$), while larger sampled acoustic leverage accompanies larger homogeneous $G_{\text{div},s}/\text{Planck}$ -factor drift. A -5.55% sound-horizon node has $|\gamma_{\text{PPN}} - 1| = 0.0174$ in the distinct unscreened local-response proxy, whereas the only sampled node satisfying the displayed zero-centred absolute-deviation proxy changes r_s by -0.00323% ; this is not a full Cassini likelihood. The divided coefficient and the PPN proxy must not be identified. The finite scan is a useful action-owned benchmark. It neither proves a continuous-family no-go nor supplies a JWST-abundance or Hubble-tension solution.

Separately, an owner-specific five-row early-response envelope tests a constant coordinate ζ_{ER} without identifying it with the named two-slope orbit or with a universal ECT parameter. The radiation term is deliberately left unmodified as an imported guard, not as derived radiative protection. At $z = 10$ the five prescribed rows give

ζ_{ER}	D/D_0	$(1 + z_{\text{eq}})/(1 + z_{\text{eq},0})$	δ_c	$R_{\text{PS}}^{\text{top-hat}}$
1.5612556×10^{-6}	1.000005	1.000025	1.726760	1.000131
0.00750044	1.024950	1.131953	1.728110	1.830827
0.02958834	1.097679	1.668466	1.732341	8.515859
0.03300000	1.108822	1.777309	1.733025	10.502280
0.03755500	1.123667	1.936589	1.733951	13.755949

The background/equality algebra and the linear-growth integration are exact or reproducible only inside this supplied envelope. The top-hat barrier and Press–Schechter rare-tail ratios are Level-C

sensitivity estimates. The five rows are not posterior samples, a JWST abundance prediction, or a likelihood; in particular, the boosted rows also shift the declared equality coordinate by about 13–94%.

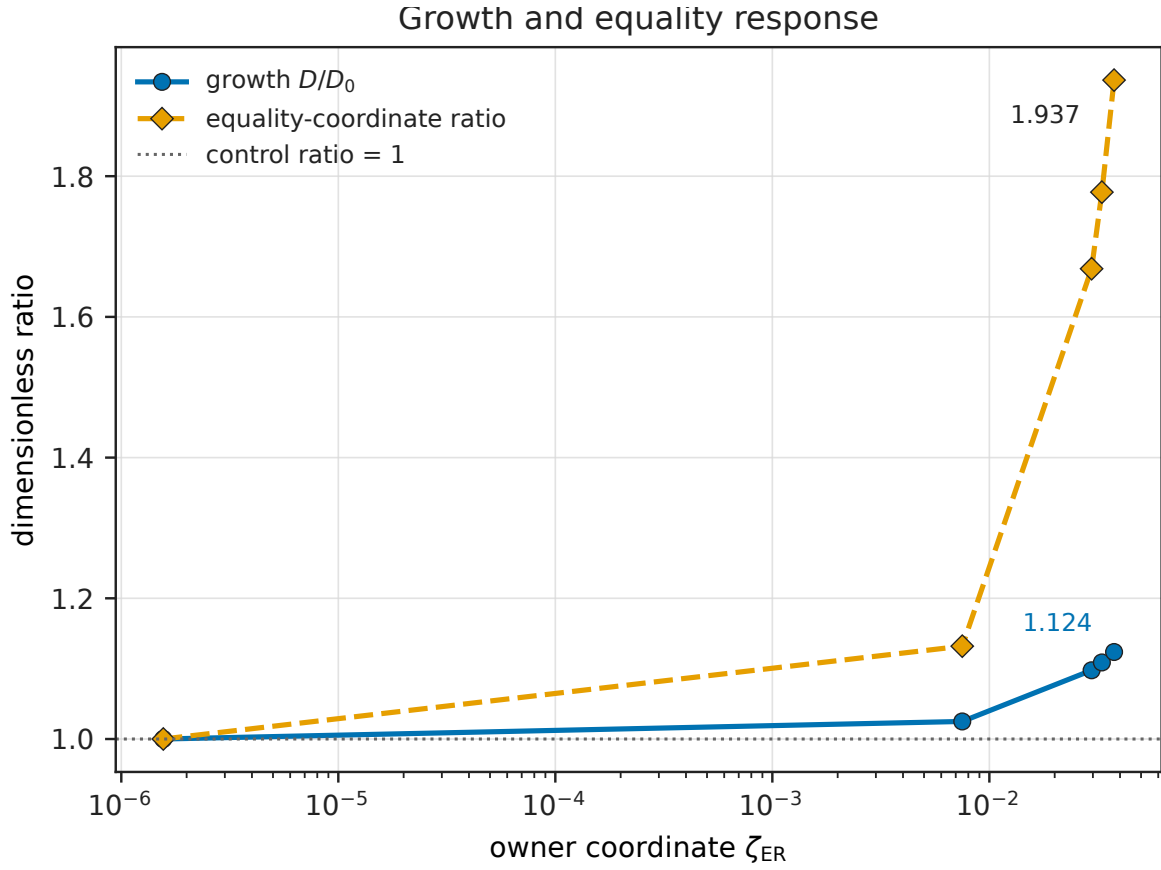
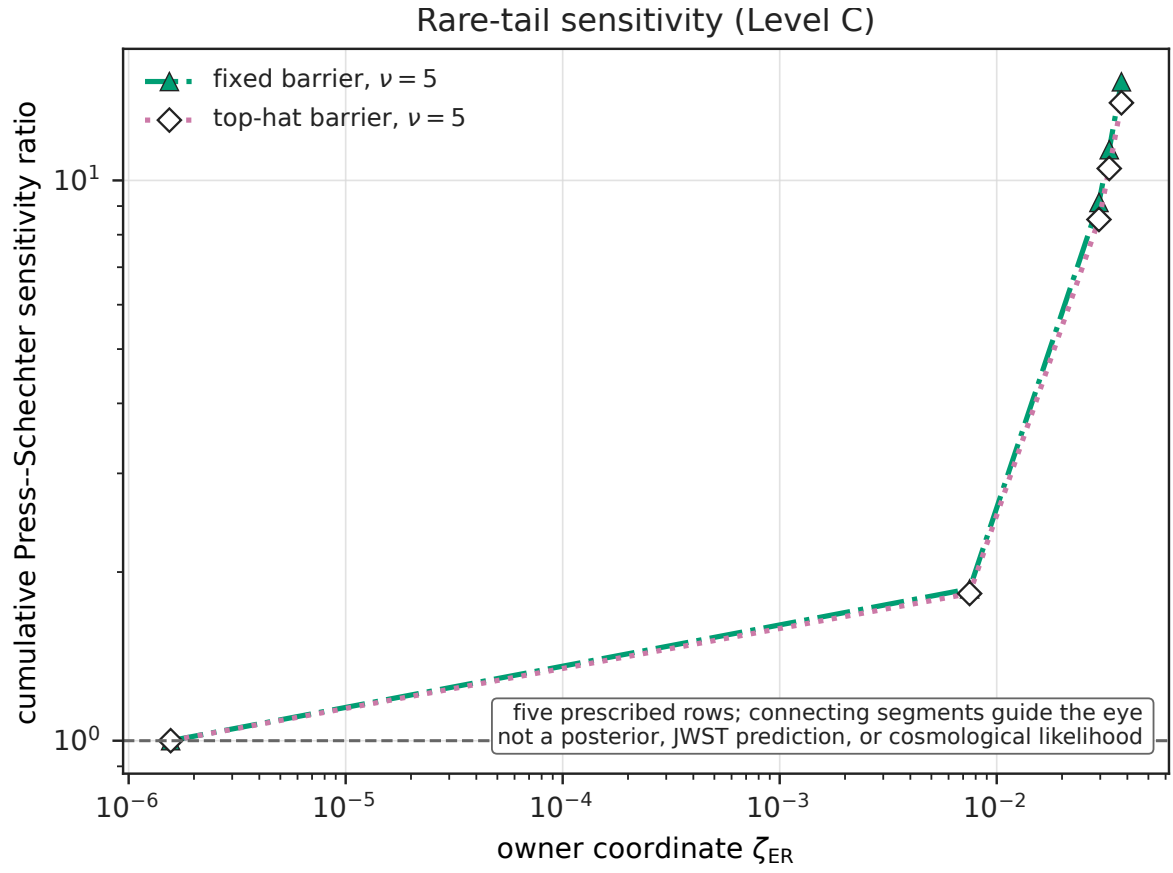


Figure 8: Owner-specific early-response sensitivity envelope: growth and equality response for the five prescribed stress-test rows. This is an enlarged crop of the frozen left panel; no data or styling have been recomputed. The response is conditional on the supplied envelope. Lines guide the eye between the five rows; they do not define a posterior, JWST prediction, or likelihood. Colour is redundant with marker and line style for grayscale-safe reading.



Readable detail of Fig. 8: collapse and rare-tail response. This is an enlarged crop of the frozen right panel; no data or styling have been recomputed.

HRC addresses a separate quasistatic galactic response after an additional gravity bridge. It does not provide the homogeneous stress tensor or the cosmological perturbation equations, and a possible relation $a_M \propto cH$ remains a matching hypothesis until one action yields both.

19. The JWST early-galaxy problem

Early massive objects cannot be predicted by changing one background response factor. ECT must derive the time and volume elements, primordial spectrum, transfer and scale-dependent growth, non-linear halo abundance, baryonic mass-to-light relation and survey selection. None of these may be replaced by an independently tuned common parameter. The present result is therefore an explicit programme and falsifier list, not a claimed JWST resolution: an accepted completion must reproduce the observed abundance and morphology with the same action and state that pass Hubble, lensing, growth and local-gravity gates.

20. HRC galactic response and rotation curves

The current galactic programme uses the HRC response hierarchy. Its logical chain is deliberately layered:

$$\begin{array}{ccc}
 \underbrace{\text{S11a/S11b: local deformation / finite paths}}_{\text{kinematical / candidate}} & \not\Rightarrow & \underbrace{\text{ERP-}\Phi}_{\text{separately supplied constitutive specialisation}}, \\
 & & \\
 \underbrace{\text{ERP-}\Phi}_{\text{ideal constitutive specialisation}} & \Rightarrow & \underbrace{\text{HRC}}_{\text{dimensionless response}} \\
 & \xrightarrow{\text{separate matching}} & \underbrace{\text{weak-field gravity}}_{\text{conditional}}.
 \end{array} \tag{22}$$

S11a states only the guarded local tangent-space property. S11b separately allows finite admissible branch paths but supplies neither a dissipative law nor dynamical accessibility. Neither contains gravity, force, metric or acceleration. ERP- Φ is narrower: in an ideal quasistatic, zero-temperature and non-hysteretic response sector, equivalent normalised channels receive one common scalar shift and a channel is recruited when its stability eigenvalue crosses zero. Positive inverse-DtN compliance supplies its static weight, and a positive semifinite trace adds recruited channels. These clauses are a Level-B candidate constitutive extension, not a result of the frozen P1–P6 action. In particular, the P4 background is not stationary under the frozen P3 action, so a physical constrained response Hessian still requires a stationary ordered-phase action.

For the homogeneous three-dimensional spectral model the primitive and response are

$$W_0(s) = \sqrt{s(1+s)} - \text{arsinh } \sqrt{s}, \tag{23}$$

$$\hat{\mu}_0(s) = \frac{dW_0}{ds} = \sqrt{\frac{s}{1+s}}. \tag{24}$$

This HRC-0 result is exact inside the stated augmented inverse-DtN/recruitment model. Algebraically, with $x = \sqrt{s}$, it is exactly the conventional MOND-standard interpolation $x/\sqrt{1+x^2}$; HRC-0 therefore introduces no new galaxy-level interpolation shape. The proposed response owner and the Open scale/gravity bridges, not this algebraic function, are the distinctive ECT

programme. A supplied equal-three-channel ($n_{\text{ch}} = 3$), rank-one response gives

$$\hat{\mu}_3(s) = \hat{\mu}_0(s) \left[1 - \frac{4}{3} \frac{s}{(1+s)^2} \right]. \quad (25)$$

Here $n_{\text{ch}} = 3$ and $r_{\text{ch}} = 1$ are respectively the channel count and selected-projector rank, so the matrix coefficient is $4r_{\text{ch}}/n_{\text{ch}} = 4/3$. These integers belong to the supplied matrix model; they are not fitted galaxy parameters. HRC-3 is Level A inside that supplied matrix model and a Level B proof-of-class, but its three-channel owner is Open in P1–P6. It was constructed after the crossover residuals were inspected and therefore is *not* an independent observational prediction.

Finite coupling need not shift the threshold in one precisely defined factorised class. Let $\hat{\Lambda}$ be the dimensionless constrained stability operator, let $C > 0$ be a static compliance satisfying $[C, \hat{\Lambda}] = 0$, and let τ be the positive semifinite trace. Define

$$\tilde{\Omega}_s[\mathcal{Q}] = \tau \left[C^{1/2} (\hat{\Lambda} - sI) C^{1/2} \mathcal{Q} \right], \quad 0 \leq \mathcal{Q} \leq I, \quad P_s = \mathbf{1}_{(-\infty, s)}(\hat{\Lambda}), \quad (26)$$

Then the crossing set is unchanged, $\mathcal{Q}_{*,s} = \arg \min_{0 \leq \mathcal{Q} \leq I} \tilde{\Omega}_s[\mathcal{Q}] = P_s$, and $-d\tilde{\Omega}_*/ds = \tau(CP_s)$. This is Level A inside the supplied factorised commuting model and a Level-B closure route. It is not unique: any positive $h(\hat{\Lambda})$ has the same threshold property, while a generic non-commuting readout/source term shifts crossings. The physical choice $C = (1 + \hat{\Lambda})^{-1/2}$ and its action owner remain Open in P1–P6.

A second mandatory guard concerns the infrared limit. Compactness of an interface alone does not imply a discrete or gapped spectrum. If the constrained stability operator is self-adjoint with compact resolvent, all physical zero modes have been removed, and its first retained eigenvalue is strictly positive, then recruitment on the finite interface is a staircase with an infrared gap. Under those additional hypotheses, smooth deep-HRC behaviour requires the continuum/thermodynamic limit before $s \rightarrow 0$, or a noncompact or gapless internal fibre. No spherical cosmic boundary is assumed. The HRC formulas below describe that stated continuum response model, not a generic finite interface.

Only after this internal response has been defined is a gravitational layer introduced. With

$$Y_g = \frac{|\nabla \Phi_{\text{gal}}|^2}{a_M^2}, \quad s = f(Y_g), \quad \mu_g = A_g \hat{\mu}_\Phi(f) f', \quad (27)$$

the computations below declare the identity bridge $f(Y_g) = Y_g$, $A_g = 1$. That bridge, the baryonic matter vertex, the photon/metric map and the origin of a_M are Open. The static conditional functional gives

$$\nabla \cdot [\mu_{g,k}(Y_g) \nabla \Phi_{\text{gal}}] = 4\pi G_N \rho_b, \quad k \in \{0, 3\}. \quad (28)$$

The present matching value $a_{M0} = cH_0/(2\pi) \simeq 1.08 \times 10^{-10} \text{ m s}^{-2}$ for $H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$ is an explicit input, not a medium postulate or derived coefficient. The stronger law $a_M(z) \propto H(z)$ is a separate hypothesis and is not assumed here.

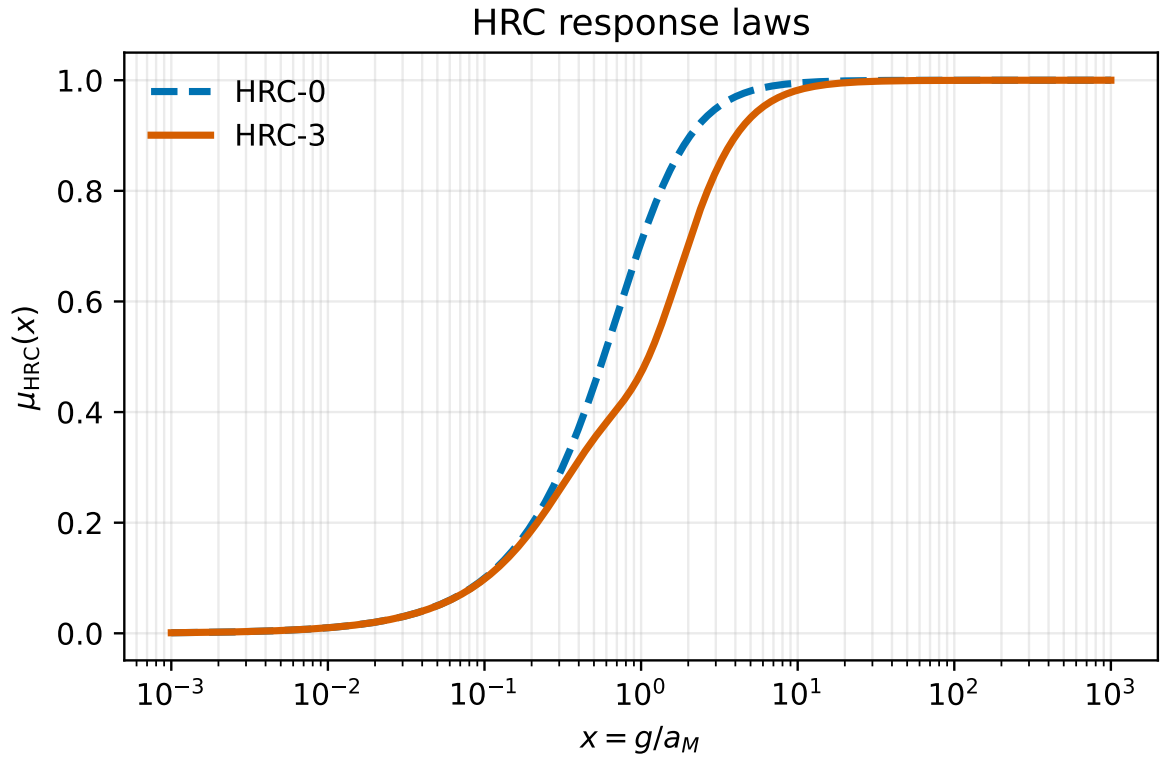
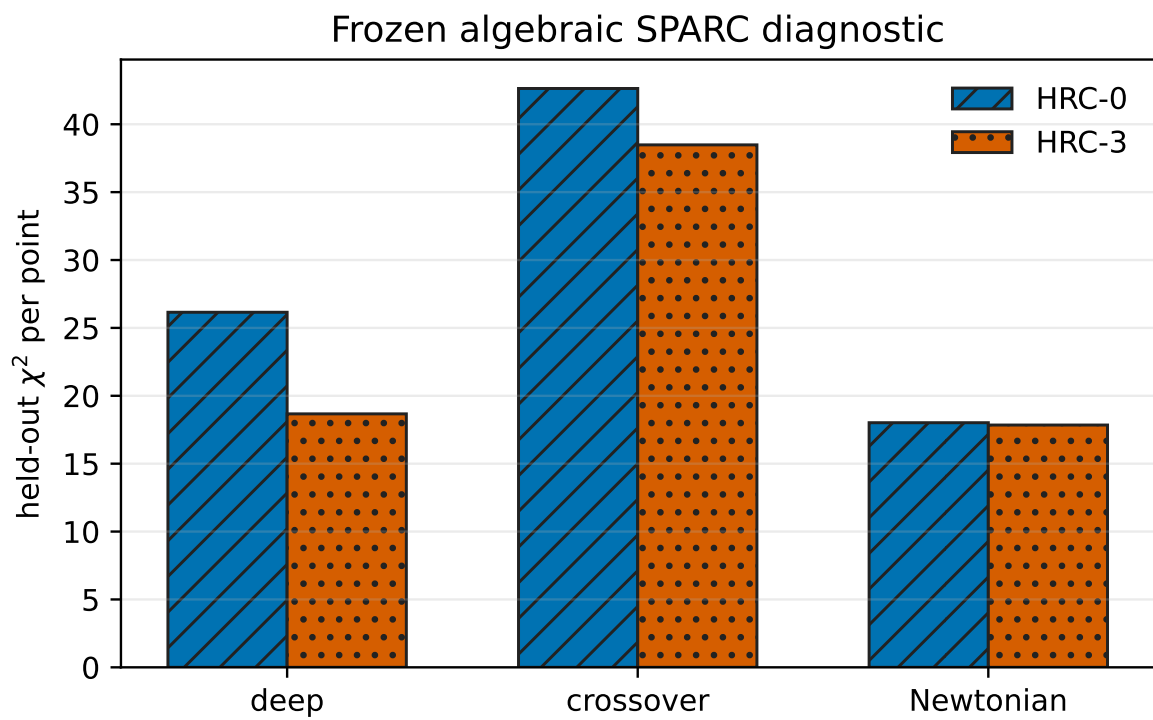


Figure 9: HRC-only response functions. HRC-0 is exact inside the stated augmented response model; HRC-3 is conditional on the supplied three-channel owner. This is the enlarged frozen left panel of the response-and-regime diagnostic; no curve or fitted quantity has changed. Colour is redundant with line style and direct labels.



Continuation of Fig. 9: regime and held-out residual diagnostics. The absolute held-out residual scale is displayed rather than hidden. This is the enlarged frozen right panel; no curve or fitted quantity has changed.

SPARC comparison I: diffuse systems

Black circles: SPARC. Smooth traces: residual-gated display curves; faint nodes: frozen model ordinates.
 Fits and diagnostics use the unsmoothed nodes. HRC-0 $\equiv$ MOND-standard at equal scale.
 Fixed scales: $a_M = 1.0824 \times 10^{-10}$ and $a_0 = 1.2 \times 10^{-10} \text{ m s}^{-2}$; NFW fitted per galaxy.

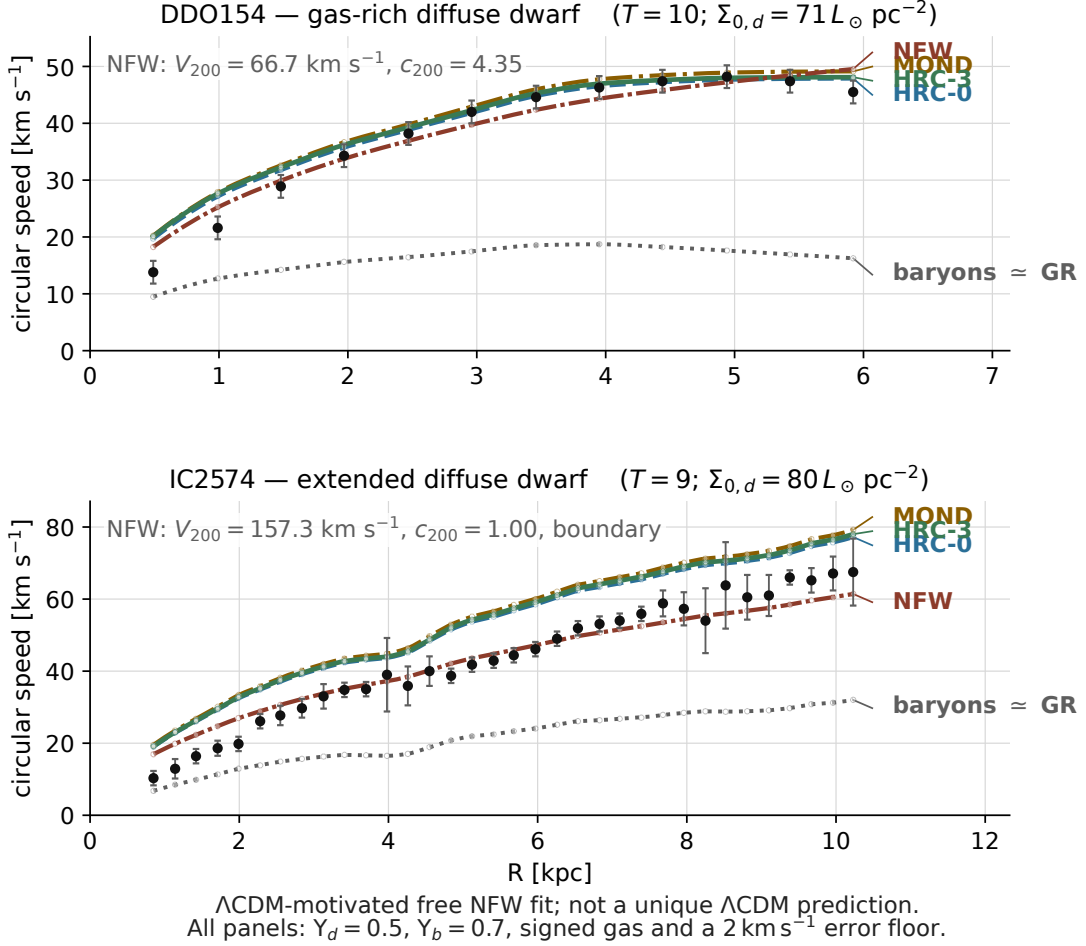


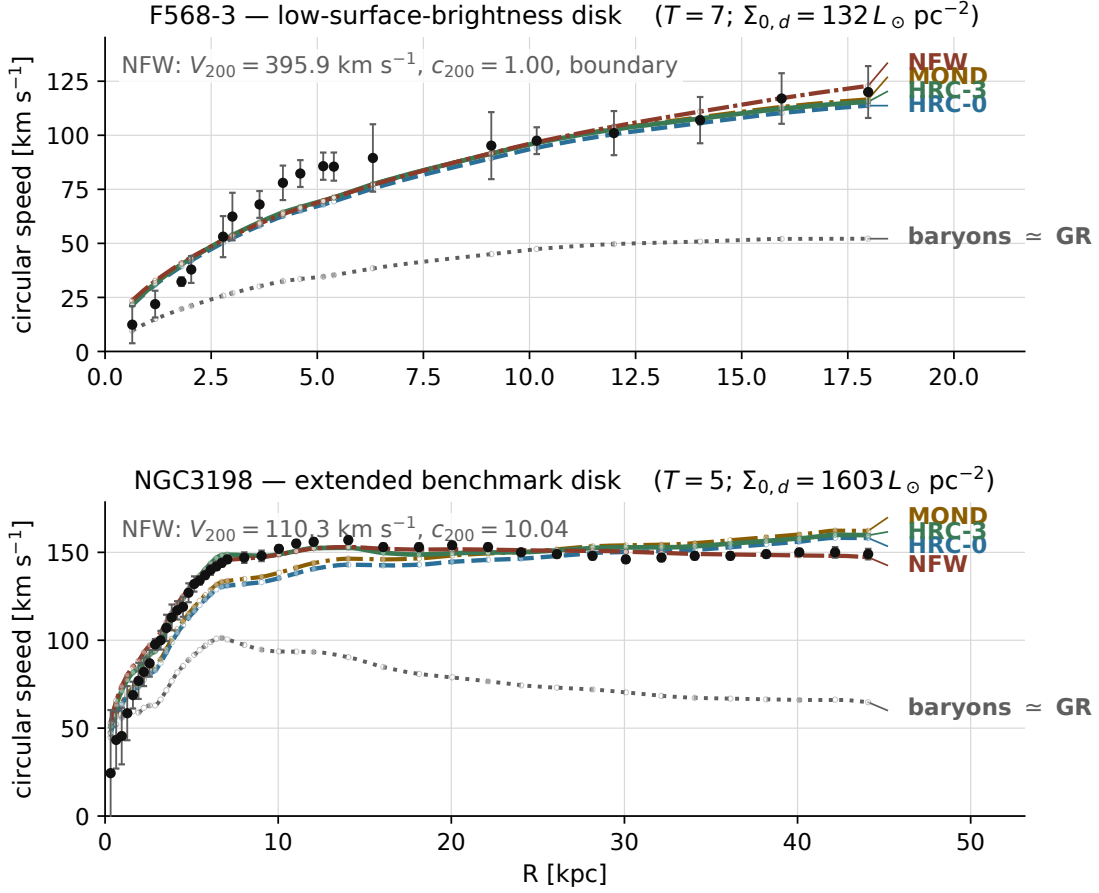
Figure 10: Representative SPARC external comparison, set I: DDO 154 and IC 2574. Fixed baryonic inputs are compared with baryon-only weak-field GR, MOND-standard, HRC-0, conditional HRC-3 and a separately fitted two-parameter NFW halo. HRC-0 and MOND-standard have the same response shape at equal scale. The NFW curve is a Λ CDM-motivated free NFW fit, fitted to the displayed rotation curve; it is not a unique Λ CDM prediction and not a population model-selection result. This is a Level-C local algebraic diagnostic, not a population likelihood. SPARC measurements are unconnected error-bar markers, and source model ordinates are faint, unconnected nodes. Smooth traces are residual-gated, shape-preserving display approximations through curvature-regularised display nodes; the maximum node-displacement tolerance is $0.10\text{--}0.25 \text{ km s}^{-1}$. All fit and diagnostic metrics use the original nodes, and no continuous physical owner is implied.

SPARC comparison II: low-surface-brightness and benchmark disks

Black circles: SPARC. Smooth traces: residual-gated display curves; faint nodes: frozen model ordinates.

Fits and diagnostics use the unsmoothed nodes. HRC-0 $\equiv$ MOND-standard at equal scale.

Fixed scales: $a_M = 1.0824 \times 10^{-10}$ and $a_0 = 1.2 \times 10^{-10} \text{ m s}^{-2}$; NFW fitted per galaxy.



Λ CDM-motivated free NFW fit; not a unique Λ CDM prediction.
All panels: $Y_d = 0.5$, $Y_b = 0.7$, signed gas and a 2 km s^{-1} error floor.

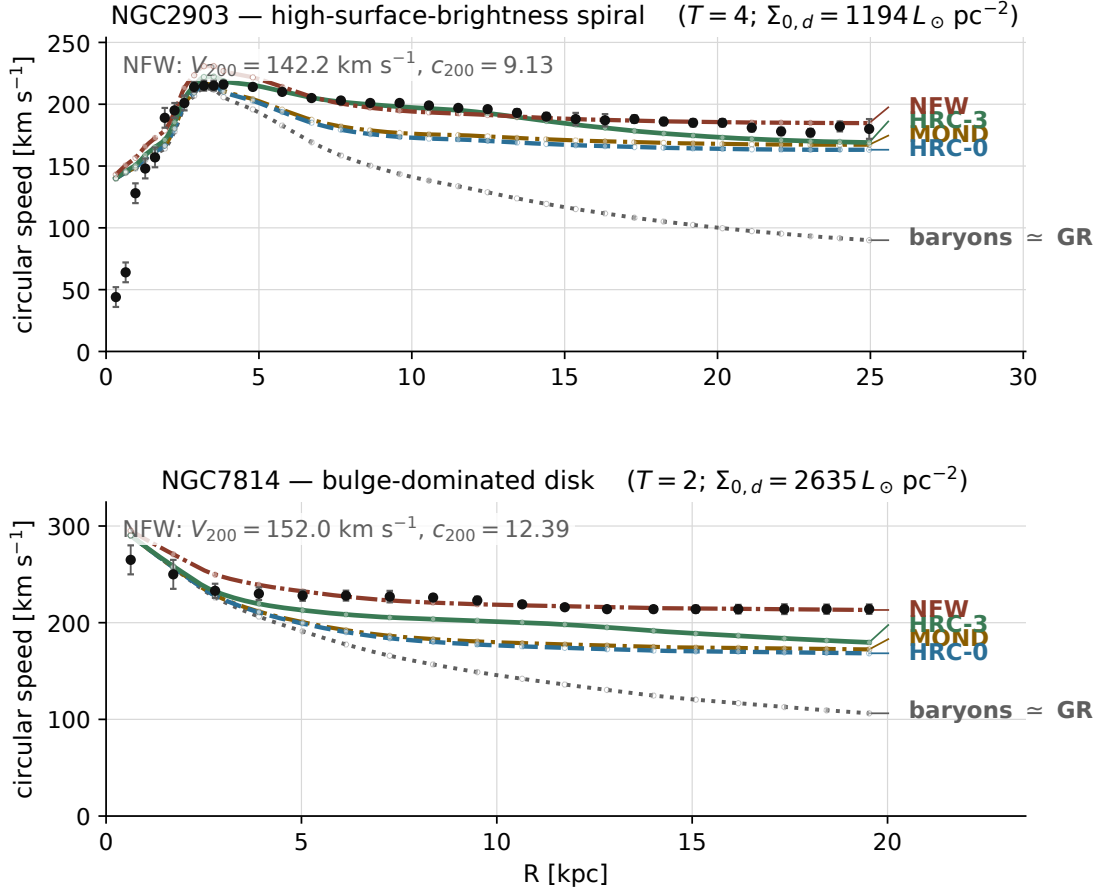
Figure 11: Representative SPARC external comparison, set II: F568-3 and NGC 3198, with the same fixed baryonic inputs and status guards as Fig. 10. The separately fitted NFW curve is a Λ CDM-motivated free NFW fit; it is not a unique Λ CDM prediction and not a population model-selection result. The reference-ordered technical-preprint appendix *Visual atlas: HRC diagnostics and rotation comparisons* keeps held-out, frozen acceleration-stratified and post-hoc residual-stress families separate; none is a survey-level comparison or ECT-discrimination result.

SPARC comparison III: bright and bulge-dominated disks

Black circles: SPARC. Smooth traces: residual-gated display curves; faint nodes: frozen model ordinates.

Fits and diagnostics use the unsmoothed nodes. HRC-0 $\equiv$ MOND-standard at equal scale.

Fixed scales: $a_M = 1.0824 \times 10^{-10}$ and $a_0 = 1.2 \times 10^{-10} \text{ m s}^{-2}$; NFW fitted per galaxy.



Λ CDM-motivated free NFW fit; not a unique Λ CDM prediction.
All panels: $Y_d = 0.5$, $Y_b = 0.7$, signed gas and a 2 km s^{-1} error floor.

Figure 12: Representative SPARC external comparison, set III: NGC 2903 and NGC 7814. The five displayed curves use the same frozen fixed-baryonic inputs and status guards as Fig. 10. The separately fitted NFW curve is a Λ CDM-motivated free NFW fit, fitted to the displayed rotation curve; it is not a unique Λ CDM prediction and not a population model-selection result. Together, the three companion figures provide reader-scale views of the six-galaxy external-comparison owner without upgrading this Level-C diagnostic to a survey likelihood.

The frozen algebraic registry uses 165 SPARC galaxies, signed gas and five repeated whole-galaxy held-out folds. Its absolute statistics are

Response	fixed- M/L $\langle\chi^2\rangle$	fixed- M/L SD	common- M/L $\langle\chi^2\rangle$
HRC-0	117598.72	3645.86	109439.95
HRC-3	99501.19	2967.18	103763.46

The fixed- M/L fits use 3343 points per seed; the common- M/L mode uses 3342. Thus the absolute scale is approximately $\chi^2/\text{point} \sim 30$, which is *not* an acceptable survey likelihood. The calculation omits covariance, intrinsic scatter, distance and inclination marginalisation, hierarchical stellar populations, the disk curl field and the external-field response. These results are Level C diagnostics, not ECT-specific evidence. They show that HRC can be evaluated consistently, but not that the galactic sector is closed.

How to read the rotation-curve lines. Every baryonic, MOND/HRC or NFW source ordinate is evaluated at the corresponding tabulated SPARC radius and retained as a faint, unconnected node. The coloured traces are residual-gated, shape-preserving display approximations through curvature-regularised display nodes. This presentation step is not a second physical or statistical fit: all fitted parameters, likelihoods, residuals and diagnostic metrics use the original frozen ordinates.

Reader guide. The three preceding pages are one six-galaxy fixed-baryonic external-comparison sequence. They retain the Level-C local-diagnostic status stated above and are not a survey-level or population-model-selection result. The complete reference-ordered visual atlas remains in the technical preprint.

For a declared exponential Milky-Way disk and 14 representative velocity points, the HRC sensitivity calculation gives

$$\begin{array}{c|cc}
 & a_M/a_{M0} & \chi_{\text{red}}^2 \\
 \hline
 \text{HRC- 0} & 2.798 & 1.813 \\
 \text{HRC- 3} & 1.703 & 2.287
 \end{array} . \tag{29}$$

This is not a modern Milky-Way posterior: gas, bulge, correlated systematics and baryonic-model uncertainty are absent.

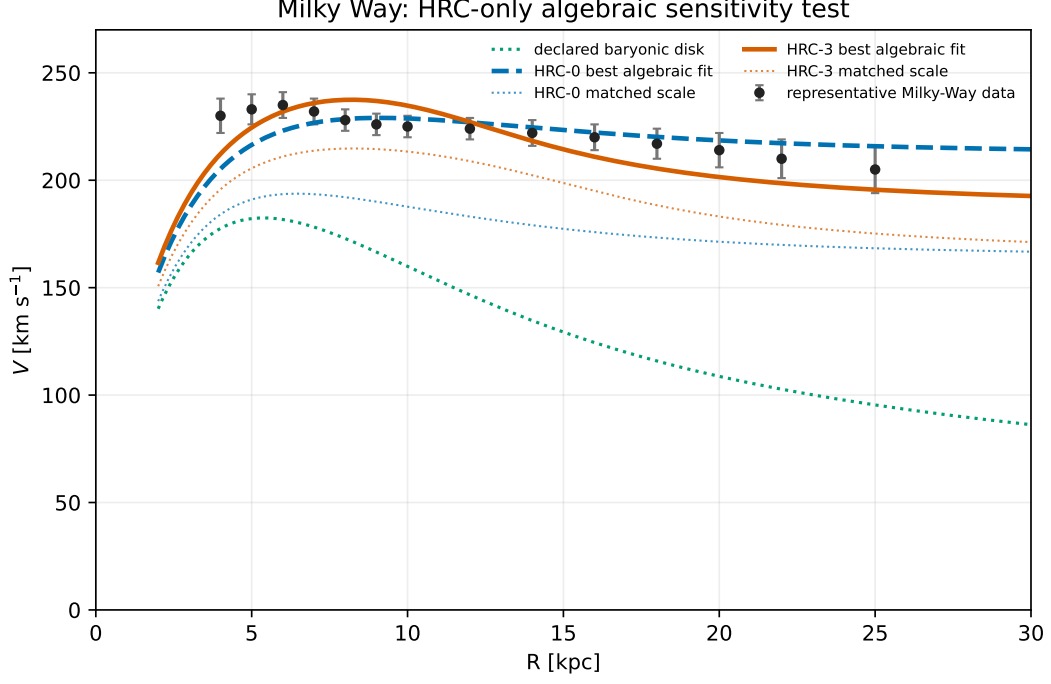


Figure 13: Representative Milky-Way HRC sensitivity test. Dotted curves use the matched scale; solid/dashed curves use the best-fit scale for the declared exponential disk. This is not a joint Milky-Way posterior.

21. The baryonic Tully–Fisher relation, RAR, and UDG stress test

Both HRC responses share $\mu_g \sim g/a_M$ in the deep branch. For an isolated spherical source, conditional on the separate gravity bridge,

$$g^2 \simeq g_N a_M, \quad v_{\text{flat}}^4 = G_N M_b a_M. \quad (30)$$

The slope 4 is therefore Level A algebra inside the supplied static functional and Level B/Open after the bridge. Its normalisation and scatter are observational questions and do not identify a microscopic owner for a_M .

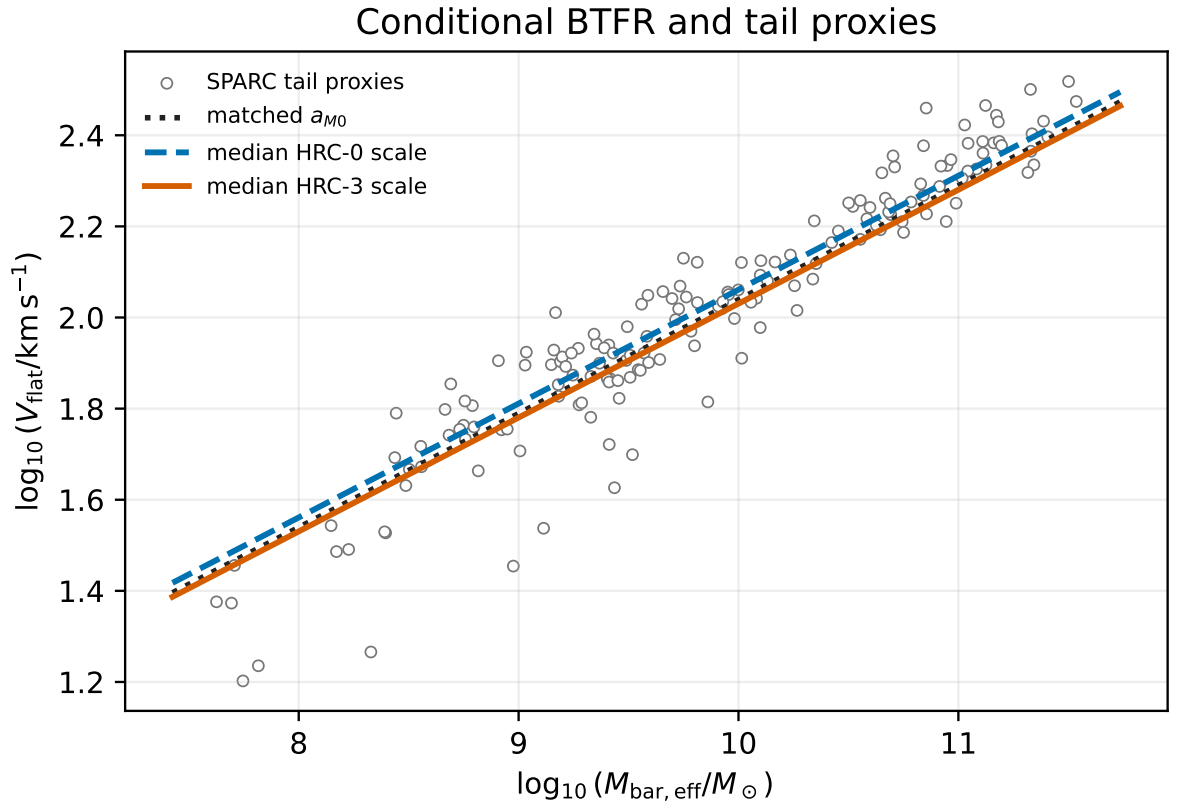
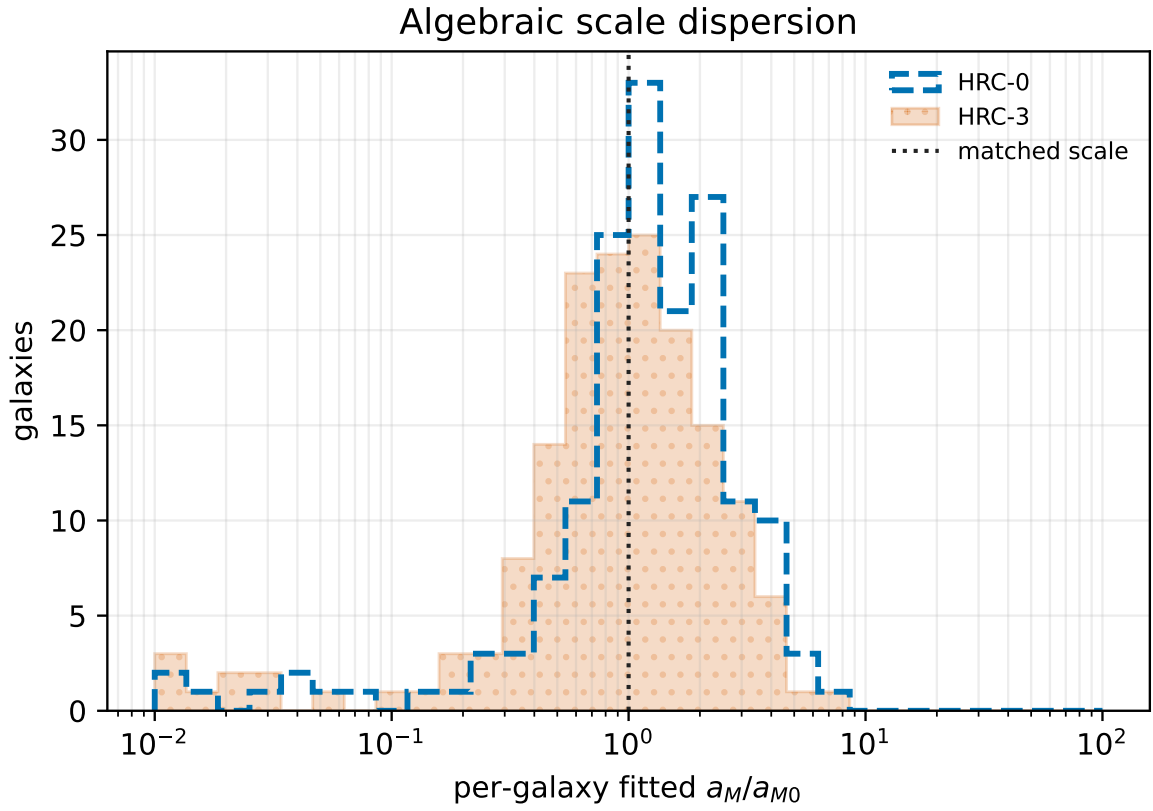


Figure 14: Conditional BTFR proxies in the HRC-only diagnostic. The slope-four lines are conditional on the identity gravity bridge; the mass proxy is taken from the same SPARC decomposition and is not an independent catalogue likelihood. This is the enlarged frozen left panel; no mass proxy or line has changed.



Continuation of Fig. 14: fitted-scale dispersion. This is the enlarged frozen right panel; no fitted value or bin has changed.

The symmetry-reduced RAR diagnostic solves

$$g_N = \mu_{g,k}(g_{\text{obs}}^2/a_M^2)g_{\text{obs}}, \quad k \in \{0, 3\}. \quad (31)$$

For a generic disk this local relation omits the curl field in $\mu\mathbf{g} = \mathbf{g}_N + \nabla \times \mathbf{H}$. Consequently the RAR plot is a pipeline diagnostic, not an independent prediction of P1–P6.

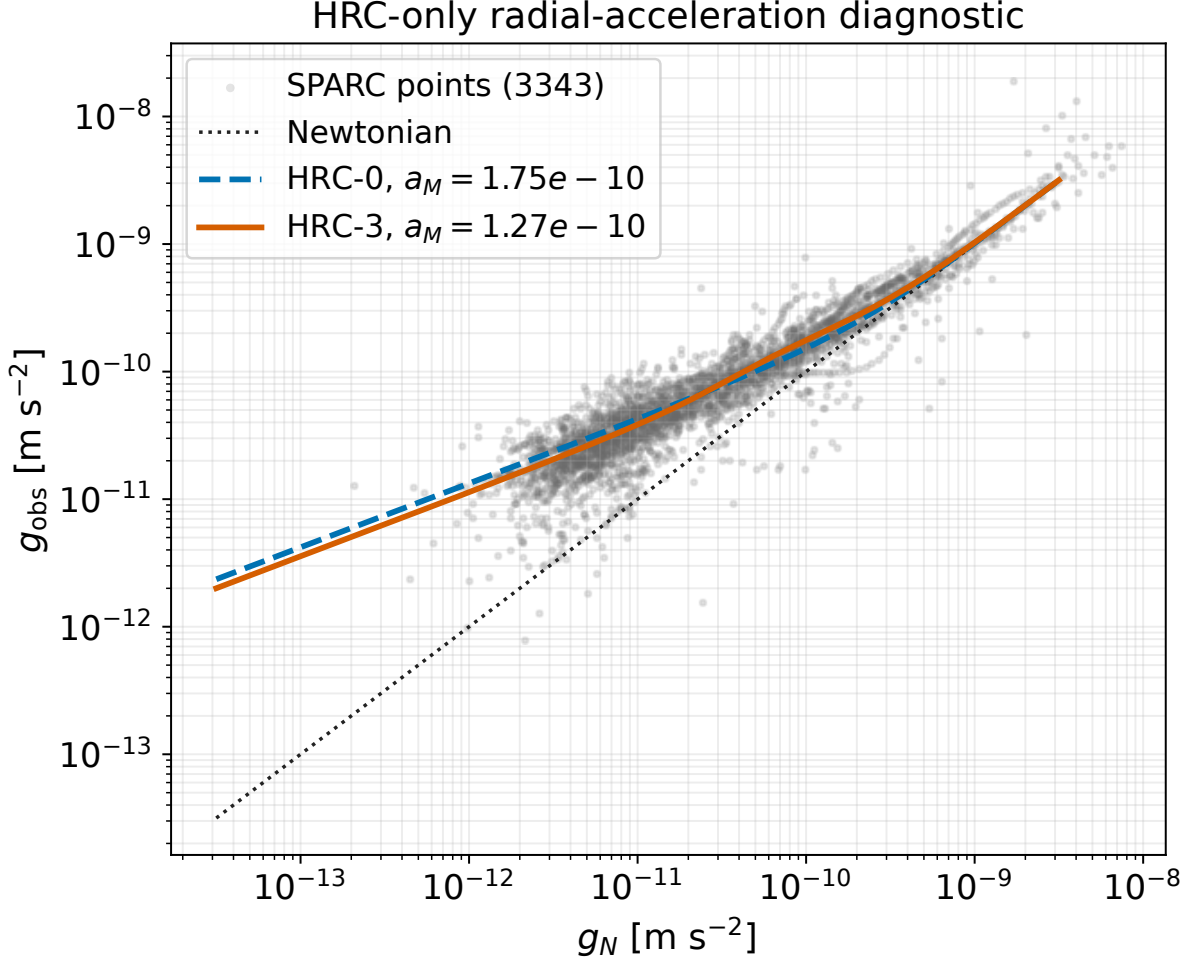


Figure 15: HRC-only RAR diagnostic for 3343 SPARC entries. The curves use the mean held-out fitted scales of the frozen local algebraic pipeline; covariance and the full disk field are absent.

For pressure-supported ultra-diffuse galaxies define $\mathcal{R}_{\text{dyn}} = g_{\text{obs}}/g_N$. A positive HRC enhancement exists only for $\mathcal{R}_{\text{dyn}} > 1$. HRC-0 gives

$$\frac{a_M^{(0)}}{g_{\text{obs}}} = \sqrt{\mathcal{R}_{\text{dyn}}^2 - 1}, \quad (32)$$

and HRC-3 is inverted by monotone bracketing. Central Wolf-proxy values give: DF4, no positive solution; FCC 224, $a_M^{(0)}/a_{M0} = 0.0155$ and $a_M^{(3)}/a_{M0} = 0.00780$; DF2, 0.0476–0.470 and 0.0278–0.457; NGC 5846-UDG1, 0.947 and 0.935; Dragonfly 44, 4.951 and 4.944. These heterogeneous central proxies are not a common-confidence Jeans sample and do not establish an environment law.

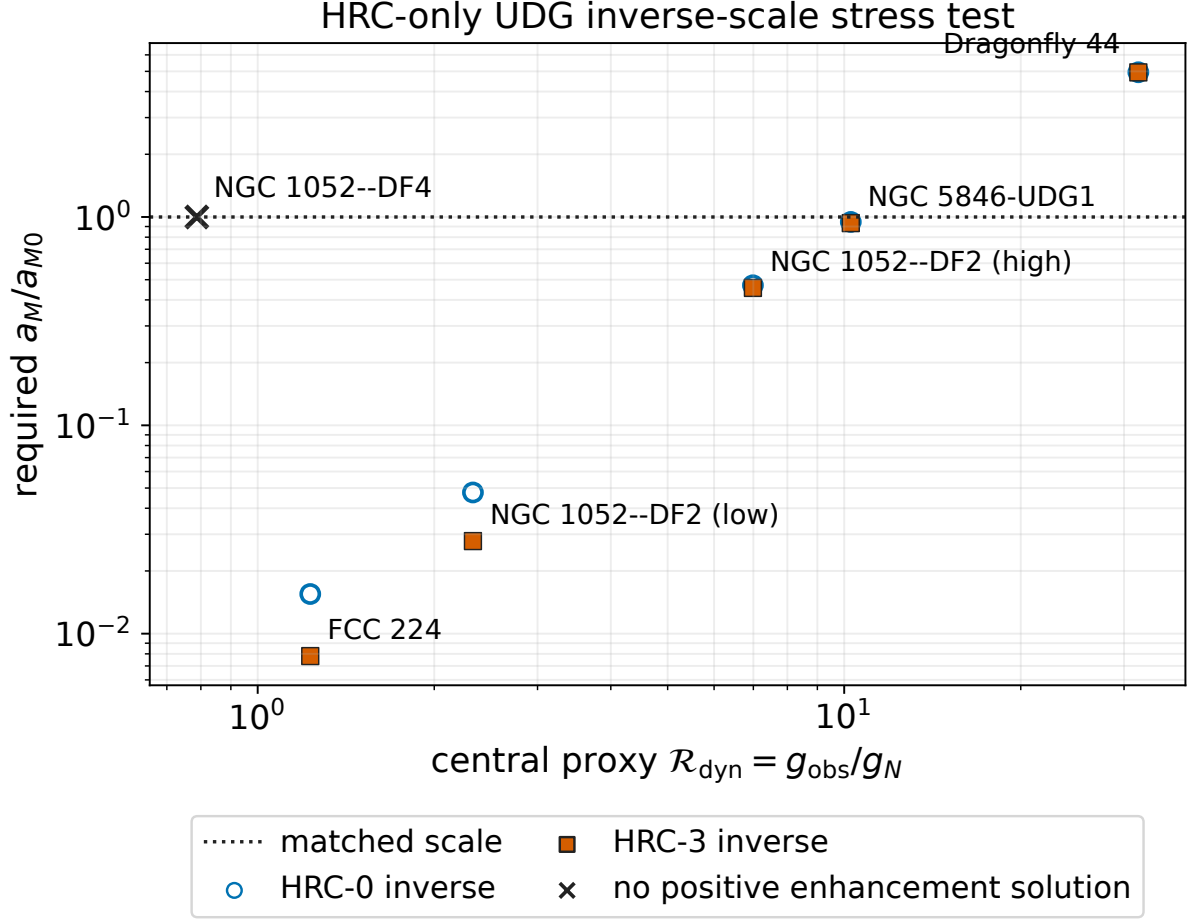


Figure 16: HRC-only inverse-scale UDG stress test. The cross marks a central proxy outside the positive-enhancement domain. The matched-scale line is not an environment fit.

22. Audit of the projected cluster proxy

The existing cluster script is not a physical HRC cluster solver. It applies a local surface-density map paired with the conditional HRC-0 point-source response to hand-set baryonic maps. For any strictly increasing positive local map, preservation of ordering and of the possibly empty set-valued total-map argmax is a Level-A theorem inside that supplied map. Whenever peaks are attained, the transformed maximizers equal the input maximizers. No synthetic replay or finite-box amplitude is retained as active phenomenology. The physical matter source, response kernel, photon metric, retarded collision dynamics and observational map are Open. Hence the identity does not explain residual cluster mass, lensing amplitude or Bullet-like offsets; those require independently constrained baryonic maps, a nonspherical dynamical solution and a prespecified joint lensing likelihood.

More explicitly, the retained proxy uses

$$\Sigma_{\text{cl}} = \Sigma_b \nu_{\text{cl}} \left(\frac{2\pi G_N \Sigma_b}{a_{M,\text{cl}}} \right), \quad \nu_{\text{cl}}(y) = \sqrt{\frac{1 + \sqrt{1 + 4/y^2}}{2}}. \quad (33)$$

Because $d[\Sigma_b \nu_{\text{cl}}(K \Sigma_b)]/d\Sigma_b > 0$ throughout the positive value interval, the transformed and input maps have the same set-valued argmax. Whenever peaks are attained, their maximizer

sets coincide. This exact identity is why peak agreement is not morphology evidence.

Restricted transport no-go and figure status. The local-map theorem cannot generate a collision offset. Independently, if one assumes one constant real Debye pole, $\tau_{\text{pole}} = 2\pi/H_0$, together with ballistic transport $\ell = v\tau_{\text{pole}}$, the static matching time is about three orders of magnitude longer than the 70–100 kpc merger-lag time. This is a conditional Level-A no-go only for that joint one-pole/ballistic model; a multi-pole, dispersive, non-ballistic or separately owned causal kernel remains Open. A physical cluster figure still requires an action-owned retarded collision and lensing calculation; no hand-set proxy map is presented as a physical cluster model.

23. Matter–antimatter asymmetry

The observable universe contains far more matter than antimatter: $\eta_B^{\text{obs}} \approx 6 \times 10^{-10}$. A conventional dynamical baryogenesis mechanism requires baryon-number violation, C/CP violation and departure from thermal equilibrium. The Standard Model contains sphaleron processes and CP violation, but with the measured parameters it does not supply sufficient CP and out-of-equilibrium dynamics to reproduce the observed asymmetry.

P4 supplies an ordered branch, but not a derived real-time transition or nonequilibrium state. Topology does not define physical baryon number, and the director term does not supply a nonzero CP coefficient, baryon-violating vertex, transport or washout. The observed η_B is therefore an external target for an Open completion programme, not an ECT result. No imported thermal-leptogenesis parameter tuple is retained as ECT phenomenology.

24. Conclusions of Part II

24.1. Status-separated outputs, benchmarks, and Open targets

The current preprint records the following geometric-branch results and closure benchmarks:

1. Lorentzian kinematics as the homogeneous limit of the scalar/phase characteristic model (*A inside that model*); physical special relativity requires the clocks/rods, matter, and photon maps (B/Open).
2. Scalar-only metric normalization derived inside its ansatz; an orientation stress $T_{\mu\nu}^{(n)}$ requires a covariant $S_n[g, n, \lambda_n]$ and remains Open.
3. Structural or conditional non-inflationary routes to horizon-scale homogeneity and a flat baseline branch, plus the A-math conditional statement $\pi_2(S^3) = 0$ for the supplied orientation target and an exact flat *formal local seed* only in a supplied pure-quartic kernel; the physical vacuum manifold, relic conclusion, finite band, state, transfer and observed primordial spectrum remain Open.
4. Late-time acceleration and $w(a)$ require an action-owned stable background and joint likelihood (Open).
5. Cosmological background and age: one supplied two-slope action and legitimate state datum give conditional $H(z)$, distances, high-redshift times and $\tau_{2s} \equiv H_{2s}(0)\Delta t = 0.96368678$; a

dimensional age additionally uses one declared calibration. Whether microscopic Φ dynamics constrains this completion, and the physical metric/redshift/lapse/front/congruence map, remain Open.

6. JWST abundance and morphology require an end-to-end background, perturbation, halo, baryonic and selection forward model (Open).
7. 165 SPARC galaxies: HRC-only held-out totals are disclosed with an absolute scale $\chi^2/\text{point} \sim 30$; this is a Level-C stress diagnostic, not a precision fit.
8. BTFR slope exactly 4 conditional on the adopted deep-branch closure (B/C).
9. RAR fit with $a_M \approx cH_0/(2\pi)$ (C); the physical matching and environment law are Open.
10. Strictly increasing local-map order/argmax theorem (Level A inside the supplied map); no active synthetic replay or finite-box amplitude; physical source, metric, retarded collision and lensing map Open. One constant real Debye pole with $\tau_{\text{pole}} = 2\pi/H_0$ plus ballistic transport is conditionally excluded, while general causal kernels remain Open.
11. Baryogenesis: only an external observational target and an Open completion inventory are available; no real-time transition, baryon current/vertex, CP coefficient, transport, washout or ECT-native η_B is derived.
12. A Level-C no-halo galactic fit and a separate calibrated late-time scalar closure; their ECT matching, particle content, and common microscopic origin remain Open.

24.2. Key equations and principles of Part II

Scalar-only metric equation (undivided): $FG_{\mu\nu} = 8\pi G_N T_{\mu\nu}/c^4 + \mathcal{T}_{\mu\nu}^{(q)}$; division by F introduces no orientation term, and $T_{\mu\nu}^{(n)}$ requires a separate covariant $S_n[g, n, \lambda_n]$ (Open).

HRC algebraic relation: $g_N = \mu_{g,k}(g^2/a_M^2)g$, $k \in \{0, 3\}$.

BTFR: $v_{\text{flat}}^4 = G_N M_{\text{bar}} a_M$ (slope 4, algebraic).

$t_{0,\text{ECT}} = \int_{a_{\text{ord}}}^1 da/[aH_{\text{ECT}}(a)]$; the named two-slope state gives $\tau_{2s} \equiv H_{2s}(0)\Delta t = 0.96368678$ conditionally. Only after declaring $H_{2s}(0) = H_0^{\text{cal}}$ may this be written as $H_0^{\text{cal}}t_0$; the physical observable map and microscopic selection remain Open.

24.3. Conditional completion tests and external observational gates

The macroscopic sector currently supplies a mixed set of conditional closure tests, fitted diagnostics, external observational gates and Open programmes. These items are not uniformly ECT predictions:

- BTFR slope = 4 conditional on the adopted galactic closure (B/C); fitted $a_M \sim cH_0/(2\pi)$ (C).
- The homogeneous director-vector term has no spin splitting and no source-generated Eötvös force; external spin/WEP data constrain only separately declared completions. The physical late-time source and $w(a)$ remain Open.

- Exploratory environment/history law Ξ (*Open*); physical tensor polarisation and graviton mass remain *Open*, with two TT modes only in a supplied massless Fierz–Pauli completion.

The full status-separated list is given in Part V, Section 53.

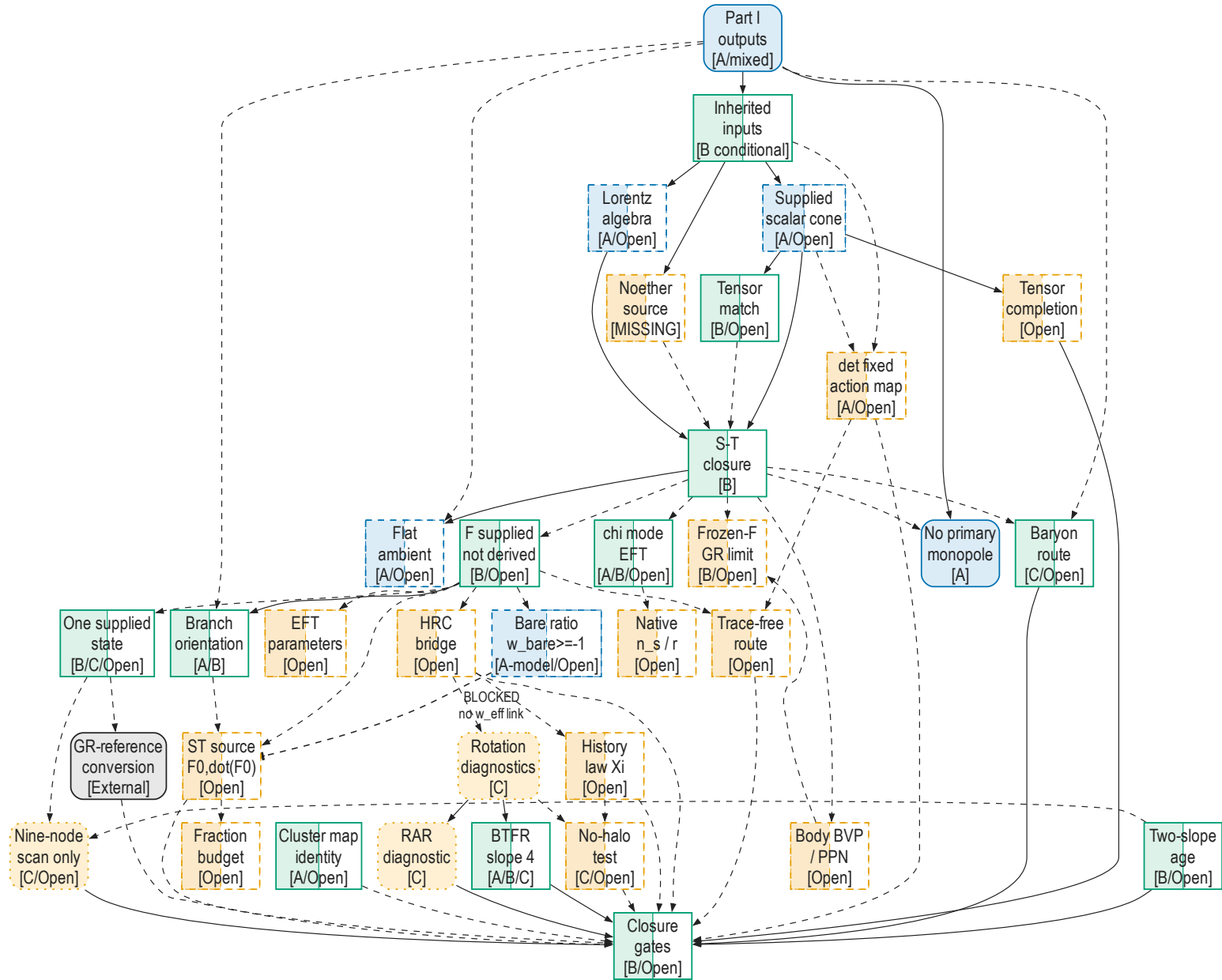
Figure 8 now visualises a frozen five-row sensitivity envelope; it is deliberately not called a cosmological prediction. A genuine prediction image remains deferred until one exact ECT background and observable pipeline passes all closure gates.

24.4. Part II status and dependency logic

Figure 17: Complete connected Part II status and dependency map. The supplied ordered-branch EFT, conditional macroscopic completions, owner-specific cosmology gates and *Open* targets are kept distinct. The routes separate the exact orientation determinant, conditional trace-free $V + C_I$ and two-slope results, and the *Open* physical ECT action/state/metric map. All canonical nodes and edges appear in one colour-coded A4 reader map; arrows do not certify derivation. The deterministic exact-key and machine-topology records preserve the canonical source labels and edge inventory.

The status and dependency logic of Part II is shown in Fig. 17. It is deliberately kept whole so that the gravity, cosmology, galactic and vacuum-offset branches remain visibly connected. The PDF is vector content for reader zooming; no edge upgrades the literal status of either endpoint. It explicitly marks F as supplied, limits $w_{\text{bare}} \geq -1$ to the bare positive-kinetic ratio, preserves the false observable- w_{eff} implication as a blocked edge, separates the GR-reference density conversion from the *Open* scalar–tensor late source, and restricts the JWST result to the nine computed nodes.

Selected Part II status and dependency map



A exact | B conditional | C fit/benchmark | Open/MISSING explicit. Solid border = A/external; striped = conditional/Open; dotted = C; dashed + tee + BLOCKED = no implication. Colour is redundant.

Part III

Quantum Sector: Coherent Branch

Why does quantum mechanics exist? In ECT, the answer has three layers.

First: the Euclidean functional integral. Because the fundamental arena is a four-dimensional Euclidean manifold with no built-in time, the P3 model distinguishes the dimensionless reduced functional $\mathcal{I}_E^{\text{P3}}$ from the physical classical action $S_E^{\text{P3,phys}} = A_{\text{P3}} \mathcal{I}_E^{\text{P3}}$, where $A_{\text{P3}} > 0$. A separately supplied P3 measure has the weight $\exp[-\kappa_{\text{P3}} \mathcal{I}_E^{\text{P3}}] = \exp[-S_E^{\text{P3,phys}}/\Sigma_{\text{PI,P3}}]$, with $\Sigma_{\text{PI,P3}} > 0$ and $\kappa_{\text{P3}} = A_{\text{P3}}/\Sigma_{\text{PI,P3}} > 0$. Neither action scale, their ratio, nor a map to S_0 , Σ_θ , Σ_ϑ or $\hbar$ is derived; $\kappa_{\text{P3}} = 1$ would be only an explicit normalisation convention. These action symbols are also unrelated to the protocol entropy production $\Sigma[\gamma]$. The integral sums over all field configurations, including “closed histories” where the field returns to itself along the fourth coordinate; there are no events, temporal ordering or cause–effect sequences at this level. For supplied Gaussian/free kernels satisfying the complete applicable corrected OS-II hypothesis package, the positive-semidefinite form can be quotiented by its null space and completed to a conditional Hilbert-space reconstruction. The declared checks establish only their specified kernels; the interacting ECT measure, observable algebra, continuation and generator, physical clock/metric and theory-wide unitarity remain B/Open.

There is one useful same-channel closure gate. For a supplied positive free mode in the infinite-time vacuum, if the OS and canonical descriptions use the same normalised q , calibrated m, ω and pole residue, with no field/source renormalisation, their full correlators are respectively $\Sigma_{\text{PI,ch}} e^{-\omega|\tau|}/(2m\omega)$ and $S_{0,\text{ch}} e^{-\omega|\tau|}/(2m\omega)$. Equality for that same channel forces $\Sigma_{\text{PI,ch}} = S_{0,\text{ch}}$. This is standard conditional free-mode mathematics and a B/Open ECT consistency gate, not a P3 identification, $A_{\text{P3}} = \Sigma_{\text{PI,P3}}$, cross-sector universality or a derivation of $S_0 = \hbar$. Field/residue maps, finite temperature, interactions, constraints or non-vacuum states require a new comparison.

Second: supplied action-scale candidates and their possible match to $\hbar$. After BR1 supplies a compact complex field, single-valuedness gives $\oint \partial_A \theta dX^A = 2\pi n$, while a phase-only fixed-core closure defines $\iota_{\text{elem}}^{(\text{fixed core})} = 2\pi \iota_0^{\text{EFT}}$. CDE- Φ is only a post-order acceptance criterion. A distinct reparametrisation-invariant texture-worldline route defines $\iota_{0,\text{bal}}^{(\vartheta)} = \sqrt{2K_\vartheta^{1\text{D}} \mathcal{T}_{\text{loop}}}$ only after a 4D owner supplies the carrier, compact modulus and proper-time ensemble. Under the separate bending-free $n = 1$ matching this gives $\Sigma_\vartheta \mathcal{I}_{1,\text{min}} = 2\pi \hbar = h_{\text{P}}$; the relation is physically motivated but does not derive the measured value. Profile/Hessian, bending/barrier and cross-sector universality remain Open (*Level-A reduced algebra / Level-B route / Open ownership and matching*).

Third: reduced-state records. A declared environment can suppress interference between specified alternatives, but the pairwise record exponent does not create Lorentzian signature, define a universal quantum–classical boundary, or select one realised outcome. Deriving the ECT record operators, state, and detector update remains Open.

25. ECT versus standard quantum mechanics

Standard quantum mechanics supplies a tested operator and probability framework. ECT offers reconstruction routes from selected Euclidean coherent subsectors, but their statuses differ:

the quadratic RP/OS bridge is controlled, whereas interacting states, Born weights without a supplied physical probability owner and the complete representation-theorem hypotheses, detector updates, and solution-selection measures remain B/Open. A Euclidean boundary-value formulation is therefore an architectural hypothesis, not a proof that quantum probability is merely coarse-graining or that one unique global solution exists.

25.1. Detailed structural comparison: ECT vs standard QM

The following examples compare the ECT coherent-branch interpretation with standard quantum mechanics in specific situations. Each case states what the standard formalism already predicts, what additional owners a future ECT reconstruction would require, and the present status; no extra explanatory success is assumed.

The hydrogen atom. The standard Coulomb boundary/eigenvalue problem yields discrete levels. ECT is compatible with embedding that effective problem in an ordered medium, but compact winding is not a universal explanation of every discrete spectrum. The Coulomb Hamiltonian, $-13.6/n^2$ levels, fine structure, Lamb shift, hyperfine splitting, and α_{fs} have not been derived from the minimal ECT action (B/Open).

The double-slit experiment. Admissible Euclidean solutions or an ensemble of solutions may depend on the declared apparatus boundary data. ECT has not established uniqueness of that nonlinear boundary-value problem, a physical measure over solutions, the map to slit/detector settings, or the observed $|\psi_1 + \psi_2|^2$ probabilities. The global-boundary reading is interpretive compatibility (B/Open), not an ECT-native derivation of the fringe pattern.

The Aharonov–Bohm effect.

Standard QM.

A charged particle acquires an observable phase shift $\Delta\varphi = (e/\hbar) \oint A_\mu dx^\mu$ even in regions where the electromagnetic field strength vanishes, provided the path encircles a non-trivial gauge flux.

ECT (conditional picture).

The compact phase $\theta(X)$ can support a winding label, but an electromagnetic holonomy follows only after a physical Abelian connection and charged matter representation have been constructed.

Required ECT owners.

A future ECT calculation would need an unbroken $U(1)_{\text{em}}$ connection, a charge/matter map, a candidate-to-operational action owner, and the separately adopted Level-B calibration $S_0 = \hbar$.

Status.

Interpretive compatibility (*B/Open*), not an ECT derivation of the electromagnetic AB phase. The physical connection, charge and matter map remain Open.

Measurement and outcome selection. A specified influence-functional reduction can suppress interference and form stable records. Pointer-basis selection, a single outcome, exact Born weights, and a detector update theorem are not established. For a complex Hilbert space of dimension at least three, the ordinary projector Gleason theorem constrains only an already admitted normalized, noncontextual, countably additive probability measure on orthogonal projector decompositions to trace form. In dimension two, ordinary projector Gleason does not apply. Under the separately supplied condition C2q—a generalized measure v on all effects $0 \leq E \leq I$, with $0 \leq v(E) \leq 1$, $v(I) = 1$, and σ -additivity for every countable effect sequence satisfying $\sum_i E_i \leq I$ (a normalized POVM with $\sum_i E_i = I$ is the special case)—the Busch extension [12] may instead be applied; without C2q it may not. Neither result creates probability or selects the prepared state. The separate application condition C3 must also be supplied: a preparation together with a projector (ordinary C2) or an effect/POVM outcome (C2q) must be mapped to a laboratory event. ECT has not derived the physical owner of that map; it remains Open. Dephasing and record formation belong to a distinct application layer and are not theorem hypotheses. The Hilbert/projector setting, measure assumptions and origin of probability therefore remain B/Open rather than following from bare P1–P6.

Conditional field-level determinism. The Euclidean field equation defines a classical non-linear boundary-value problem once a branch and admissible boundary data are specified. Global existence, uniqueness, stability, and a measure/selection rule over multiple solutions have not been proved. The no-fundamental-collapse ontology is an architectural assumption; subsystem probabilities, Born weights, and the measurement update retain their B/Open reconstruction status. Thus ECT is not presently a theorem that one boundary data set yields one unique global configuration or that probability is absent at the deepest level.

Boundary dependence and delayed-choice compatibility. For a well-posed elliptic problem, interior solutions can depend on all specified boundary data. Mapping laboratory settings to ECT boundary data, showing that later settings enter the physical problem, and reproducing delayed-choice/eraser probabilities and no-signalling are all Open. The boundary-value picture is therefore an interpretive compatibility (B/Open), not a derivation of future-setting influence or retrocausality.

Common-architecture hypothesis for the coherent and geometric branches.

Standard physics.

General relativity describes gravity; quantum mechanics describes microscopic coherence; how the two fit together is the canonical problem of quantum gravity.

ECT (physical picture).

Both branches are organised within the same ordered-condensate architecture. The controlled long-wavelength equation used here is scalar-only; a geometric orientation stress is a separate Open $S_n[g, n, \lambda_n]$ extension. The shorter-wavelength coherent sector has a supplied local KG-form equation and a separately supplied complex law; only under the stated clock, positive-frequency, mass, interaction and slow-envelope assumptions does their leading NR relation admit a Schrödinger-type reading.

What this suggests.

The common architecture makes a coupled gravity/coherent-sector completion conceivable. A covariant orientation action, back-reaction vertex, and interacting quantum kernel are still required before quantitative back-reaction is claimed.

Status.

The scalar mode decomposition is Level A conditional mathematics only inside the supplied scalar ansatz. Its interpretation as a common physical origin of tensor gravity and quantum dynamics is a Level B/Open architectural hypothesis; the common action, state, back-reaction vertex and observable map remain Open.

What ECT conditionally organises rather than derives in full.

- (i) A leading Schrödinger-type envelope as conditional asymptotic algebra after a norm-preserving complex law, Hamiltonian/domain, state/stable positive-frequency component, the same physical clock, mass matching, weak interaction, slow-envelope regime and residual control are supplied; the ordered real-scalar branch alone does not derive it;
- (ii) the Klein–Gordon-form scalar sector after $d\tau = \mathcal{N}_\Phi dx^0/c_u$. The shorthand $t = \tau = w/c$ is restricted to an aligned planar branch with $\mathcal{N}_\Phi = 1, c_u = c$; independently, $\hat{c}_*$ fixes the scalar-cone slope;
- (iii) the Robertson–Schrödinger inequality as exact mathematics only after a Hilbert space, self-adjoint domains and commutator are supplied; compact phase independently gives integer winding, while the physical commutator, finite- S_0 owner and cross-sector normalisation remain Open and $S_0 = \hbar$ is the separate Level B calibration;
- (iv) reduced-state dephasing and record formation in specified channels; the ECT split, record operators, state, and kernels remain Open;
- (v) record formation without a metric-signature transition or unique-outcome theorem; pointer selection and update remain Open;
- (vi) a conditional representation-theoretic route: the connected $SO(3)$ subgroup of the supplied $O(3)$ stabiliser has spin double cover $\text{Spin}(3) \simeq SU(2)$; identifying it with physical spatial rotations and realising spinorial excitations remain Open, so this is not a derivation of the physical spin sector;
- (vii) the PES-R inventory of distinct stationarity, record, finite-time stability, and protocol diagnostics, without a universal selection theorem;
- (viii) a programme for deriving an ECT gravitational record channel; the symbolic Diósi–Penrose comparator is external [13, 14], while the physical ECT vertex, state, and noise kernel remain Open;
- (ix) separate tunnelling and entanglement owners; Euclidean connectivity supplies at most a geometric analogy, while a common dynamical mechanism remains Open;
- (x) a proposed cross-sector action-scale programme which must first derive the relation, if any, among ι_0^{EFT} , $\iota_{0,\text{bal}}^{\text{eff}}$ and the modulus-route $\iota_{0,\text{bal}}^{(\vartheta)}$, and only then test whether one independently owned coefficient maps to a universal physical action normalisation; the four-dimensional owner, finite value, matching map and universality are not established.

What remains open. The Born rule from first principles beyond the present closures; a completed detector channel, pointer basis, unique-outcome update, and ECT record kernel; Bell correlators directly from the condensate theory beyond toy visibility proxies; the detailed origin of three generations; the exact value of $\alpha_{\text{fs}} = 1/137$; the full Yukawa hierarchy. These are presented as open derivational targets, not as failures of the framework.

26. Conditional Hilbert-space reconstruction from the corrected OS-II package

One central conditional route uses the corrected Osterwalder–Schrader reconstruction theorem. Its input is not reflection positivity alone: one must supply Euclidean Schwinger functions or a measure, a physical observable algebra and time reflection, temperedness with the applicable OS-II regularity/growth condition, Euclidean covariance, reflection positivity, permutation or graded symmetry, and the applicable locality/cluster conditions. The positive-semidefinite OS form is quotiented by its null space and completed to obtain the Hilbert space. Gaussian/free checks establish only their explicitly verified kernels; the full interacting ECT measure, observables, continuation, generator, physical clock/metric map and S_0 identification remain B/Open.

The constructive Osterwalder–Schrader direction is from suitable Euclidean Schwinger data to a Hilbert/Wightman description; Euclidean continuation of an already supplied Lorentzian theory is the converse analytic route. ECT may use the constructive direction only where the complete applicable corrected OS-II hypothesis set is verified. This conditional mathematical reconstruction does not by itself identify a physical ECT time, metric, state, observable algebra or theory-wide unitary dynamics.

27. Conditional spinor-compatibility route (physical spin Open)

In nonrelativistic quantum mechanics, spin transforms in $SU(2)$ representations of spatial rotations; in relativistic quantum field theory, particle spin is classified through Poincaré representations and their little groups. Symmetry fixes the allowed representations, while which representation a particle species realises is empirical input.

ECT offers a possible representation-theoretic backdrop, not a derivation of physical fermions. The connected Euclidean rotation subgroup is $SO(4)$, whose spin double cover is $Spin(4) \simeq SU(2)_L \times SU(2)_R$. For the supplied nonzero P4 vector, the full stabiliser is $O(3)$ and its connected $SO(3)$ subgroup has double cover $Spin(3) \simeq SU(2)$. These facts make spinorial representations algebraically available. Their identification with a physical three-dimensional rotation group, and the physical metric/tetrad, spin structure, matter action and stable excitation spectrum, remain Open.

On this reading, spin is not the literal mechanical rotation of a tiny ball in ordinary three-dimensional space. Conditionally, it may be represented by a spinorial fermionic field compatible with the condensate symmetry structure. A magnetic moment follows only after a physical fermion, an unbroken $U(1)_{\text{em}}$, a matter vertex and its coefficient are supplied; it is not an output of rotational group theory alone.

Important caveat. In the current formulation, fermions are introduced as fields compatible with the condensate symmetry, not yet derived as collective condensate excitations. The electron

g -factor, the fine-structure constant α_{fs} , and the hydrogenic fine structure are not yet calculated from condensate microdynamics. These remain open derivational targets (*Level Open*).

Superfluid analogy. A useful but limited analogy: in superfluid $^3\text{He-A}$ (Volovik 2003), Weyl fermions emerge as low-energy excitations near topological Fermi points—the spinorial character is a property of the quasiparticle spectrum, not an external input. ECT does not yet reach this level of derivation for fermions. The structural compatibility route developed in the main text should be understood as the present limit of what the framework can establish. The superfluid example is an external existence example only. It supplies neither evidence for an ECT Fermi point nor an ECT spinor derivation.

28. Superposition, interference, exchange statistics, and the Pauli exclusion principle

Linear superposition route (*A inside the quadratic subsector; B/Open physically*): solutions of the verified linearised equation can be added. Identifying this with the full interacting quantum-state principle requires the RP/OS and operator completion.

Interference route (*Level B route / Open physical ECT*): after a quantum measure/coherent structure, composable kernel, state/preparation and observable map are supplied, standard constructive/destructive amplitudes can be represented. Compact phase, topology, linearity or elliptic boundary dependence alone derives neither the quantum superposition rule nor arbitrary laboratory patterns.

Indistinguishability of same-species particles (*Open physical owner*): membership in the same emergent mode sector does not by itself exclude unresolved core, orientation, internal or environmental labels. If two physical ECT excitations are independently supplied as identical unlabeled point excitations in defect-free $\mathbb{R}^3$, the coordinate configuration space is the quotient by exchange used below. ECT has not yet derived that excitation/identity map or excluded additional discrete fibres and sectors.

Exchange statistics (*A topology conditional / Open physical statistics*): For

$$\mathcal{M}_2 = ((\mathbb{R}^3 \times \mathbb{R}^3) \setminus \Delta)/S_2$$

of two supplied identical point excitations in defect-free $\mathbb{R}^3$, $\pi_1(\mathcal{M}_2) = \mathbb{Z}_2$ gives two homotopy classes. Factors $+1$ and -1 appear only after choosing the corresponding one-dimensional unitary character; topology does not choose the character or tie it to spin. For $N \geq 3$, higher-dimensional S_N representations are also mathematically available. A separate convention-fixed calculation in the preprint shows that, for a supplied free minimal massive Euclidean Dirac kernel with $m > 0$, the Grassmann construction passes the displayed quadratic RP test whereas the commuting assignment of the same kernel has an explicit negative mode. The massless/IR case is outside that test. This is Level A conditional mathematics for the declared kernel, not a derivation of physical ECT fermions, CAR or a universal interacting spin–statistics theorem.

Pauli route (*B/Open for ECT matter*): antisymmetric fermionic quantisation/CAR enforces exclusion once the physical ECT spinor kernel is obtained. The free-kernel check is stronger, but the interacting ECT fermion reconstruction remains Open.

Conditional local-CAR normalisation result (*Level A inside the supplied local algebra;*

physical ECT matter Open): if a local first-order fermionic EFT with the standard CAR algebra and a smooth, strictly positive temporal kinetic factor is supplied, an invertible local canonical field normalisation removes that factor while preserving the supplied algebra. Condensate-dependent masses and local interaction terms then modify mode dynamics rather than the assumed anticommutator. This statement does not derive physical ECT fermions or CAR and implies no curvature scaling, compact-object estimate or astrophysical exclusion window. Curvature operators and Wilson coefficients, the state and the observable map remain Open (OP-Q11, OP-Q21).

In textbook nonrelativistic quantum mechanics, Hilbert-space linearity and the Born rule are axiomatic, while identical-fermion antisymmetry is imposed by the symmetrisation postulate. Relativistic local quantum field theory derives the spin–statistics connection only under its standard hypotheses. ECT organises conditional geometric, topological and representation-theoretic routes, but the full operator-level fermionic reconstruction, spin–statistics theorem and physical Pauli-exclusion closure remain Open.

29. PES-R: response, retention, and records

PES-R is an organising taxonomy, not a universal selection theorem. First solve the sector’s Hamiltonian, retarded, Liouvillian, Floquet, or constrained problem. Then test finite-time retention and pairwise record robustness for a declared protocol. Boundary, charge, representation, and compact-phase constraints apply only in their owning sectors.

The owners must remain separate. Euclidean action stationarity belongs to the variational problem; D^R belongs to causal response; an established nonnegative $\Phi_{ab}(T)$ belongs to pairwise reduced-state records; finite-time singular values or γ_a belong to stability/decay diagnostics; and $\Sigma[\gamma]$ belongs only to a named forward/reverse stochastic protocol. No universal quietness ratio or necessary-and-sufficient intersection theorem has been derived.

An additive one-history action–noise stationarity equation is not the generic Feynman–Vernon mean equation. The Gaussian bulk formulas require a named joint initial preparation and closed-time-path contour, with the unresolved-sector one-point function centred or absorbed into the declared system force. A compatible factorised preparation is sufficient for the usual bulk form; a correlated preparation requires an explicit initial-boundary term $S_{ic}[q_+, q_-; t_0]$, an equivalent extended contour, or a proof that the extra term vanishes or has been absorbed consistently. Thus the complete coarse-grained action is $S_{CGEA}^{corr} = S_{bulk} + S_{ic}$, and the full mean equation and full $-\ln|F|$ may receive initial-correlation contributions. A finite translation susceptibility $\Gamma_{tr}[q; \eta]$ exists only when the directional record derivative belongs to the noise quadratic-form domain with finite $N^{1/2}$ -norm; otherwise only extended liminf/limsup diagnostics, or NOT IDENTIFIABLE, are justified. It is not a universal force or lifetime. PES-R does not derive Born weights, energy spectra, least action, or a unique outcome. ECT-specific record operators, preparation/state, and kernels remain Open.

For convention closure define $d_{ab}(\omega) = \int_0^T dt e^{i\omega t} \Delta O_{ab}(t)$ and $\Xi_{ab}^{full}(T) := -\ln|F[a, b]|$. Whenever the full influence magnitude has been shown contractive, set $\Phi_{ab} := \Xi_{ab}^{full} \geq 0$. Only in a stationary bulk regime, with initial-boundary/preparation transients absent or explicitly separated, may one write $N(t - t')$ for the action-normalised time-domain noise kernel, $N^{(S)}(\omega) \equiv \tilde{N}(\omega)$ for its spectral Fourier transform, and $S_H(\omega) \equiv S_0 N^{(S)}(\omega)$ for the action-scaled spectral record kernel;

otherwise the two-time kernel $N(t, t')$ must be retained. Under those premises,

$$\Phi_{ab}^{\text{bulk}}(T) = \frac{1}{2S_0} \int_0^T dt dt' \Delta O_{ab}(t) N(t - t') \Delta O_{ab}(t') = \frac{1}{2S_0^2} \int \frac{d\omega}{2\pi} d_{ab}^\dagger S_H d_{ab}. \quad (34)$$

When S_{ic} is absent or has no modulus contribution, $\Xi_{ab}^{\text{full}} = \Phi_{ab} = \Phi_{ab}^{\text{bulk}}$. Otherwise Ξ_{ab}^{full} also contains the explicitly retained preparation contribution; a nonnegative full Φ_{ab} is NOT IDENTIFIABLE until full contractivity is proved. Both bulk forms are dimensionless by definition. An exactly bulk-quiet direction satisfies $S_H^{1/2} d_{ab} = 0$ almost everywhere; it need not be a stationary wavefunction. The separate objects are D^R (causal response), Ξ_{ab}^{full} (generic full log modulus), Φ_{ab} (established nonnegative pairwise records), Γ_{tr} (tangent diagnostic), γ_a and $\Gamma_{E,a} = S_0 \gamma_a$ (rate/energy width). For a named forward/reverse protocol let $Q := P_R^\Theta$ be the reverse measure transported to the forward path space, $Q(A) = P_R(\Theta A)$. Under $P_F \ll Q$, define $\Sigma[\gamma] = \ln(dP_F/dQ)(\gamma)$ P_F -a.s.; the reciprocal/unit-IFT forms require $P_F \sim Q$. The shorthand $\ln[P_F[\gamma]/P_R[\tilde{\gamma}]]$ is valid only when common densities have been specified.

Same-channel response/noise protocol and terminal PES guards. For one supplied stationary physical channel, Wightman positivity and the canonical KMS/FDT identities are Level-A standard conditional mathematics; using them as a response/noise experiment is only a Level-B protocol. The same Hermitian operator, source, projector, normalisation, preparation and detector transfer must be used in both arms. A finite window or broad bin mixes frequencies, so the thermal model must be forward-convolved through the measured leakage kernel; evaluating the coth factor at a bin centre is not valid without a narrow-bin bound. Direct resolved or correctly forward-modelled ordered spectra determine only T/S_0 , whereas reconstruction through S_H and D^R requires an independent S_0 match or a joint fit with its degeneracy reported. Negative-frequency symmetry, retarded analyticity, Kramers–Kronig with subtractions/tails/contact and Ward terms, preparation transients, and a full matrix or preregistered informationally complete complex projections are mandatory. The KMS hypothesis must name its time generator; generalised Gibbs, charged and rotating states require their modified detailed balance. Agreement in one channel does not prove global KMS, and disagreement rejects only the declared two-point model on its nonzero tested support. For ECT the physical operator/vertex, state and common transfer remain MISSING VERTEX / MISSING STATE / NOT IDENTIFIABLE; M1 is therefore an Open protocol, not an ECT prediction.

The M5 quantity $B'_+(0)/B(0)^{3/2}$ is only a convention-declared local static diagnostic and does not derive a retarded or metric bridge. M6 retains stationary physical Wightman positivity, but neither the arbitrary matrix absolute-value inequality nor abstract complete positivity supplies a frequency-local ECT kernel. P28 is likewise Open: acoustic data could test only a fully specified gauge-invariant primordial reduced channel after its state, generally two-time kernel and Boltzmann/recombination transfer are given; the CMB does not directly measure Φ_{ab} , purity or a unique pointer basis.

30. Probability, the Born rule, and measurement

In standard quantum mechanics, the Born rule $P = |\psi|^2$ is a foundational axiom. ECT has not derived a physical probability measure. For a complex Hilbert space with $\dim \mathcal{H} \geq 3$, the ordinary projector Gleason theorem gives conditional Level-A mathematics only after a

normalized, noncontextual, countably additive measure on orthogonal projector decompositions has been supplied: its representation is $\mu(P) = \text{Tr}(\rho_\mu P)$ for a supplied positive unit-trace state operator. The projector theorem does not cover $\dim \mathcal{H} = 2$; an explicitly supplied full effect/POVM framework may instead use the Busch extension [12], or a separately stated composite-system extension. Neither theorem supplies probability, state preparation, an event map, dephasing, outcome or update. The physical ECT owner remains Open.

The Born-weight route, reduced-state record formation, and measurement update have distinct owners. The complete corrected OS-II package conditionally constructs a Hilbert space only in verified subsectors; ordinary Gleason at complex dimension at least three constrains an independently admitted normalized, noncontextual, countably additive projector measure, while dimension two requires the separately supplied Busch effect/POVM domain or another named extension; a specified reduced-state channel can suppress selected coherences only after its coupling, environment state, kernel, trace and finite timescale are supplied, but it supplies neither that measure nor a unique outcome. The origin of probability, pointer selection, unique outcome and update remain Open. PES-R creates no derivation edge among these statements.

The measurement problem is therefore not declared solved. Supplied open-system source models provide conditional examples of interference suppression and record formation; the physical ECT channel, pointer records and timescale remain Open, but a complete detector-level theorem for single-outcome selection, pointer-basis selection, and exact Born-rule weights remains open (*Level Open*).

31. Quantum entanglement and Bell correlations

A common ordered medium offers a candidate nonfactorising architecture, but a shared substrate alone does not entail entanglement, Bell violation, no-signalling, or the Tsirelson bound. A physical nonfactorising state/channel, the appropriate vertices and kernels, and a detector/measurement model are required. These quantitative constructions remain Open. ECT is therefore compatible with a common-medium reading at Level B/Open; it has not derived the laboratory Bell or delayed-choice correlations from the minimal action.

32. Tunnelling

In standard quantum mechanics, tunnelling is the ability of a system to cross a region that is classically forbidden in Lorentzian language. For a particle with energy E facing a barrier $V(x) > E$, the Lorentzian kinetic energy would be negative, so no classical trajectory exists through the barrier.

The standard WKB construction may be written with a Euclidean saddle only after the analytic continuation, contour/thimble, measure, state, clock, boundary data and fluctuation determinant are supplied. Bare Euclidean DP does not by itself produce a Lorentzian amplitude or decay rate. Under those standard conditional inputs the forbidden Lorentzian segment is represented by a Euclidean saddle contribution. The barrier does not disappear. It is re-expressed as the physical under-barrier action of that external source model:

$$S_{\text{WKB}}(E) := \int dx \sqrt{2m [V(x) - E]}.$$

The tunnelling probability is then exponentially suppressed,

$$P_{\text{tunnel}} \sim e^{-2S_{\text{WKB}}(E)/\hbar},$$

as in the standard WKB result. Here $\hbar$ and S_{WKB} belong to the externally calibrated quantum-mechanical source model; this formula neither identifies a P3 path-integral denominator nor derives the canonical ECT S_0 . The Euclidean picture therefore does not say that the particle freely rolls through the barrier. It says that the forbidden Lorentzian process is represented by a classically allowed Euclidean contribution whose weight is small because its Euclidean action is large.

This is the same logic used in standard instanton and WKB methods. A specifically supplied physical ECT action, saddle, state, continuation and detector map would permit an ontological Euclidean-medium interpretation, but those owners are not presently derived. At current status ECT is compatible with the standard Euclidean-saddle description; the exponent remains external standard physics, and there is no distinct numerical prediction or demonstrated fourth-coordinate bypass.

Wick rotation: when does it apply. Wick rotation, $t \rightarrow -i\tau$, is a standard formal tool when the analytic continuation and boundary conditions are well defined. Its physical content depends on the question being asked. For ordinary WKB tunnelling it converts the classically forbidden Lorentzian segment into a Euclidean action cost. For false-vacuum decay, when two classically distinct configurations are separated by a barrier and a suitable Euclidean saddle exists, it gives the bounce / instanton exponent controlling the semiclassical transition. For a system with a single classical minimum it does not create a bounce saddle; it instead yields the Euclidean structure underlying ordinary stationary states such as ground-state localisation around the minimum. The Euclidean machinery is therefore a powerful reconstruction tool, not a recipe for turning every Lorentzian problem into a barrier-crossing problem.

33. Coherent coupling versus irreversible records

Coherent interactions can exchange energy and momentum and generate entanglement while record channels remain weak. Decoherence is a distinct, channel-specific suppression of off-diagonal reduced-state terms after unresolved records are traced out. No universal scalar threshold separates interaction from non-interaction or quantum from classical dynamics.

34. Influence of boundary data

Elliptic boundary-value solutions can depend on data across the mathematical boundary when the problem is well posed. The present theory has not derived a map from later laboratory detector settings to those boundary data, nor the specific delayed-choice, eraser, Bell, or no-signalling probabilities. This is an interpretive B/Open compatibility statement and implies no Lorentzian retro-signalling.

35. Quantum vacuum effects

The programme asks whether a physically realised ordered medium could respond to boundaries, acceleration and cosmological expansion. The present manuscript supplies three conditional or external benchmark routes; it does not yet derive an observable response of the P4 chart itself:

The Casimir route (*A mathematics/Open ECT*): the standard Dirichlet calculation for N independent massless scalars is exact inside that supplied boundary-value problem. For three independently supplied scalar channels with the same propagation speed as light, its force is arithmetically $3/2$ of the two-polarisation QED result. The present integrable ECT soft sector does not supply those three particles, a material boundary operator, or a stress observable. Thus the $3/2$ identity is a toy-BVP check, not an ECT correction or laboratory prediction.

Unruh route (*A external/Open ECT*): Euclidean–Rindler periodicity reproduces the standard temperature $T = \hbar a / (2\pi c k_B)$ only after the physical Lorentzian state/cone, detector coupling, KMS condition, and $S_0 \leftrightarrow \hbar$ matching are supplied. The present P1–P6 action does not independently derive an observable ECT Unruh response or detector rate.

Cosmological particle-production route (*A external/Open ECT*): a supplied non-adiabatic quadratic history may Bogoliubov-mix modes, but real ECT particle production requires the time-dependent action, transition clock/history, in/out vacua, physical particle map and state. Those owners are not yet derived; no Planck-time transition or three-Goldstone population is claimed.

36. Decoherence and the quantum–classical boundary

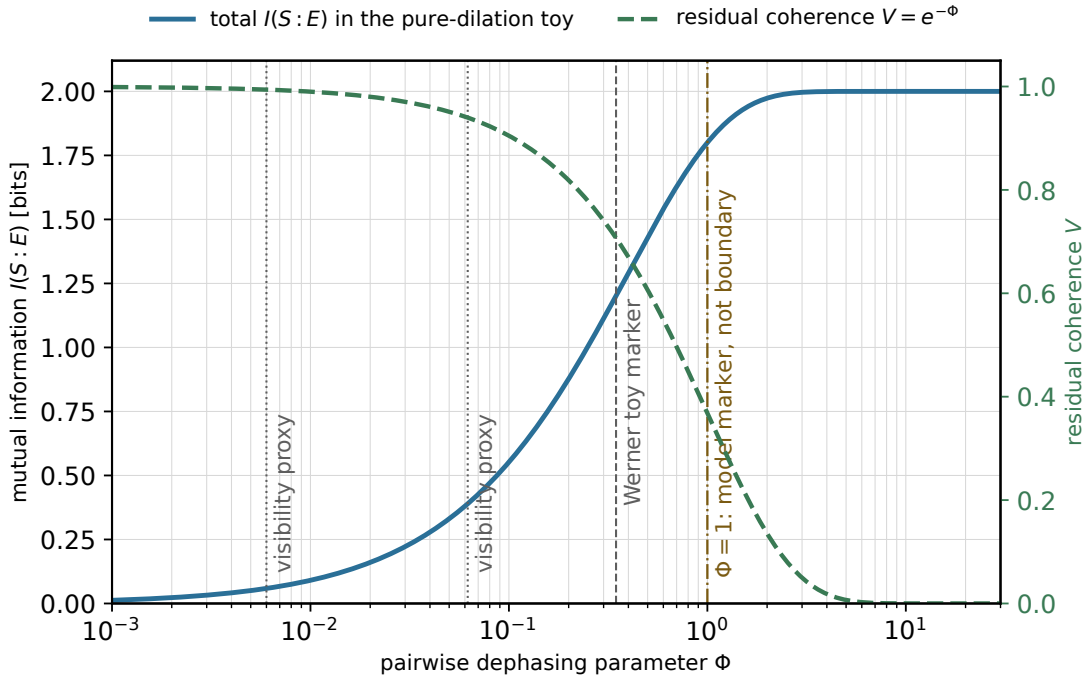


Figure 18: Declared pure-dilation qubit toy. The mutual-information and visibility curves are exact within that model. The visibility and Werner markers are source/model proxies, not ECT calibrations, collapse thresholds, or metric-signature boundaries. *Display semantics: Both continuous traces are analytic relations of the declared pure-dilation toy; no sampled-data markers are implied.*

Specified environmental channels can suppress interference and form stable reduced-state records. The physical record operator, environmental state, and protocol determine the pairwise exponent; neither the total mode count nor one universal dephasing scalar fixes the crossover. Decoherence does not select one ontologically realised outcome, and the pointer/update problem remains Open.

Environmental decoherence is the standard reduced-state loss of visibility caused by specified channels such as thermal photons, gas collisions or phonons after their unresolved records are traced out. This is a physical open-system process, not an observer-induced projection postulate.

For gravity, the symbolic Diósi–Penrose formula remains an external comparator, not an ECT-derived rate. At fixed α, β , about homogeneous $\bar{n} = w$, the projected tangent-director linear density vertex is missing; quadratic, nonlinear, composite and additional-field routes remain parametric/Open, the scalar trace vertex is only parametric, and the relevant ECT state/noise kernel is Open/Not Identifiable.

DP-02 owner firewall. The tested candidate channels do not supply the physical ECT projector, vertex, state-dependent noise kernel or reduction-rate map needed to identify the external Diósi–Penrose infrared comparator. No ECT coefficient or collapse time follows; the state, pole, vertex and normalisation remain Open.

C_{60} and macromolecular interference are standard environmental- decoherence source-model anchors. Quantitative visibility times must be computed from the declared collision/radiation channel, state, coupling-rate convention and observation protocol; no universal numerical object list is published.

Partial recoherence is possible in specified echo or control protocols. Reversing a declared reduced channel does not produce a universal backward probability law: the reverse dynamics, preparation and path measure must be defined separately.

37. Quantum gravity in ECT

In standard approaches, quantum gravity — the unification of GR and QM — is one of the central unsolved problems of physics. Loop quantum gravity, string theory, causal dynamical triangulations, and other programmes attempt to quantise the gravitational field on a pre-existing spacetime or to construct spacetime from more fundamental elements.

ECT currently supplies quantum/open-system reconstruction routes and a scalar characteristic geometry inside one ordered-medium architecture. It has not derived a physical graviton, universal gravity metric, common canonical action scale or gravitational record channel. Quantum gravity is therefore an Open completion programme, not an already unified pair of derived sectors.

The scalar running and its Landau-pole estimate do not constitute graviton-loop power counting and do not prove improved ultraviolet gravity. Newton’s constant is only matched inside a supplied gravity completion; the bridge from condensate stiffness to its canonical tensor normalisation is Open. Any UV advantage must be established from the constrained physical tensor propagator, vertices, counterterms and RP/OS reconstruction.

The open sequence begins earlier: derive a constrained physical tensor action, source, positive residues and canonical normalisation; only then attempt tensor-loop power counting, coupled

RG and RP/OS reconstruction. OS reconstruction yields a Hilbert space only after the complete applicable corrected OS-II package—including correlators/measure, observable algebra and reflection, regularity/growth, covariance, RP, symmetry and applicable locality/cluster conditions—is proved; it does not create the missing physical tensor dynamics, clock or metric. These are open problems, not solved theorems (*Level Open*). Thus ECT reframes the question “how do you put gravity and quantum mechanics together?” through a single-medium architecture, while the full UV/spin-2 closure remains open.

38. Reduced-state reformulation of the gravity–matter problem

A long-standing difficulty, often called the *hybrid-consistency problem*, is that combining quantum matter with any non-quantised or emergent description of gravity runs into a characteristic set of obstructions: reduced states must remain positive; no classical channel should transmit quantum information; energy–momentum bookkeeping must stay controlled; relativistic compatibility and diffeomorphism invariance must be preserved. Historical semi-classical proposals by Møller and Rosenfeld already exhibited these tensions; later thought experiments and effective-field-theory analyses sharpened them further. Modern hybrid frameworks, such as post-quantum classical gravity (PCCG), accept these tensions as part of a fundamentally stochastic and irreversible law of gravity (*Level B for the general landscape*).

ECT takes a candidate architectural route (Level B/Open). Information preservation is a requirement on a future global interacting completion, not an established postulate or theorem of P1–P6. Reduced coherent and classical-looking descriptions may arise after tracing specified channels; however, a shared medium alone does not imply a nonfactorising physical state, entanglement, or a gravity-mediated quantum channel. Those require an explicit state, vertex, kernel, and reduction map and remain Open.

39. Gravity-induced entanglement: can gravity be “classical”?

One of the sharpest near-term tests of the quantum character of gravity is the proposal, due independently to Bose, Mazumdar and Morley and to Marletto and Vedral [1, 2], to look for entanglement between two small mesoscopic masses held in spatial superposition and interacting only through their mutual gravitational field. In its standard reading, *Gravity-Induced Entanglement* (GIE) in such a setup would force the conclusion that the gravitational mediator itself carries non-classical degrees of freedom, because local classical channels cannot transmit quantum information. A refinement due to Christodoulou and collaborators showed that the effect admits a consistent locally mediated reading within linearised quantum gravity, with proper causal retardation.

This standard inference has more recently been stress-tested. A proposal by Aziz and Howl argued that, once matter is treated at the level of quantum field theory, a locally classical-looking gravitational coupling might still generate entanglement; several critical responses argued that in the specific nonrelativistic limit employed, the relevant unitary factorises and no genuine gravity-mediated entanglement is produced. On the constructive side, Trillo and Navascués established that a Diósi–Penrose-type classical gravity, treated as a *correlated-noise* mechanism, *can* generate gravity-induced entanglement in that specific model class. In other words, once one admits common-medium or correlated-noise structure between the probe masses, the slogan

“classical gravity cannot entangle” is no longer generally valid.

Three disciplined claims about ECT. (A) ECT does not take the observation of gravity-induced entanglement in a mesoscopic protocol, by itself, to force a fundamentally quantised independent metric field; once common-medium loophole classes are admitted, the standard no-go inference applies most directly to strictly local factorising classical channels. (B) ECT-specific gravity-induced entanglement is currently NOT IDENTIFIABLE; an external loophole class may allow it only if a derived reduced channel is shown to arise whenever the experimentally relevant reduced dynamics still retains common-medium non-factorisable structure — that is, when the protocol probes a Level IIa regime rather than a Level IIb regime. (C) ECT does not yet establish, from first principles, which exact experimental protocols realise Level IIa, nor does it yet derive the explicit reduced mediator channel through which entanglement would be generated (*Level Open*).

Operational criterion. The architectural position of ECT in the BMV debate can be stated sharply. ECT falls under the standard no-go logic only if, in the actual regime realised by the experiment, the effective reduced mediator is well approximated by a strictly local classical factorising channel with no residual common-medium correlations. ECT *escapes* that logic if the experimentally relevant reduced dynamics still retains correlated common-medium structure — i.e. belongs to the Trillo–Navascués loophole class. Different protocols probe different reduced regimes of one underlying condensate theory; ECT does not claim both descriptions for one and the same operational regime.

Imported benchmark formula and required apparatus inputs. Inverting the external unit-coefficient Diósi–Penrose scaling $\tau_{\text{DP}}^{\text{ext}} \sim \hbar R / (G_N m^2)$ at a supplied fixed radius and declared coherence time defines the guide

$$m_{\text{guide}}^{\text{DP}} \sim \sqrt{\hbar R / (G_N \tau_{\text{exp}})}.$$

It is not an environmental crossover and is not a scalar-sector or entangling-channel ECT prediction. If one separately supplies the hard-sphere collisional rate $\lambda_{\text{coll}} = n_{\text{gas}} v_{\text{th}} \pi R^2$ for a homogeneous sphere $m = 4\pi\rho R^3/3$, and the unit-coefficient external rate $\lambda_{\text{DP}}^{\text{ext}} = G_N m^2 / (\hbar R)$, equality gives

$$R_{\text{cross}}^{\text{coll-DP}} \sim \left(\frac{9\hbar n_{\text{gas}} v_{\text{th}}}{16\pi G_N \rho^2} \right)^{1/3}, \quad m_{\text{cross}}^{\text{coll-DP}} \sim \frac{3\hbar n_{\text{gas}} v_{\text{th}}}{4G_N \rho}.$$

This is exact algebra inside the two displayed approximate source models, not a universal collapse law or an ECT crossover. It deliberately uses a unit coefficient. A separate external continuum-sphere orientation fixes the factor-1/2 functional and explicitly adopts $\tau = \hbar / E_G$ for two identical sharp homogeneous spheres, centre-to-centre separation $d = R$, and

$$R = 1 \mu\text{m}, \quad \rho = 2000 \text{ kg m}^{-3}.$$

For that nominal point only,

$$C_{\text{DP}}(1) = \frac{51}{160}, \quad \tau_{\text{DP}}^{\text{ext}} = 70.6 \text{ ms}.$$

This is Level-A geometry inside an imported source model and an external Level-C benchmark; it is not a data test, physical uncertainty statement or ECT collapse time. Profiles, smearing, separation, regularisation and normalisation change it. Inserting 51/160 into the unit-coefficient guide

or crossover would change their coefficients and therefore would define a different comparator.

A physical apparatus comparison requires the gas species and equation of state, P and T , a thermal-speed convention, total or differential cross section, target and chamber geometry, spatial-superposition profile, and the coherence-time/visibility protocol. A completed ECT calculation would additionally require the physical projector, vertex, state-dependent noise kernel, finite-body charge and absolute normalisation, all presently Open/Not Identifiable.

40. How gravity may talk to matter: candidate record channels

The following split is an operational taxonomy, not a derivation of two physical ECT baths.

Scalar trace-response candidate. The scalar-only action yields a parametric canonical coupling to the matter trace T . It does not by itself determine a state-dependent noise kernel, a finite-body charge, or an absolute decoherence rate. The fixed-radius unit-coefficient $m_{\text{guide}}^{\text{DP}} \sim \sqrt{\hbar R / (G_N \tau_{\text{exp}})}$ is therefore only a benchmark relation after the required apparatus inputs and coherence protocol are supplied; it must not be confused with $m_{\text{cross}}^{\text{coll-DP}}$, which additionally depends on the supplied gas and density model, or with the separate geometry-resolved $C_{\text{DP}}(1) = 51/160$, 70.6 ms sphere point. None is an ECT-derived rate.

Orientation/tensor candidate. An orientation channel could in principle provide a common retarded mediator, but the displayed scalar action contains no n^A sector and does not derive its matter-density projector. A covariant orientation action, its metric and matter vertices, the physical state, and the Hadamard/noise kernel are all required before this candidate can be classified quantitatively.

Status. Response, noise, and entangling capacity must be kept distinct. Reusing a static gravitational response as a stochastic noise kernel would double-count the same physics unless a subtraction is specified. The ECT gravitational record channel is therefore Open/Not Identifiable. The Diósi–Penrose functional, the declared geometry-resolved sphere point, fixed-radius unit-coefficient $m_{\text{guide}}^{\text{DP}}$ relation and conditional unit-coefficient collisional $m_{\text{cross}}^{\text{coll-DP}}$ comparison are distinct external comparators; no numerical D–P/environment crossover or ECT rate is published.

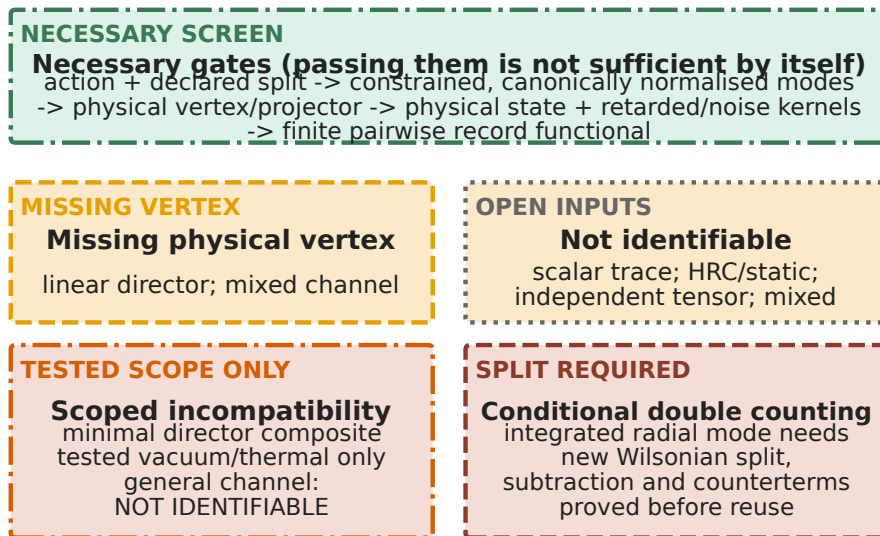
41. The infrared sky: asymptotic symmetries, memory, and soft gravitons

Modern general relativity has sharpened an old intuition: at very long wavelengths, near null infinity, there lives a rich structure of *asymptotic symmetries*, *gravitational memory*, and *soft gravitons* that goes far beyond the naive Minkowski-at-infinity picture. The asymptotic symmetry group discovered by Bondi, van der Burg, Metzner, and Sachs (BMS) is an infinite-dimensional extension of Poincaré symmetry; gravitational memory is the permanent shift left behind in the metric at null infinity after a burst of gravitational radiation passes; soft graviton theorems connect zero-energy gravitons to both BMS supertranslations and memory through the modern “asymptotic symmetry / soft theorem / memory” triangle.

ECT offers a candidate infrared interpretation (Level B/C compatibility only) in asymptotically flat ordered-branch settings. The asymptotic data $(\phi_0, n_A^{(0)})$ provide a possible stage for a BMS analysis, and a persistent Δn_A shift is a candidate memory variable. However, the covariant orientation action, its TT spin-2 mode, asymptotic charges, soft theorem, and memory map have not been derived. Long-wavelength δn_A modes therefore cannot yet be called physical ECT soft

Record-channel necessary-screen summary

10 detailed ledger rows -> 7 reader-facing families -> 0 identified physical PES channels



The seven-family graphic aggregates - and does not replace - the detailed ten-row ledger.

Static HRC compliance is a response, not a PES noise kernel;
 imported D-P numbers are external Level C.

- missing physical vertex
- incompatible in tested scope
- not identifiable / Open inputs
- double counting unless split proved

Figure 19: Reader-facing necessary-screen summary of the bounded record-channel audit. The graphic aggregates, but does not replace, the authoritative ten-row candidate/subcase ledger. No tested row supplies the complete action-owned physical modes, vertex/projector, state, retarded/noise kernels and finite pairwise record functional. The general channel remains **NOT IDENTIFIABLE**; reuse of an integrated radial mode is **DOUBLE-COUNTED** unless a new Wilsonian split and subtraction ledger are proved; the Diósi–Penrose functional remains an external comparator. Passing this screen would be necessary, not sufficient, for a physical PES channel. The complete connected Part III A4 reader map and its deterministic exact-key and machine-topology records preserve the canonical source labels, edge inventory and status provenance without duplicating them here.

HYPOTHETICAL COMPLETION · LEVEL C / OPEN

No ECT shell action, location, stability, coupling or dynamics has been derived.

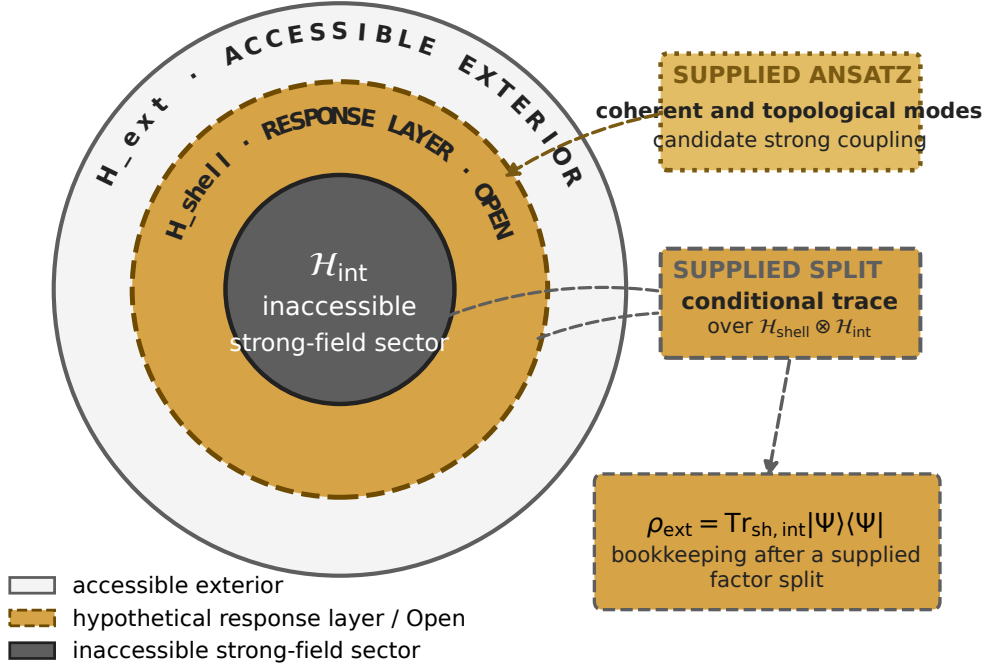


Figure 20: Candidate Level-C/Open critical-shell picture. The amber callout is the supplied strong-coupling ansatz for coherent and topological modes. Two gray leader lines select shell and interior factors for a conditional trace; they are not dynamical arrows. The sole gray arrow runs from the trace operation to the reduced exterior state. The factor split, shell geometry, correlations and Euclidean-core continuation remain Open. Reduced-state thermality is compatible with global purity, but P1–P6 do not yet prove globally unitary interacting strong-field evolution or outgoing-correlation recovery.

gravitons, and no first-principles BMS/memory/soft triangle is claimed.

Caveat. The existence of soft modes does not, by itself, solve any information problem; it only identifies a potentially relevant low-energy bookkeeping sector. Whether the long-wavelength δn_A modes suffice to store and recover the information content required by black-hole evaporation is not established here and is not claimed.

42. Black-hole thermodynamics and the information problem

ECT investigates a C/Open completion in which the black-hole horizon is represented by a condensate interface. The current action does not derive that interface, Hawking emission, or controlled corrections; this section separates generic reduced-state mathematics from the supplied shell toy.

If information falls into a black hole and the black hole evaporates via Hawking radiation, is the information lost? This is the information paradox — one of the deepest problems in theoretical physics.

Premise and ownership audit. The information problem arises from the simultaneous use of semiclassical effective field theory, quantum unitarity, a regular horizon, evaporation and an inside/outside factorisation (together with the applicable locality and causal assumptions).

Merely declaring Lorentzian geometry or the horizon “emergent” does not identify which premise fails and does not solve the problem. ECT has not yet derived the physical shell, metric, subsystem factorisation or evaporation channel. Conditional on supplying them, Hawking-like thermality may be represented as reduced-state thermality,

$$\rho_{\text{ext}} = \text{Tr}_{\text{shell,int}} |\Psi\rangle\langle\Psi|,$$

not by itself evidence for destruction of a full condensate state. The question “is information lost?” therefore splits into two questions. The general reduced-state identity is exact; whether ECT possesses a globally unitary interacting strong-field completion, and whether outgoing radiation recovers the correlations, remain separate Open questions.

ECT addresses it at two levels (Fig. 20).

Conditional global-unitarity hypothesis: If a completed ECT strong-field theory supplies a global Hilbert space, physical subsystem split and isometric interacting evolution, then $\rho_{\text{ext}} = \text{Tr}_{\text{shell,int}} |\Psi\rangle\langle\Psi|$ can be mixed while the global state stays pure. This excludes only a complete state-erasing final channel; one-body thermal spectra can remain exact with information in phases or higher correlations. P2 does not construct the Hilbert space, Hamiltonian, factorisation, evaporation channel or Page curve.

Level C/Open (shell ansatz and optional H_{sh} completion): A candidate critical shell is supplied as a possible transition between effective regimes. The named optional stationary passive-scattering completion H_{sh} uses natural units $\hbar = c = 1$, a dimensionless shell field $[\chi_{\text{sh}}] = 1$, and local coefficients $a_{\text{sh}} > 0$, $\lambda_{\text{GL}} > 0$, with $[a_{\text{sh}}] = E^2$ and $[\lambda_{\text{GL}}] = E^4$. The symbols a_{sh} and λ_{GL} belong only to this shell closure and are not the P3 gradient or quartic coefficients. The completion also supplies an independent self-adjoint Sturm–Liouville problem on $I_\rho \subset (\rho_c, \infty)$ with positive coefficient and weight. A declared flux-normalised channel may be written $\mathcal{T}_{\omega\ell}[\bar{\chi}_{\text{sh}}] = \exp(i\delta_{\omega\ell} - \Gamma_{\omega\ell})$, with $\delta_{\omega\ell} \in \mathbb{R}$, $\Gamma_{\omega\ell} \geq 0$ and $0 < |\mathcal{T}_{\omega\ell}| < \infty$; passivity gives $|\mathcal{T}_{\omega\ell}| \leq 1$ only in that flux convention. After fixing a logarithm branch and radial mode projection, its supplied Fréchet kernel is $K_{\omega\ell}(\rho) := \delta \ln \mathcal{T}_{\omega\ell}[\chi]/\delta \chi(\rho)|_{\bar{\chi}_{\text{sh}}}$, with $\delta \ln \mathcal{T}_{\omega\ell} = \int_{I_\rho} d\rho K_{\omega\ell}(\rho) \delta \chi_{\text{sh}}(\rho)$ and $[K_{\omega\ell}] = E$. One supplied local vertex may be $\mathcal{L}_{\text{int}}^{H_{\text{sh}}} = -g \delta \chi_{\text{sh}} (\partial \varphi)^2 / 2$, where a canonically normalised four-dimensional probe has $[\varphi] = E$ and the independent coefficient g is dimensionless and may vanish. The background, state, Hamiltonian coefficients, Green function, interval, boundary data, flux normalisation, kernel and detector map are completion inputs; mode mixing, resonances, superradiance or nonstationarity require another representation. Neither this closure nor the area-cell and finite-memory toys derive a physical shell, Hawking spectrum, entropy coefficient, evaporation channel, Page curve, echo, ringdown or interior geometry.

Three-regime structure. The supplied thermal-shell toy labels three candidate regimes: (i) an exterior region where standard effective geometry is assumed; (ii) a candidate shell/interface regime, where a toy ordered-branch amplitude is assumed to change; and (iii) a proposed non-geometric core. No physical crossover temperature, metric breakdown or universal shell thickness is derived; the Tolman toy contains only $\rho_* = \hbar c / (2\pi k_B T_*)$.

Topological consistency. The identity $\pi_2(S^3) = 0$ says only that a supplied S^2 shell map need not carry π_2 charge; it does not derive formation, dissolution, stability, or dynamics of a shell. The nontrivial sector $\pi_3(S^3) = \mathbb{Z}$ admits possible topological configurations, but the present work does not derive a stabilising Skyrme-type term or a finite-energy gravitating topological black-hole solution.

What is not yet derived. The present shell analysis does not yet derive the full interior profile, first-principles horizon thermodynamic coefficients, or a constructive Page curve.

Comparison with the islands / quantum-extremal-surface language. The modern holographic programme approaches the same black-hole information question through a different language: the *islands formula* and the notion of *quantum extremal surfaces* (QES). In that language, the entropy of Hawking radiation is computed as a generalised entropy $\text{Area}(\partial\Sigma)/(4G_N) + S_{\text{bulk}}(\Sigma)$, with the applicable extremisation and minimisation prescription. In representative semiclassical evaporating-black-hole models, the trivial saddle dominates before the Page time, while a nontrivial island saddle can dominate after it and produce a Page-curve turnover. The candidate shell and islands/QES pictures admit a C/Open analogy only after global unitarity and physical subsystem algebras are supplied. ECT derives neither a shell/interior factor nor a QES, so no physical equality of roles is asserted. Crucially, the critical-shell boundary is *not* identified here with a derived quantum extremal surface, and ECT does not derive the generalised-entropy formula or carry out a replica-wormhole-type calculation on its underlying Euclidean condensate path integral; those remain open tasks. The point of the mapping is compatibility of structural logic, not an ECT derivation of the Page curve.

43. Area-law entropy: a conditional medium programme, not yet an ECT theorem

Black-hole entropy and gravitational entropy-bound arguments motivate area scaling in specified gravitational regimes; they do not imply a universal area law for arbitrary regions or states. Bekenstein’s black-hole and entropy-bound work, and the later holographic proposals of ’t Hooft and Susskind, have shaped much of modern theoretical physics. In string theory it appears as the AdS/CFT duality; in loop quantum gravity it emerges from a counting of quantum-geometric states. In both cases, however, area scaling is tied to specific features of the gravitational theory.

ECT offers a possible medium interpretation, but an important distinction is essential. The radial mass m_σ sets a correlation length and the breakdown scale of a reduced EFT. It does *not* by itself provide a microscopic lattice spacing, a hard UV cutoff, or a finite-dimensional cell. A massive continuum field still has arbitrarily short-wavelength modes, and its entanglement entropy is UV divergent.

Nor does locality plus a finite two-point correlation length prove an area law for every state: volume-law and data-hiding states are not excluded by those facts. The honest surviving result is therefore a conditional counting lemma.

Conditional boundary-cell count. Draw a closed surface Σ only after a physical subsystem split has been supplied. Also supply (i) a coarse-graining or microscopic resolution ℓ_{cg} , (ii) a finite entropy capacity s_{cg} per retained cell under a stated truncation or energy restriction, and (iii) a state class whose cross-interface entropy is localised, up to ΔS , in a layer of thickness ξ_{ent} . The last input may follow from a separately proved finite-depth, Markov, or area-law property; it does not follow from finite correlation length alone. Counting cells then gives

$$S_{\text{red}}(\Sigma) \leq C_{\Sigma} s_{\text{cg}} \frac{A(\Sigma) \xi_{\text{ent}}}{\ell_{\text{cg}}^3} + \Delta S. \quad (35)$$

Only with the additional matching $\xi_{\text{ent}} = O(\ell_{\text{cg}})$ and sub-area ΔS does this reduce to $S_{\text{red}} = O(A/\ell_{\text{cg}}^2)$. The implication is Level A mathematics under its supplied assumptions; ownership of those assumptions by ECT is Open.

Why this is not just a restatement of the QFT result. In local quantum field theory, the leading ultraviolet-sensitive term in the vacuum entanglement entropy of many low-energy states often obeys an area law. Its coefficient is regulator- and renormalisation-scheme dependent. This standard result does not by itself supply a physical ECT cutoff or entropy coefficient.

The present continuum ECT action does not upgrade or remove that divergence. Identifying its regulator with m_σ^{-1} would confuse an EFT breakdown/correlation scale with a UV completion.

Black-hole entropy as a special case. The shell-entropy result discussed earlier, $S_{\text{shell}}^{\text{toy}} \sim A \ln q / \ell_{\text{cg}}^2$, is one instance of the conditional count only after a physical shell, factorisation, cell scale, finite local dimension q , and suitable state have all been supplied. The exact Bekenstein–Hawking coefficient $1/(4\ell_{\text{Pl}}^2)$ is merely a target matching condition inside that toy; it does not derive a black-hole capacity or the microscopic scale.

Strengthening the Page-curve programme. The general area bound has an important consequence for the information paradox. The boundary-layer bound does not derive a physical shell, its Hilbert capacity, or an evaporation channel. Accordingly it neither proves a finite information reservoir nor strengthens the conditional isometry identity. Thermal marginals and the Page-curve shape remain unfixed. A complete Page-curve derivation still requires additional dynamical ingredients (information return mechanism, purification dynamics), which remain open.

Connection to decoherence. An interface may also enter a supplied open-system model, but the cell-counting lemma gives no decoherence kernel, no entropy-production theorem, and no PES result.

From area law to holography. It is important not to overclaim. ECT currently establishes only the conditional finite-cell implication. It does not yet establish:

- a physical ECT area law for the completed state;
- a covariant entropy bound analogous to the Bousso bound for dynamical horizons;
- a full bulk-boundary reconstruction (a gravity/boundary duality of the AdS/CFT type);
- a complete holographic principle in the strongest reconstruction sense.

The correct summary is:

ECT provides an interface-motivated programme and an explicit list of missing microscopic/state inputs. A physical area law and full holography remain Open.

How ECT differs from other approaches. In AdS/CFT, area scaling arises from a duality between a gravitational theory in the bulk and a conformal field theory on its boundary. In loop quantum gravity, it arises from counting discrete quantum-geometric states. In ECT, a boundary-cell picture is a candidate only. It becomes an alternative physical explanation if a

microscopic completion derives the split, finite local capacity and area-law state class instead of assuming them.

What remains open. The microscopic resolution and local capacity, the physical subsystem split, the relevant state theorem, soft/gauge-edge sectors, the black-hole shell and coefficient, and any covariant null-surface bound all remain Open.

44. What the compact phase does—and does not—establish about charge

The coherent branch contains a compact coordinate,

$$\theta \sim \theta + 2\pi.$$

This proves the existence of integer *winding sectors* of that coordinate. It does not by itself prove that the coordinate is the physical electromagnetic gauge fibre, that its redundant directions have been quotiented correctly, or that an unbroken massless connection couples to matter.

There is a useful conditional theorem. If a completed ECT sector provides a compact physical $U(1)$ principal bundle, a gauge-reduced kinetic operator, and matter representations of that group, then the representation weights are integer relative to a chosen elementary normalisation:

$$q = n e_0, \quad n \in \mathbb{Z}.$$

Likewise, the compact generator has a discrete spectrum only after a physical Weyl representation, its action normalisation and a self-adjoint quasiperiodic domain have been supplied. For twist ν_{tw} the spectrum is $S_0(\mathbb{Z} + \nu_{\text{tw}}/2\pi)$; the lattice $S_0\mathbb{Z}$ is only the periodic special case. This generator is not thereby the electric-charge operator, and compact field winding is a separate topological integer. These are standard consequences *inside* a completed compact- $U(1)$ representation; ECT has not yet derived the physical charge map ($B/Open$). A charged condensate can also Higgs the candidate connection, so an unbroken photon must be shown rather than assumed.

Open physical owners. ECT must still derive the gauge volume or reduced Hessian, Maxwell normalisation, the identity of $U(1)_{\text{em}}$, its unbroken photon, the charge operator, and the matter map. The observed Standard-Model pattern additionally requires electroweak embedding,

$$Q = T_3 + \frac{Y}{2},$$

the colour sector, anomaly-consistent matter representations, and the global gauge-group quotient. The recorded colour-interlock layer is a consistency condition on completions: with the standard $\mathbb{Z}_6$ -type global quotient, hypercharge, colour and weak data must satisfy an explicit congruence, which fixes the fractional electric pattern per colour class ($Q \equiv -t/3 \pmod{1}$) and reduces the residual freedom to an explicitly characterised discrete family. This is a consistency requirement on completions, not a derivation of the observed hypercharges.

45. Why the post-Big-Bang Universe developed the way it did

ECT offers a candidate Level B/Open branch-boundary interpretation of one part of the Big-Bang question: a P1 Euclidean configuration class and a separately supplied P4 ordered class may be related by ordering dynamics. Calling that relation the physical Big-Bang boundary requires a state, clock/lapse, observable metric and transfer to matter and radiation; it is not a chronology theorem of P1–P6.

But this still leaves a second question: once the ordered branch has formed, why does the Universe then evolve in the way we observe?

ECT does not yet solve the full transition dynamics. In the main preprint this open problem is now decomposed into four parts of very different difficulty: the ordering mechanism itself, the orientation of the branch (why expansion), the thermal completion into a hot Big-Bang history, and the observable relics of the ordering event. The orientation question has recently been sharpened in two steps. First, the homogeneous background equations turn out to be blind to the sign of expansion on their own: until a thermodynamic arrow of time is fixed, an “expanding” and a “contracting” description are two parametrisations rather than two physically distinct options, so part of the old question dissolves. Second, once the arrow of time is fixed, a conditional no-bounce result applies in the spatially flat positive-potential/no-ghost sector: a smooth solution cannot change the sign of H . This does not determine the sign of $\dot{H}$, monotonic growth of density or curvature, or a breakdown timescale. Contracting de Sitter supplies an explicit counterexample to that stronger inference. A separately conserved component with $w > -1$ does grow on a contracting branch, but promoting this to total breakdown requires a complete action, state, energy-exchange law and threshold. The physical branch orientation, ordering mechanism and thermal completion therefore remain Open rather than being selected by persistence alone.

Within the explicitly supplied scalar actions one can classify conditional post-transition histories. The two-slope completion provides a regular radiation–matter–de-Sitter proof of class. Separately, its bare positive-kinetic scalar pieces obey $w_\phi^{\text{bare}} \geq -1$ when $K_\phi^{\text{phys}} + U > 0$; non-minimal $F, \dot{F}, \ddot{F}$ terms leave the observable w_{eff} unconstrained. These are action-internal statements: they do not show that the physical ECT condensate follows that history, preserves standard early cosmology, or owns the observed late acceleration.

In condensed-matter systems, ordered domains often grow into a disordered environment. Applying that picture to ECT is only a conditional analogy: the present action does not derive the relevant nonequilibrium front, its clock, or the selection of an expanding physical branch over a contracting one.

Other branches require their own complete action and state. A contracting branch is a kinematically admissible acceptance case; the growth analogy is not evidence sufficient to disfavour it. The named positive-potential two-slope action tends to its stable de-Sitter root and does not produce recollapse. A contracting or recollapsing branch remains an Open acceptance case for a different explicitly supplied action/state; no ECT recollapse timescale has been derived. The inequality $w_\phi^{\text{bare}} \geq -1$ excludes only a phantom *bare ratio* inside the supplied positive-kinetic decomposition when $K_\phi^{\text{phys}} + U > 0$; it does not exclude an observable $w_{\text{eff}} < -1$ history generated by non-minimal terms.

The theory therefore supplies a programme for comparing ordered-branch cosmologies once an

action, state and clock/metric completion are specified. It does not yet select the observed expansion history or derive a geometric origin of the Big-Bang branch, and its hot radiation history remains Open.

46. The grandfather paradox

ECT does not yet solve the grandfather paradox; it provides a conceptual programme for asking how causal worldlines could arise from a Euclidean boundary-value ontology.

Classical causal paradoxes of time travel arise from trying to maintain two incompatible assumptions: that one can move backward along the physical time axis, and that causal chains remain linearly irreversible. The contradiction appears because the traveller is imagined as re-entering an already fixed causal sequence from within that sequence.

The coordinate w is not a fundamental causal axis. However, P1–P6 do not yet derive a physical worldline map, a clock, controllable motion in w , or a rule connecting global Euclidean configurations to observable causal histories. Therefore statements about a traveller moving backward in w or crossing an earlier worldline are not current consequences of ECT. The defensible claim is narrower: ordinary Lorentzian paradoxes need not be primitive at the Euclidean level, while any physical resolution must be demonstrated in a completed clock/metric/matter theory with operational no-signalling and consistency tests.

47. Analogue laboratory programme

Superfluids, Bose–Einstein condensates, and related platforms are analogue systems for selected mechanism classes and conditional closures. They do not instantiate the full ECT vacuum: success is not evidence for ECT, and failure rejects only the explicitly mapped analogue implementation. Useful platforms include:

Superfluid ^4He and Bose–Einstein condensates: phase-winding quantisation, vortex dynamics, Goldstone spectrum, and the condensate-mediated correlation structure.

Acoustic black holes (“dumb holes”) in BECs: analogue Hawking effect and critical-shell structure.

Boundary-response experiments can be specified only after a physical reduced ECT bundle, material boundary operator and stress response are derived. At present no condensate-sector Casimir force is predicted.

Atom and mesoscopic interferometry: constraints on standard Diósi–Penrose benchmarks and on any future spectrally specified ECT record kernel.

48. Conclusions of Part III

48.1. Status-separated outcomes and Open owners

The coherent-branch discussion contains the following conditional mathematics, external targets, structural routes, and Open owner problems:

1. Conditional Hilbert-space route only through the complete applicable corrected OS-II

package in the declared Gaussian/free kernels (A-math for those kernels); the interacting ECT measure, observable algebra, continuation/generator, clock/metric and canonical completion remain B/Open.

2. A supplied local KG-form scalar equation and a separately supplied Schrödinger-type complex law; their leading NR envelope relation is conditional A-level asymptotic mathematics only with the stated clock, mass, stable-frequency, weak-interaction, slow-envelope and residual-control inputs, while a common ECT descent and physical owner remain B/Open.
3. Linear superposition is A only in the verified quadratic subsector and B/Open physically; interference is B/Open; Pauli is A-math for the free antisymmetric kernel and B/Open for physical ECT matter.
4. Integer compact-phase winding sectors exist at Level A; action normalisation and physical spectrum reconstruction are B/Open.
5. Compact-phase winding sectors (A) and a conditional Abelian connection/charge-lattice route (B/Open); physical $U(1)_{\text{em}}$, photon, charge operator and matter map remain Open.
6. PES-R is a status-separated inventory of stationarity, record, finite-time, and protocol diagnostics (B); no universal selection theorem.
7. Conditional probability-representation route: a complete corrected OS-II package supplies a Hilbert space in verified subsectors; ordinary projector Gleason is Level-A mathematics only for complex dimension at least three with a supplied normalized, noncontextual, countably additive measure, while a qubit needs a separately supplied Busch effect/POVM or composite extension. Dephasing is separate, and the physical ECT probability/Born owner remains Open.
8. Reduced-state dephasing only inside specified source models with declared coupling, state, kernel, trace and timescale (B/Open); the physical ECT channel, probability measure, pointer records, outcome and update remain Open.
9. Common-medium nonfactorisation as a candidate entanglement route (B/Open); Bell, no-signalling, and Tsirelson correlators remain Open.
10. Euclidean/WKB tunnelling as interpretive compatibility (B); no distinct ECT tunnelling derivation is claimed.
11. Supplied scalar Dirichlet BVP (A mathematics/Open ECT); no physical Casimir coefficient is predicted.
12. Standard Euclidean–Rindler/Unruh target (A external/Open ECT); physical state, cone, detector response and rate Open.
13. The external Diósi–Penrose functional and one scoped 51/160, 70.6 ms sharp-sphere orientation point are retained; neither supplies an ECT coefficient or collapse time, and the ECT gravitational record channel remains Open/Not Identifiable.
14. Thermal marginals are compatible with global purity under a supplied split/isometry; global ECT black-hole unitarity is Open.

15. Bekenstein–Hawking area law as an external target; ECT coefficient and physical state owner Open.
16. Conditional boundary-cell counting lemma (A mathematics under supplied coarse-graining/state inputs; Open ECT).
17. Illustrative Page-curve-like turnover in a finite-memory shell toy model ($C/Open$); a constructive ECT Page curve remains Open.
18. Quantum gravity structurally reframed by the single-condensate architecture; full UV and spin-2 closure remain open ($B/Open$).

48.2. Key equations and principles of Part III

Winding condition: $\oint \partial_A \theta dX^A = 2\pi n$.

Loop-scale candidates: $\iota_{\text{elem}}^{(\text{fixed core})} = 2\pi\iota_0^{\text{EFT}}$ in the phase-only model, which has no nonzero global minimum as L varies; a BR1 phase-tension reduction may define $\iota_{0,\text{bal}}^{\text{eff}}$; and the separate texture-worldline modulus route defines $\iota_{0,\text{bal}}^{(\vartheta)}$ only after its owner and proper-time ensemble are supplied. CDE- Φ is an acceptance criterion rather than that owner. The calibration $S_0 = \hbar$ is Level B; profile/Hessian, bending/barrier, stability, microscopic ownership and universality remain Open.

PES-R inventory: stationarity, pairwise record stability, and compact-phase admissibility are separate filters; a universal intersection theorem remains Open.

Source-model visibility proxy: $\tau_{\text{vis}} = 1/(N_{\text{eff}}\gamma)$ only after the channel, state, rate convention and observation protocol are specified; it is not a universal ECT decoherence time or sector boundary.

Casimir: supplied N -scalar Dirichlet algebra only; for separately supplied $N = 3$ the $3/2$ toy identity is retained, while the physical ECT bundle, boundary map, stress and force coefficient remain Open and no ECT correction is predicted.

BH entropy target: $S_{\text{BH}} = A/(4\ell_{\text{Pl}}^2)$; the current ECT shell toy derives neither this scaling nor its coefficient.

Conditional cell count: $S_{\text{red}} \leq C_{\Sigma} s_{\text{cg}} A \xi_{\text{ent}} / \ell_{\text{cg}}^3 + \Delta S$; an A/ℓ_{cg}^2 form needs separately supplied $\xi_{\text{ent}} = O(\ell_{\text{cg}})$ and a sub-area remainder.

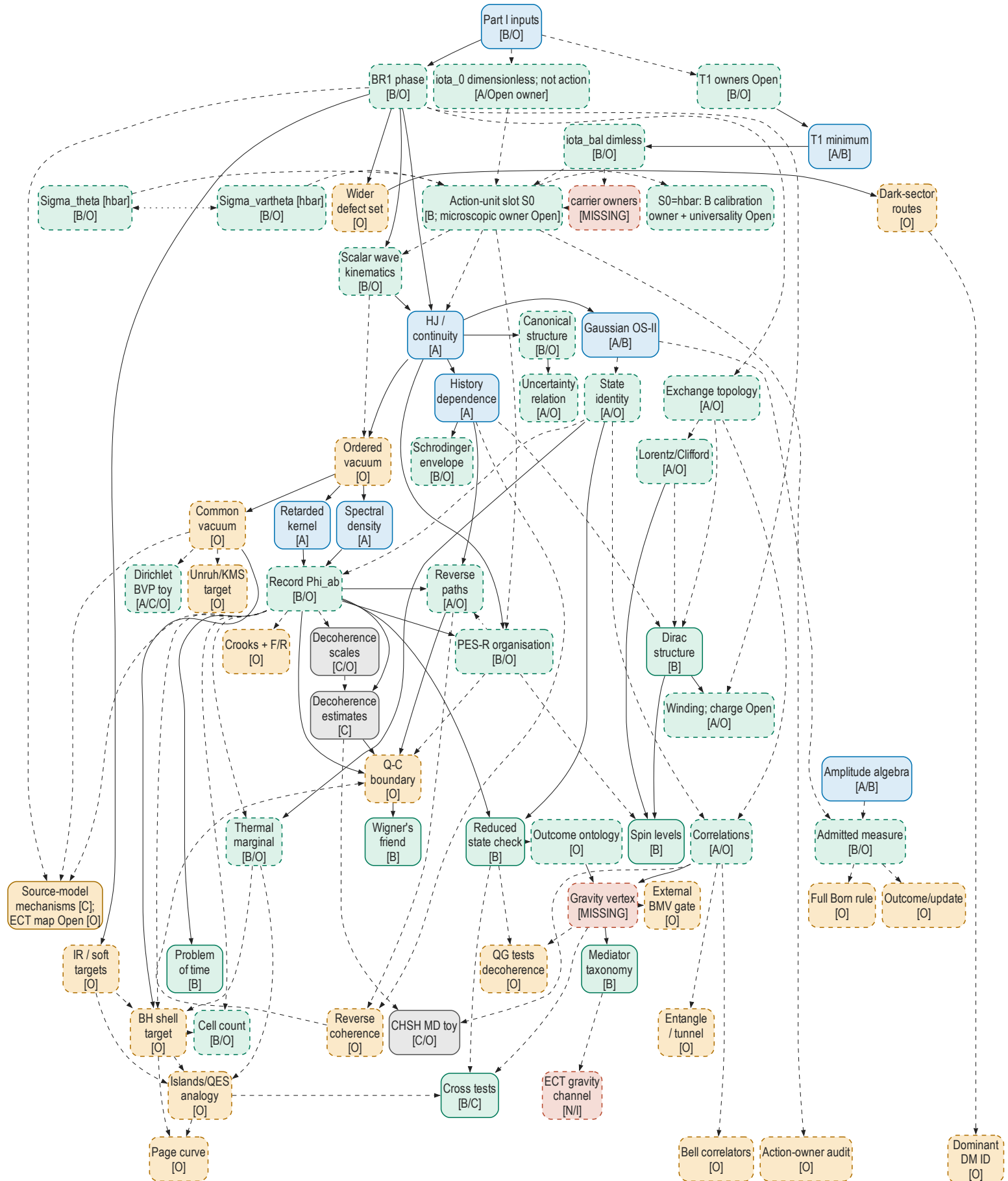
48.3. Part III status and dependency logic

Figure 21: Complete connected Part III status and dependency map. The single colour-coded A4 reader map retains every canonical node and directed edge; node text and adjacent prose carry the scientific qualifications, while arrows organise declared dependencies and never upgrade status. The deterministic exact-key and machine-topology records preserve the canonical source labels and edge inventory.

Figure 21 is kept as one connected graph, avoiding the false impression that its quantum, PES-R, probability and application branches are independent derivations. The PDF remains vector content for zooming and its literal status text is authoritative.

Selected Part III status and dependency map

Status focus: dimensionless ι_0 is not an action; S0 is an action-unit slot; S0=hbar is a Level-B calibration with owner and universality Open.



P postulate/input | A exact in stated model | B conditional/calibration | C fit/toy | O Open | MISSING owner absent. Solid: exact declared step; dashed/dotted: conditional or Open dependency. Literal node status controls.

Part IV

Summary, Comparisons, and Outlook

49. Architectural comparison with neighbouring programmes

ECT is not the only attempt to address the foundational questions of physics. The following comparison locates it among its closest architectural neighbours.

General Relativity + Standard Model: Time and Lorentzian structure are supplied. The Standard-Model gauge/matter content is input. In standard Λ CDM applications, cold dark matter is an additional non-SM component and late acceleration is modelled by Λ . The black-hole information problem remains open.

Einstein–aether theory (Jacobson, 2004): Lorentzian structure is fundamental; a preferred frame is added via a postulated unit vector u^A . In ECT, $n^A = \partial^A \Phi / |\partial \Phi|$ is a kinematic identity wherever the gradient is nonzero, but P4 supplies its aligned background value; a physical state and its selection are not derived.

Hořava–Lifshitz gravity: Lorentzian spacetime is supplied and high-energy Lorentz invariance is broken explicitly. ECT instead starts from Euclidean $O(4)$ and P4 supplies a nonzero-vector direction whose stabiliser is $O(3)$; the physical ordered state and spontaneous selection remain Open; the scalar principal tensor is Lorentzian only when the supplied coefficients obey $\beta(\beta - \alpha) < 0$; on the retained $\beta > 0$ branch this is $\alpha > \beta > 0$. A physical metric remains Open.

Volovik’s “Universe in a Helium Droplet”: Condensate analogies motivate emergent geometry. ECT conditionally classifies one supplied scalar principal tensor as hyperbolic, while physical branch selection and gravity remain Open.

Causal Dynamical Triangulations: Lorentzian structure is imposed as a lattice constraint. In ECT, residual symmetry classifies the supplied scalar principal tensor as Lorentzian iff $\beta(\beta - \alpha) < 0$ (equivalently $\alpha > \beta > 0$ only on the retained $\beta > 0$ component); SSB does not select that inequality or a physical universal cone.

Hartle–Hawking no-boundary proposal: The proposal defines a quantum state through a path integral (or semiclassical saddles) over regular compact positive-definite or complex four-geometries. It does not posit ECT’s Euclidean scalar medium.

MOND (Milgrom, 1983): The critical acceleration a_M is introduced as a universal free constant. In practical functional G the corresponding scale is fitted and compared with $cH_0/(2\pi)$; its microscopic origin and any environment/history law remain Open. The characteristic acceleration and the fitted HRC response are Level C inputs; the BTFR slope is conditional B/C algebra inside that functional, while a universal environment law remains Open.

Superfluid dark matter models: Spacetime is pre-existing; a superfluid component is added as dark matter. In ECT, physical spacetime descent and medium response are investigated within one architecture, but the physical metric remains Open.

ECT places these sectors within one condensate architecture while classifying every link separately as structural, closure-level, fit-level, or Open; it does not derive every compared structure from the minimal action.

50. Distinctive features of the framework

Features emphasised by the present ECT programme include:

Single-condensate architecture: Spacetime, gravitational response, quantum-theoretic structure, gauge-sector routes, and dark-sector phenomenology are all placed within one scalar-condensate architecture, but with different logical status: some parts are strict structural consequences, some are closure-level reconstructions, and others remain open completion targets.

No external time in the bare model: P4 adopts the physically relevant ordered direction, and the declared scalar EFT admits a candidate coordinate $x^0 = w$. Physical clocks, rods and time observables are not derived by symmetry breaking alone and remain Open.

Mechanism programme: compact phase, verified reflection-positive subsectors and the Euclidean boundary-value formulation supply distinct conditional inputs; they do not jointly derive the complex law, operator algebra, probability measure or physical clock behind selected quantum equations; exact Born, measurement, interacting operator, and matter completions remain Open.

Honest epistemology: Every result is classified as Level A, B, C, or Open; the status ceiling controls the prose.

Cross-sector action-scale programme: The phase-only fixed-core ι_0^{EFT} , BR1 phase-tension $\iota_{0,\text{bal}}^{\text{eff}}$ and texture-worldline modulus coefficient $\iota_{0,\text{bal}}^{(\vartheta)}$ are not interchangeable definitions. CDE- Φ is only a post-order acceptance criterion; it does not choose among them. Reuse in independently owned wave and canonical closures is provisional. Compact winding derives none of their finite physical values, common normalisation or identification with $\hbar$. The candidates do not generate probability: the complete applicable corrected OS-II package conditionally supplies a Hilbert space only in verified subsectors, ordinary Gleason constrains a supplied normalized noncontextual countably additive projector measure only at complex dimension at least three (with a separate Busch effect/POVM or named qubit extension), and only a specified source-model channel can suppress selected coherences. A completed derivation must first establish one 4D action/state/profile/subtraction/quotient owner, proper-time ensemble, Hessian and barrier, then prove any relation among the reduced coefficients, and finally show that independently normalised sectors use the same physical action scale. Incompatible normalisations would falsify universality, not the algebraic existence of a sector-specific minimum.

Dark-sector application scope: The adopted galactic fit has no particle-halo term and the late-time diagnostic closure has no rigid external cosmological constant. These are closure choices: they neither derive all dark-sector phenomena from the scalar action nor exclude hidden-sector particles.

Common pre-geometric research programme: A conditional scalar characteristic geometry, an Abelian-connection programme, algebraic weak-sector ingredients, the adopted Level-B P7 rank-three bundle postulate with fibrewise $SU(3)$ reduction and an Open physical-colour map, and a candidate director EFT operator are organised in one framework but have different status. P7 is not a necessary or unique physical colour completion; that identification remains Open. Physical gravity, local gauge bundles and actions, matter representations, weak chirality, the director coefficient and observable magnitudes remain Open. This broad scope is an organising feature, not a derivation or unification theorem.

Director coupling: The stated EFT basis conditionally admits a lowest-dimension homogeneous vector-current operator (B), whose coefficient may vanish. It is a density term, not a spin

interaction or source-generated fifth force. Relative vector-current observables require a non-universal/gradient/curvature map; spin and composition-dependent force signals require separate operator and source/body completions. All physical magnitudes remain Open.

Minimal displayed parameter inventory: P3 contains the two potential couplings (μ^2, λ) , equivalently (ϕ_0, λ) ; P4 separately supplies the dimension-two gradient datum u_0 ; and the scalar principal operator separately uses the dimensionless coefficients (α, β) . The choice $(\alpha, \beta) = (2, 1)$ combines a Level-B unit-slope benchmark with a normalisation convention; it does not reduce the complete ECT input/owner count to two scales. Action magnitudes, states, matching coefficients, sector operators and observable maps remain additional completion data. No irreducible parameter count is claimed until those owners and their relations are derived.

51. Status map: standard structures, conditional ECT reconstructions, and Open programmes

This section juxtaposes standard structures with several logically different ECT entries: theorem-level results inside supplied models, conditional reconstruction routes, imported calibration or source-model relations, phenomenological benchmarks, and Open physical owners. Inclusion in one table does not make every row an ECT consequence or give the rows uniform evidential strength. The following compact status map retains the most representative cases. A full version with all entries and section references is given in the technical preprint (§51 corresponds to the preprint’s “Comprehensive map”; see there for the extended seven-group classification). Status labels: A = an exact result inside the explicitly declared model and assumptions with a reproducible derivation; B = a structural/conditional derivation or closure route with an explicit Open matching, identification or physical-owner step; C = a phenomenological fit, proxy, benchmark or application-level estimate; Open = not established.

Standard structures, conditional ECT reconstructions, and Open programmes (mixed-status map).

Standard structure	Conventional status	ECT result, route, or missing owner	Level
Foundations and spacetime			
Lorentzian scalar signature $(-, +, +, +)$	Fundamental input	Classified iff supplied $\beta(\beta - \alpha) < 0$; retained shorthand $\alpha > \beta > 0$ only for $\beta > 0$; P4/SSB does not select the branch	A-math conditional / Open
Time as a distinguished direction	Fundamental	P4 supplies the ordered direction w ; a physical time/clock requires the branch, lapse and matter/metric maps	postulate/Open

Standard physics (cont.)	Status there	ECT origin	Level
Arrow of time / second law	Open / empirical	Retarded response does not fix entropy sign; ECT reduced channel, entropy functional, and detailed-balance/contractivity hypotheses remain Open	Open
Scalar slope / physical speed	Fundamental speed constant	$\hat{c}_*^2 = \beta/(\alpha - \beta)$ is dimensionless; $c_{\text{char}} = c_u \hat{c}_* / \mathcal{N}_\Phi$ uses the clock map and $c_{\text{char}} = c$ is the adopted scalar-clock calibration, while photon/tensor cone equality remains Open	A-math / B calibration / Open cone universality
Newton's constant G_N	Fundamental constant	Conditional EFT identity $G_N^{\text{nat}} = (8\pi M_G^2)^{-1}$ after a physical tensor normalisation is supplied; microscopic map and response matching Open	B conditional / Open owner
Reduced Planck constant $\hbar$ and ordinary Planck constant h_P	Fundamental constants related by $h_P = 2\pi\hbar$	Supplied reductions distinguish ι_0^{EFT} , $\iota_{0,\text{bal}}^{\text{eff}}$ and the texture-worldline $\iota_{0,\text{bal}}^{(\vartheta)}$; a conditional owner map plus the explicit Level-B physical calibration $S_0 = \hbar$ give $\Sigma_\vartheta \mathcal{I}_{1,\text{min}} = h_P$ only in the bending-free $n = 1$ branch. The common 4D owner, proper-time ensemble, stability, absolute derivation and cross-sector universality remain Open	A-math in supplied models / B route and calibration / Open ownership and derivation
Quantum mechanics postulates			
Complex Hilbert space / superposition	Axioms	Corrected OS-II reconstruction only after the full measure/correlator, algebra/reflection, regularity/growth, covariance, RP, symmetry and applicable locality/cluster hypotheses are supplied; Gaussian/free kernels verify only their declared subset	A-math conditional / Open ECT

Standard physics (cont.)	Status there	ECT origin	Level
Schrödinger equation	Axiom	Leading conditional NR envelope after a separate norm-preserving complex law, Hamiltonian/domain, state/stable KG positive-frequency component, the same physical clock, mass matching, weak interaction, slow-envelope regime and residual control are supplied; no generic descent from the real scalar action	A-math asymptotic / B-Open owner
Planck–Einstein relation $E = \hbar\omega$	Quantum postulate / empirical identification	A supplied stationary eigenproblem gives $E = S_0\omega$; the Hamiltonian, clock and finite coefficient require an owner, and $S_0 = \hbar$ is the separate Level B empirical calibration	B calibration / Open owner
Canonical commutator $[\hat{q}, \hat{p}] = i\hbar$	Postulate	Conditional only after a Hilbert representation, self-adjoint domains and commutator are supplied; compact winding does not fix the same S_0 normalisation	B/Open
Projective measurement / update rule	Textbook operational axiom; ontological collapse is interpretation-dependent	Specified source-model dephasing can suppress selected coherences after coupling, state, kernel, trace and timescale are supplied; physical ECT channel, probability, pointer, outcome and update remain Open	Open
Born rule $P = \psi ^2$	Postulate	Ordinary Gleason: complex $\dim \mathcal{H} \geq 3$ plus supplied normalized noncontextual countably additive projector measure; qubit requires supplied Busch effect/POVM or named composite extension. Physical measure/state/event map, outcome and update remain Open	A-math conditional / Open ECT

Standard physics (cont.)	Status there	ECT origin	Level
Pauli exclusion principle	Nonrelativistic fermionic antisymmetry postulate; relativistic spin-statistics consequence under standard hypotheses	Free-kernel antisymmetric OS consistency is mathematical A; application to a physical ECT spinor/CAR sector is B/Open	A-math / B-Open
Dirac equation	Postulated fermion dynamics	Conditional only in the supplied irreducible four-complex-component Clifford-factorisation class with physical metric/tetrad, spin structure, local linear Lorentz covariance and exact KG factorisation; reducible, constrained/gauge and non-Clifford systems are outside	A-math conditional / physical ECT spinor Open
Gauge sector			
Abelian gauge sector	Postulated gauge principle	Compact phase supplies winding and a conditional connection/Maxwell candidate; gauge reduction, physical $U(1)_{\text{em}}$, unbroken photon, charges and matter map remain Open	B/Open
Weak-sector structure	Postulated chiral $SU(2)_L$	Orientation-sector $SU(2)$ lift plus Hopf-fibered phase-orientation locking gives a candidate pre-gauge geometry with three broken directions; local chiral $SU(2)_L$, hypercharge, g, g' and θ_W remain open	B/Open

Standard physics (cont.)	Status there	ECT origin	Level
Neutrino masses and mixing	Seesaw / Dirac / Majorana (model-dependent)	Standard one-flavour seesaw and Weinberg-operator normalisations hold after their masses, coefficients and flavour content are supplied; ECT derives none of those owners; PMNS/MSW may be supplied	A-math conditional / Open ECT
$SU(3)$ colour sector	Postulated colour symmetry	Direct injective embedding in the displayed minimal algebra is excluded by dimension; P7 is an adopted positive-definite Hermitian frame postulate, while alternative completion classes and physical colour remain Open	A restricted / B input / Open colour
Gravity, cosmology, vacuum			
Metric/gravity route	Einstein equations postulated in GR	Scalar-only metric equation holds inside its Level-B ansatz; only at fixed α, β , about homogeneous $\bar{n} = w$, and at tangent-director linear order does the fixed image have only h_{iw} and no linear TT/static-density seed; quadratic/nonlinear/composite routes and an independent tensor action, source and metric/photon map remain Open	A no-go in fixed map / Open completion
Candidate anisotropic proxy $\mathcal{X}_{\mu\nu}^{\text{cand}}[n]$	Absent in GR	External proxy only; physical $T_{\mu\nu}^{(n)}$ requires covariant $S_n[g, n, \lambda_n]$	C/Open
Late-time acceleration (Λ)	Postulated constant	Regular two-slope scalar proof of class; physical ECT source and state Open	A conditional / C benchmark / Open ECT

Standard physics (cont.)	Status there	ECT origin	Level
Vacuum-offset problem	Standard UV–IR vacuum-energy mismatch (often summarised as up to 10^{120} in a Planck-density comparison)	Exact orientation determinant does not derive decoupling; a separately supplied trace-free action is classically shift covariant through $V + C_I$	External order-of-magnitude framing / A inside supplied completion; ECT action map Open
Observed late source	Empirical fitted density	$3\Omega_{\text{late},0}^{(\text{GR ref})} \bar{M}_{\text{Pl}}^2 H_0^2$ is a GR-reference convention after the metric and Planck calibration are fixed; the generic scalar–tensor constraint also contains $F_0, \dot{F}_0$ and any orientation stress	State/matching datum; ECT owner Open
Particle dark matter (galactic fit channel)	Postulated halo	HRC-only algebraic diagnostics are recomputed (C); BTFR slope 4 is conditional on the separate bridge (B/Open)	C / B–C
Casimir / Unruh / Hawking sector	QFT vacuum/observer effects	Supplied scalar-BVP and standard QFT target calculations are A mathematics/A external; physical ECT fields, metric, state, boundary/detector coupling and strong-field owners remain Open	A mathematics/external / Open ECT
Long-standing mysteries and targets			
Fine-structure constant $\alpha_{\text{fs}} = 1/137$	Fundamental constant	The standard identity $\alpha_{\text{fs}} = q^2/(4\pi Z_A)$ is conditional on a normalised photon and charge; ECT supplies neither the physical photon nor the electroweak/matter matching owner	External identity / Open ECT
Three fermion generations	Empirical input	A supplied one-dimensional well has tunable bound-state counting, but no physical Dirac operator, family map or parameter owner is derived	A toy mathematics / Open ECT

Standard physics (cont.)	Status there	ECT origin	Level
Proton stability	Empirical lower bounds / model-dependent rules	Three conditional patterns; P7/triality supplies no $U(1)_B$, while Λ_B , mediators and coefficients remain Open; only standard conditional EFT scaling is retained	External constraint / Open ECT
Baryon asymmetry η_B	Sakharov model	+ External observational target and missing-owner inventory; the baryon current/violating vertex, nonzero CP source and state, transition, transport, washout and ECT-native derivation remain Open	Open ECT
Quantum gravity (UV/spin-2 closure)	Open problem	Scalar characteristic and quantum/open-system routes share an architecture; physical tensor, canonical scale, source and RP/OS reconstruction remain Open	Open
Measurement problem	Open problem	A specified source model can dephase selected alternatives, but dephasing is not a Gleason premise. The probability measure/state/event map, physical ECT channel, pointer, outcome and update have separate Open owners	Open

The compact table is a selective index, not a proof by unification. The full preprint adds reflection-positivity details, quantum-statistics conditions, the fixed-map tensor no-go and completion gates, conditional Klein–Gordon and Crooks routes, cosmological and lensing programmes, neutrino benchmarks, the orientation-determinant lemma, the separately supplied trace-free completion, the regular two-slope proof of class, and the Open physical action/state map. Each row must be read at its printed level: standard mathematics reproduced in a supplied model, an external result, or a fit is not automatically evidence for ECT. The useful architectural claim is narrower: these owners and missing bridges are organised in one auditable programme without presenting them as equally derived.

52. Key equations, laws, and principles of ECT

The following is a status-separated equation and owner inventory. It mixes supplied actions, conditional mathematics, external formulae, fitted diagnostics and Open targets; inclusion in the list does not make an item a derived quantitative prediction of ECT.

Foundational action: $\mathcal{I}_E^{\text{P3}} = \int d^4X \{ \frac{1}{2}(\partial_A \Phi)^2 + V(\Phi) \}$ is dimensionless; $S_E^{\text{P3,phys}} = A_{\text{P3}} \mathcal{I}_E^{\text{P3}}$ and $d\mu_{\text{P3}} \propto e^{-\kappa_{\text{P3}} \mathcal{I}_E^{\text{P3}}} \mathcal{D}\Phi$ with $\kappa_{\text{P3}} = A_{\text{P3}}/\Sigma_{\text{PI,P3}} > 0$. Here $V = -\mu^2 \Phi^2/2 + \lambda \Phi^4/4$; the two positive action owners and their maps to quantum-sector scales are Open.

Supplied P4 nonzero-gradient datum: $\langle \partial_A \Phi \rangle = u_0 \delta_{Aw}$; the nonzero vector has stabiliser $O(3)$. The brackets denote the declared coarse-grained datum; a stationary physical vacuum and spontaneous $O(4) \rightarrow O(3)$ selection remain Open.

Conditional scalar principal tensor: $K^{AB} = \beta \delta^{AB} - \alpha n^A n^B$. This local algebraic basis is unique only inside the supplied scalar-operator inventory; the coefficients, their realised sign domain and the physical action owner are independent inputs.

Scalar slope and physical speed: $\hat{c}_*^2 = \beta/(\alpha - \beta)$ and $c_{\text{char}} = c_u \hat{c}_*/\mathcal{N}_\Phi$; $c_{\text{char}} = c$ is the adopted Level-B scalar-clock calibration, while $c_\gamma = c_{\text{char}}$ remains Open.

Matching relations: $G_N^{\text{nat}} = (8\pi M_G^2)^{-1}$ is the separate tensor matching; $S_0 = \hbar$ is the adopted Level-B action calibration; and $\phi_0 = \zeta_\phi \bar{M}_{\text{Pl}}$, $\zeta_\phi > 0$, $\zeta_\phi = O(1)$, is an independent Level-B scale ansatz. Their microscopic owners and universality remain Open.

Cross-sector cone universality: The scalar and supplied phase truncations share only their common-length slope through K^{AB} ; $c_{\text{phase}} = c_{\text{char}}$ requires the same supplied clock/lapse map on the aligned frozen-coefficient patch. The identifications $c_\gamma = c_{\text{char}}$ and $c_{\text{GW}} = c$ remain Open completion gates.

Loop quantisation: $\oint \partial_A \theta dX^A = 2\pi n$ after BR1 is supplied; $\iota_{\text{elem}}^{(\text{fixed core})} = 2\pi \iota_0^{\text{EFT}}$ in the phase-only closure. A distinct texture-worldline modulus reduction can define $\iota_{0,\text{bal}}^{(\vartheta)} = \sqrt{2K_\vartheta^{\text{1D}} \mathcal{T}_{\text{loop}}}$ only after its common owner and proper-time ensemble are supplied; it equals the BR1 phase-tension coefficient only if the modulus/stiffness map is proved. With the explicit Level-B calibration, the bending-free $n = 1$ branch gives $\Sigma_\vartheta \mathcal{I}_{1,\text{min}} = h_{\text{P}}$; profile/Hessian, bending/barrier, microscopic ownership and stability/universality remain Open.

Entropy-production status: the retarded backbone and positive pairwise noise are structural, but a monotone entropy theorem requires a declared reduced channel, entropy functional, stationary state, and detailed-balance/contractivity assumptions; no universal scalar rate follows.

Generalised Einstein equations: $FG_{\mu\nu} = 8\pi G_N T_{\mu\nu}/c^4 + \mathcal{T}_{\mu\nu}^{(q)}$ in the scalar-only undivided convention; division by F adds no orientation term, and $T_{\mu\nu}^{(n)}$ requires a separate covariant $S_n[g, n, \lambda_n]$ (Open).

HRC algebraic relation: $g_N = \mu_{g,k}(g^2/a_M^2)g$, $k \in \{0, 3\}$.

BTFR: $v_{\text{flat}}^4 = G_N M_{\text{bar}} a_M$ (slope 4, algebraic).

Hubble evolution: $t_{0,\text{ECT}} = \int_{a_{\text{ord}}}^1 da/[aH_{\text{ECT}}(a)]$; the named two-slope state gives $\tau_{2s} \equiv H_{2s}(0)\Delta t = 0.96368678$ conditionally. A dimensional age uses the declared $H_{2s}(0) = H_0^{\text{cal}}$ calibration; the physical observable map and microscopic selection remain Open.

PES-R inventory: Action stationarity, pairwise record stability, finite-time sector diagnostics, and phase admissibility are separate objects; no universal intersection theorem is claimed.

Source-model visibility proxy: $\tau_{\text{vis}} = 1/(N_{\text{eff}}\gamma)$ only for a declared channel, state, rate convention and observation protocol; it is not a universal ECT decoherence time or sector boundary.

BH entropy target: $S_{\text{BH}} = A/(4\ell_{\text{Pl}}^2)$ is an external target; the current ECT action derives neither a physical black-hole area law nor the coefficient $1/4$.

Casimir (conditional): For a separately supplied three-independent-scalar Dirichlet BVP with the physical light speed, $F_{N=3}^{\text{scalar}} = \frac{3}{2}F_{\text{QED}}$ arithmetically. ECT does not derive that mode bundle, material boundary, stress observable or force coefficient.

53. Conditional completion tests, fitted diagnostics, and Open observational handles

ECT claims are separated into structural discriminants, conditional completion tests and explicitly Open observables. The named two-slope scalar–tensor action/state gives a conditional background, age and restricted diagnostic estimates, but P1–P6 do not uniquely select that completion. Physical Hubble inference, JWST forward modelling, gauge-complete growth and ISW, and lensing remain owner-specific Open programmes:

1. **BTFR slope = 4 within the adopted galactic closure** (B/C , *conditional*). This is an algebraic consequence of the assumed deep branch. A robust systematic deviation would challenge this closure rather than the entire ECT framework.
2. $a_M \sim cH_0/(2\pi)$ (*Level C matched comparison/input*). Its empirical comparison is meaningful, but its microscopic ECT origin and environment/redshift law remain Open.
3. **Director-coupling completion** (*Open*). The fixed vector spurion supplies neither a source-generated force nor an Eötvös window. MICROSCOPE constrains a future completion only after its field/profile, species/body charges, range and screening map are declared.
4. **Late-time acceleration.** Minimal P1–P6 and the positive-kinetic stress lemma do not select the physical source or $w(a)$. The named two-slope completion supplies a regular conditional history; its microscopic selection, physical metric/state map and likelihood remain Open.
5. **Exploratory environment/history law Ξ for $a_{M,\text{eff}}$** (*Open*). No sign or amplitude is predicted. A null result rejects only the declared Ξ ansatz; a correlation would motivate, but would not derive, an ECT mechanism.
6. **Casimir boundary-response programme** (*A mathematics/Open ECT*). The supplied N -scalar Dirichlet BVP is exact and its separately supplied $N = 3$ case retains the $3/2$ toy identity, but the physical ECT mode bundle, material boundary map, stress observable and force are not derived. No ECT $3/2$ correction is predicted.
7. **Tensor polarisation gate** (*Open*). At fixed α, β , about homogeneous $\bar{n} = w$, and at tangent-director linear order, the fixed image has no linear TT sector; nonlinear/composite/additional-field routes remain Open. A separately supplied massless Fierz–Pauli completion would have two TT modes; observed extra polarisations would reject that completion, not the present negative theorem.

8. **No particle-halo term in the specified galactic fit (*C*).** A confirmed particle component required by rotation-curve and RAR/BTFR data would challenge that closure; discovery of a hidden-sector particle would not by itself falsify ECT.
9. **Bare scalar-ratio identity.** With $K_\phi^{\text{phys}} = \bar{M}_{\text{Pl}}^2 \omega \dot{\phi}^2 / 2$ and a positive bare denominator, the supplied pieces have $w_\phi^{\text{bare}} \geq -1$. Non-minimal F -derivative terms leave observable $w_{\text{eff}}(a)$ unconstrained; a falsifier exists only for a complete model that predicts the same observable history.
10. **Propagation corrections (*Open*).** Photon LIV is not an operational ECT prediction: Fermi–LAT constrains specified external dispersion and source-lag models, which become an ECT gate only after a physical photon operator, coefficient, suppression scale, clock and forward likelihood map are derived. Tensor dispersion and delay likewise remain Open until their physical operator, coefficient, scale and observable map are owned.
11. **Proton stability (*Open ECT*).** Conditional on a completed gauge/flavour action, three selection-rule patterns remain possible: full low-energy protection, partial protection with $\Delta B = 1$ forbidden, or generic $\Delta B = 1$ violation. In the last case $\tau_p \propto \Lambda_B^4 / (|C_B|^2 m_p^5)$ is the usual dimension-six NDA scaling, up to channel-dependent hadronic, renormalisation-group and phase-space factors. P7 fixes neither Λ_B , C_B nor the mediator spectrum; current experimental bounds [15] constrain any named completion.

Near-future observational gates for future completions. Beyond the conditional or externally constrained handles already accessible with current data, several observational programmes expected to deliver future results can constrain named ECT completions once their action, state and observable maps are supplied. *(i)* Full-sky CMB polarisation missions such as LiteBIRD can test whether the primordial tensor spectrum is compatible with any viable condensate-based early-Universe completion of ECT. *(ii)* Combined CMB, weak-lensing and large-scale-structure measurements from programmes such as the Simons Observatory, Euclid and Rubin/LSST can probe whether a completed action-owned scalar–metric background and perturbation response, together with an explicitly external anisotropic proxy $\mathcal{X}_{\mu\nu}^{\text{cand}}[n]$, leaves a detectable signature in the lensing–growth consistency relation. The divided coefficient $G_{\text{div}}(z)$ alone is not such a prediction. *(iii)* Space-based gravitational-wave detection will constrain a future ECT tensor completion through its propagation cone, polarisation content and damping law. The tested fixed- α, β tangent-director linear map predicts neither two TT modes nor c_{GW} ; untested nonlinear/composite routes remain Open. Standard-siren determinations of H_0 remain external cosmological data rather than tests of a derived tensor prediction. A detailed table of near-future discriminants, with explicit ECT statements and falsification verdicts, is given in the technical preprint.

ISW and lensing target. An ECT prediction requires the time-dependent Weyl potential of the physical photon metric, obtained from a gauge-reduced perturbation action on the same background. A restricted same-metric, zero-slip, sub-horizon proxy gives only 0.00267–0.00292% Weyl-carrier and 0.00308–0.00813% conformal-derivative changes over $0 \leq z \leq 2$ for the named near-GR state. It is not an ISW amplitude or lensing spectrum; those full observables remain Open.

54. Colour completion: adopted P7 structure and Open physical owners

Part IV of the technical preprint does not derive physical colour from P1–P6. The exact dimension inequality $\dim \mathfrak{su}(3) = 8 > 6 = \dim \mathfrak{so}(4)$ excludes only a direct Lie-subalgebra embedding into the displayed $O(4)$ order-parameter algebra. Reducible-singlet, emergent/nonlinear, independently supplied bundle and defect-kernel routes remain possible.

P7 is an adopted structural postulate. In addition to P1–P6, P7 adopts a global complex rank-three vector bundle $\mathcal{C} \rightarrow \mathcal{M}^4$ with transition cocycle, a positive-definite Hermitian form h and a compatible nowhere-vanishing complex volume form Ω normalized by $|\Omega|_h = 1$. The fibrewise stabilizer of (h, Ω) , equivalently the reduced structure group, is $SU(3)$; this is Level-A mathematics conditional on the supplied bundle data. Global structure-preserving automorphisms form the associated gauge group of sections and are not one finite-dimensional copy of $SU(3)$. P7 is a reverse-engineered Level-B structural input, not a P1–P6 descent theorem and not a necessary or unique completion. The abstract reduction admits compatible mathematical connections under the usual bundle assumptions, but P7 selects none as the physical dynamical connection and does not identify the structure with physical colour. The physical colour connection, Yang–Mills action, measure, matter representations, couplings, spectrum and QCD map remain Open.

Scoped structural consequences. A nonzero fundamental $\mathbf{3}$ VEV has stabiliser $SU(2)$, whereas a reducible representation can contain an invariant singlet. A Proca mass is forbidden only after an exact gauge theory is independently supplied, and the P7 fields themselves provide no colour-charged Higgs. An internal oscillator tower is not identified with a physical glueball spectrum without a gauge-invariant operator, constraints and dynamics; this is non-identification, not a universal spectrum no-go.

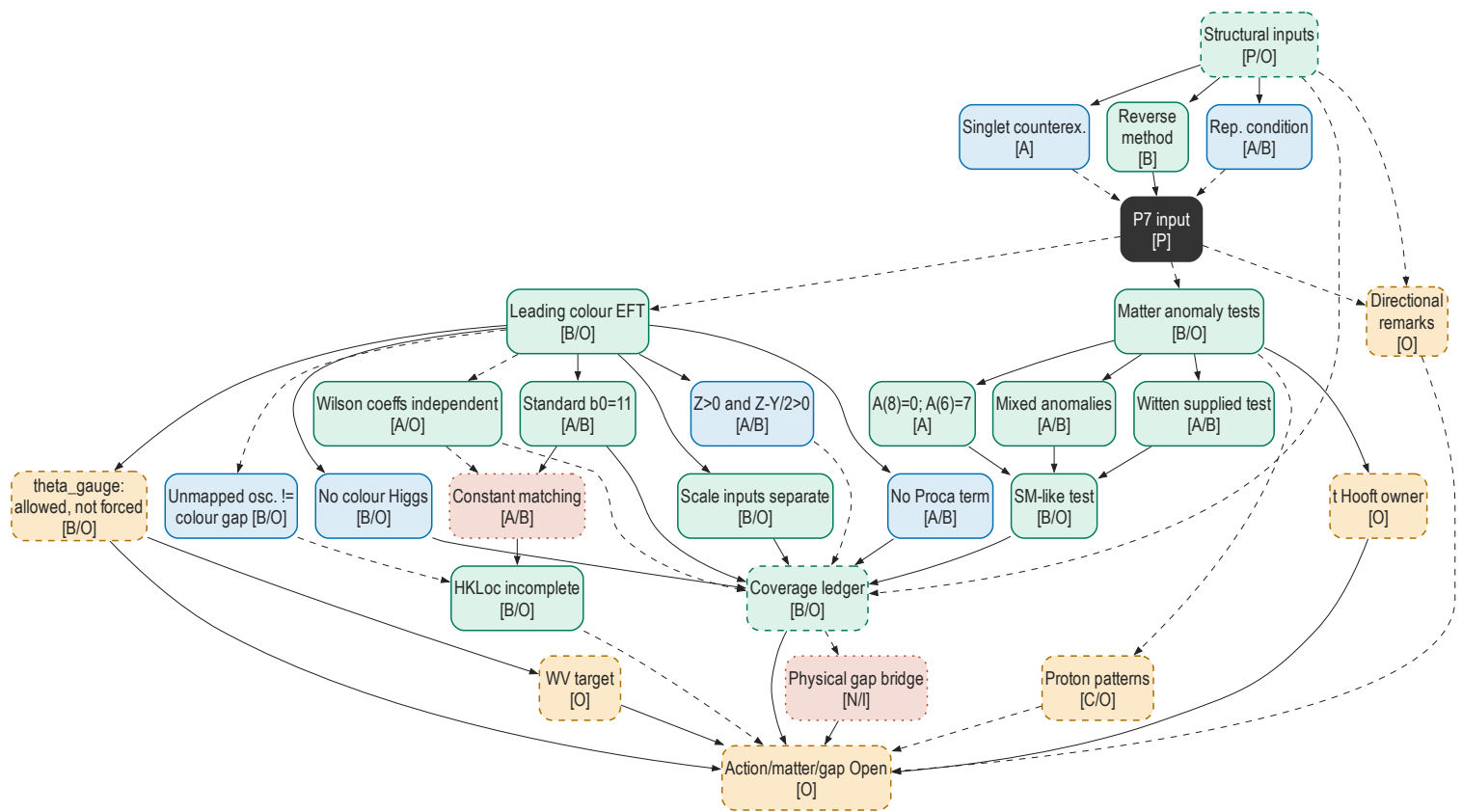
Baryon and proton firewall. The invariant $\varepsilon_{abc} q^a q^b q^c$ permits a baryon-like colour singlet, but colour triality does not derive $U(1)_B$, baryon charges, a current or a baryon-violating vertex. The three proton-selection patterns remain conditional on a full gauge/flavour action. In the generic dimension-six pattern, $\tau_p \propto \Lambda_B^4 / (|C_B|^2 m_p^5)$ is the usual NDA scaling, up to channel-dependent hadronic, renormalisation-group and phase-space factors. A named C/Open reference branch may set $\Lambda_B = c_B \phi_0$ and retain C_B as an independent coefficient; $c_B = |C_B| = \zeta_\phi = 1$ gives only the crude unit-coefficient NDA reference $\tau_p \sim 1.01 \times 10^{42}$ yr, while a protecting symmetry may give $C_B = 0$. Hadronic, RG, flavour, phase-space and normalisation owners are omitted at that point. P7 fixes none of these inputs, so no proton lifetime is predicted; experimental bounds constrain each named completion.

Source-model RG and strong CP. The familiar $b_0 = 11$ running is reproduced only on the separately declared constant, $Y_0 = 0$, CP-even, matter-free pure- $SU(3)$ source branch. General anisotropic running is multi-coefficient and Open. The physical strong-CP parameter is $\bar{\theta} \equiv \theta_{\text{gauge}} + \arg \det M_q \pmod{2\pi}$ when $\det M_q \neq 0$. In a massless-quark limit the determinant phase is ill-defined/removable and must be treated separately. P7 supplies neither mass matrix nor relaxation mechanism, so the problem is reopened, not solved.

Mass-gap statement retained conditionally and framework hypothesis H_{Clay} . A positive global gap follows from the stated external transfer theorem only after a fully specified reflection-positive colour theory has a unique vacuum, positive adjointed/reflected forms, a transfer-cyclic dense physical class, uniform exponential decay and a positive two-sided scale map. The local HKLoc bookkeeping supplies none of these global owners. Separately, ECT adopts the Level-B/Open framework hypothesis H_{Clay} that a future P7 physical-colour completion may be emergent and finite-cutoff rather than the primitive continuum object in the Clay statement. This hypothesis is not a consequence of P7, requires a colour action, measure, regulator and scale map, supplies no evidence against continuum Yang–Mills, and is neither a solution nor a dismissal of the Clay problem. Physical colour, confinement, the glueball spectrum and an ECT-owned mass gap remain Open.

Figure 22: Complete Part IV structural logic. Solid statements are scoped group/source-model algebra; dashed dependencies lead to the adopted Level-B P7 structural input and Open physical colour, baryon/proton, strong-CP and mass-gap owners. No arrow turns P7 into a P1–P6 forward derivation or into a physical-colour result.

Selected Part IV status and dependency map



P postulate/input | A exact in stated model | B conditional | C fit/toy | O Open | MISSING/N-I owner absent. Solid: exact declared step; dashed: conditional/Open dependency; dotted two-way: unproved compatibility.

Table 4 separates the conditional structural information supplied by P7 from the physical owners that remain Open.

Table 4: Conditional information supplied by P7 and the remaining owners.

Item	Before P7 input	Conditional structural result	Still Open
Physical colour	Missing	$SU(3)$ frame redundancy available	Connection, action, measure, observables
Gauge bosons	Missing colour owner	Adjoint dimension eight if gauged	Physical gauge-field identification
Meson/baryon channels	No named colour owner	Singlet operators permitted	Bound-state dynamics and $U(1)_B$
$\alpha_s, \Lambda_{\text{QCD}}$	Undefined	Source-model matching variables	ECT RG/matching owner
Proton/hadron masses	Not identified	Gauge-invariant spectral targets	Operators and dynamics
Baryogenesis	Baryon current absent	Baryon-like singlet permitted	Current, violation, state, transport
Proton stability	Selection rule absent	Three patterns can be catalogued	Λ_B , mediators, coefficients and realised pattern
Mass gap	No colour theory	Conditional theorem can be stated	Complete theory, transfer premises and scale map

55. Open problems

These are stated explicitly as unsolved targets:

1. **Three fermion generations.** A future spectral/topological route requires a specified physical Dirac operator, bundle, domain, boundary class and index problem. Topology alone gives neither chiral zero modes nor exactly three generations.
2. **Yukawa hierarchy.** The physical reference masses span from $m_e \simeq 0.511$ MeV to a top-quark mass near 173 GeV [11]; this external mass hierarchy is not a running-Yukawa dataset, and neither the Yukawa matrices nor their renormalisation-scale-dependent hierarchy are derived from ECT.
3. $\alpha = 1/137$. No first-principles prediction.
4. **Renormalisability.** Open. Scalar fixed-background power counting does not establish renormalisability or UV safety of the full ECT completion; the tensor, gauge and matter actions, constraints and counterterm closure are missing.
5. **Nonlinear GEE.** Structurally motivated (Level B); full mathematical rigour not achieved.
6. **Born rule.** Under the full corrected OS-II hypotheses a Hilbert route exists in verified subsectors. Ordinary projector Gleason is conditional A-math for complex dimension at

least three with a supplied normalized, noncontextual, countably additive measure; a qubit needs the separately supplied Busch effect/POVM or another named extension. Probability, state/event map and the physical ECT Born owner remain Open.

7. **Bell correlators.** Common-medium nonfactorisation is a candidate route (B/Open); explicit Bell, no-signalling, and Tsirelson correlators remain Open.
8. **BH evaporation.** A finite-memory shell toy gives illustrative Page-curve-like behaviour (Level C/Open); a physical ECT shell, constructive Page curve and full dynamics remain Open.
9. **Full observed electric-charge pattern.** The compact condensate phase supplies integer winding and a conditional compact- $U(1)$ representation route, not a derived physical charge lattice. The gauge reduction, unbroken $U(1)_{\text{em}}$ photon, charge operator, matter map, full Standard-Model charge spectrum, hypercharge quantisation, anomaly-consistent representations, and global gauge-group quotient remain Open; the recorded colour-interlock congruence only constrains admissible completions.
10. **Holographic bound and strong holography.** ECT currently supplies only a conditional boundary-cell counting lemma after coarse-graining, finite local capacity, a physical split and an area-law state class are assumed. A physical area law, soft/gauge-edge treatment, covariant Bousso-type formulation and full bulk–boundary reconstruction remain Open.
11. **High- z background.** Self-consistent $\phi_b(z)$ at $z \gg 1$ not derived.
12. **Gradient condensation (OP-grad).** Dynamical origin of ordered branch from bare P3: the deepest open problem. Recent sharpening localises the remaining gap (OP3.1a) to the constraint-relaxation/coarse-graining bridge, with the EFT-data derivation (OP3.1b) and the measure-concentration selection step (OP3.1c) recorded as separate subproblems; the computed candidate bridges are excluded as completions.
13. **Tensor-loop power counting (OP-UV1).** This can begin only after a physical constrained tensor propagator and vertices are derived; scalar Landau-pole arithmetic does not establish its behaviour.
14. **Coupled RG after tensor completion (OP-UV3).** Only after a constrained physical tensor propagator and vertices exist can one define condensate–tensor mixing and compute its multi-loop RG flow.
15. **OS reconstruction for a physical tensor sector (OP-OS1).** First derive the constrained tensor field, Euclidean correlators/measure and physical observable algebra; then verify the complete applicable corrected OS-II package, including reflection positivity rather than only that one member. The strict-map longitudinal coordinate, or a scalar mode inside a supplied reduced action, is not that field.
16. **Vacuum offset and late source (reformulated OP-A3).** The fixed determinant of the orientation tensor is exact algebra but does not derive a trace-free physical metric action. In a separately supplied constrained completion, $V + C_I$ is classically invariant under opposite shifts, with C_I a possible global boundary/state datum. The distinct ordinary two-slope completion gives a stable late-de-Sitter proof of class but is not shift invariant. ECT must derive the action, physical matter/photon metric, radiative completion and

state-to-observable map; the value of a particular condensate datum may then be measured rather than universally predicted.

17. **Proton stability and baryon-number selection rules (OP-proton).** Derivation of which baryon-number selection-rule pattern (full protection / partial $\Delta B=2$ only / generic $\Delta B=1$) is realised in the completed gauge/flavour sector; if generic violation, identification of decay channels and coefficient structure.
18. **Origin of neutrino masses (OP-neutrino).** The present formulation does not derive the absolute neutrino mass scale, the Dirac/Majorana character, the PMNS matrix, or the mass ordering. The standard symbolic seesaw and Weinberg maps relate separately supplied masses and coefficients, but the action selects none of them. Any future ECT derivation must own the flavour operator and absolute-scale map. A baryogenesis claim would additionally require its own baryon-number, CP, transition, transport and washout chain rather than a coincidence of hand-selected scales.
19. **Gauge-sector completion.** The compact phase fibre gives winding and a conditional bundle-covariance route. The physical connection, gauge volume/reduced Hessian, Maxwell normalisation, unbroken photon, charge and matter map, its electroweak placement, the weak sector, and the colour construction remain Open.
20. **Charge quantisation (OP-GUT3).** If the global electroweak structure group is $U(2) = (SU(2) \times U(1))/\mathbb{Z}_2$, representations satisfy $2T + Y \in 2\mathbb{Z}$; with the colour sector included, the recorded $\mathbb{Z}_6$ interlock congruence further fixes the fractional electric pattern per colour class. What remains open is the selection within the resulting explicitly characterised discrete family of charge assignments.
21. **Anomaly cancellation (OP-GUT6).** Any supplied chiral $SU(2)$ completion must pass local triangle and global Witten $\mathbb{Z}_2$ anomaly tests. These become sharp consistency checks only after the bundle and fermion representations are supplied. The preferred-direction vector bilinear $\mathcal{L}_5$ does not change representation-data anomaly coefficients, but it does not derive the missing chiral sector.
22. **Weinberg angle (OP-GUT7).** $\sin^2\theta_W$ from condensate parameters: not yet derived.
23. **Director coupling coefficient (OP-GUT8).** Microscopic derivation of μ_5 from condensate microdynamics and establishment of radiative stability.
24. **Full weak-gauge derivation (OP-GRAV-WEAK).** Whether the oriented connection sector contains independent spin-1 modes beyond those fixed by the metric, which could underlie the weak interaction as a geometric rather than purely internal gauge field.

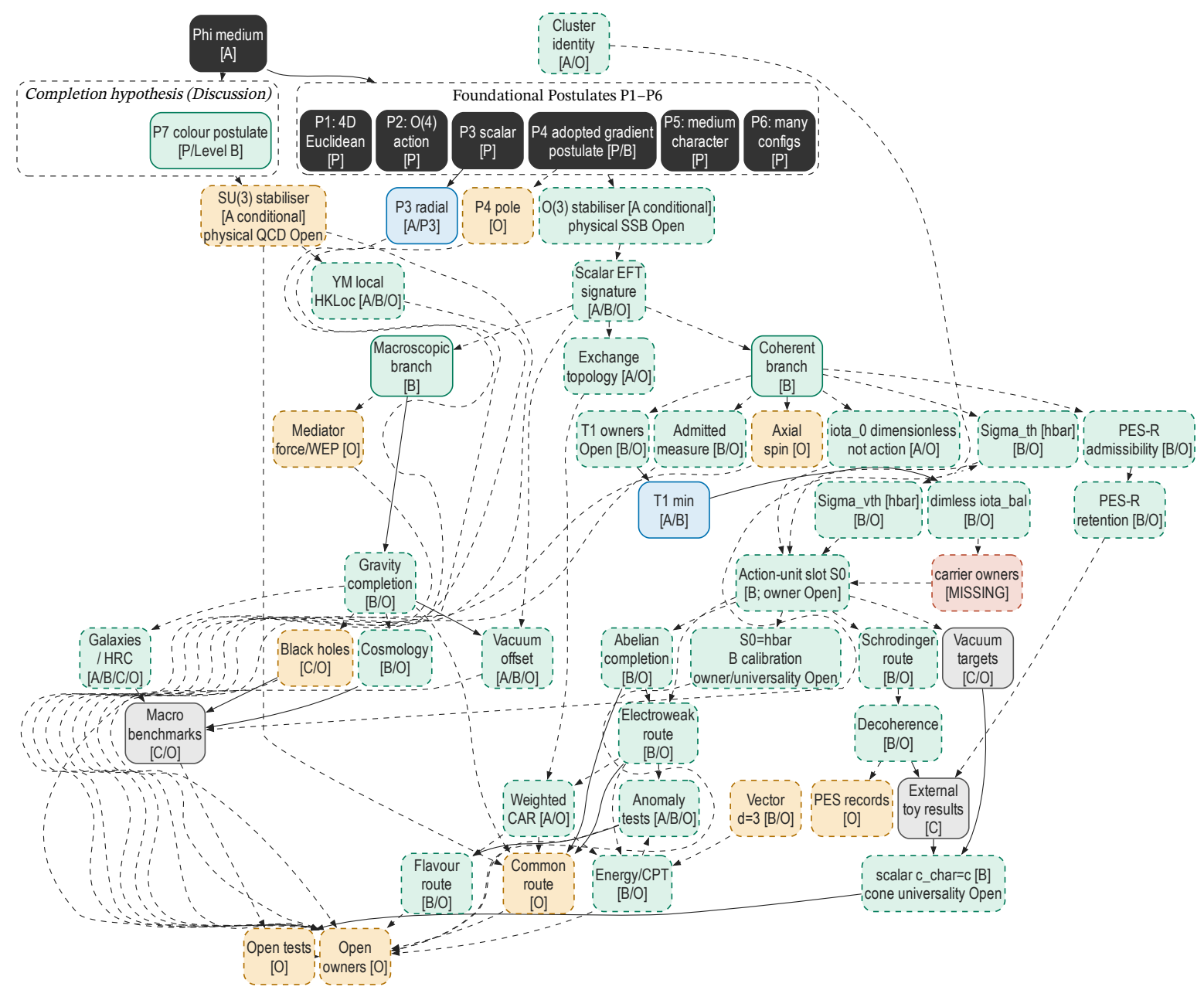
56. Selected status and dependency map

Figure 23: Complete connected ECT status and dependency map. The architecture separates postulates and supplied inputs from exact results inside declared models and assumptions (Level A), conditional closures (Level B), phenomenological diagnostics (Level C), and Open physical owners. Every canonical node and edge is present in one colour-coded A4 reader map. Colour is redundant with borders, status text and edge style; no arrow alone upgrades a claim. The deterministic exact-key and machine-topology records preserve the canonical source labels and edge inventory.

Figure 23 restores the entire dependency structure in one view. The PDF remains vector content for zooming. In particular, the probability-representation route uses separately supplied corrected OS-II/Hilbert and measure inputs; no Schrödinger, decoherence, PES or quantum–classical edge is a Born-rule derivation. The generic fifth-force node likewise belongs only to a separately supplied dynamical mediator/source/body-charge completion; it is not an output of the fixed homogeneous vector spurion.

Selected ECT status and dependency map

Status focus: P4 and P7 are adopted Level-B inputs; their stabiliser algebra is conditional; physical QCD, S0 ownership and cross-sector cone universality remain Open.



P postulate/input | A exact in stated model | B conditional/calibration | C fit/toy | O Open | MISSING owner absent. Solid: exact declared step; dashed/dotted: conditional or Open dependency. Literal node status controls.

57. Open problems of physics addressed within ECT

Within its framework, ECT addresses several well-known open problems of modern physics:

1. **Quantum–gravity completion programme.** Quantum/open-system structures and a scalar characteristic geometry share the ordered-medium architecture, but a physical tensor, universal metric and gravitational record channel remain Open.
2. **The origin of time.** Within the supplied local algebraic scalar-operator inventory, residual $O(3)$ symmetry fixes the tensor basis; the independently supplied tensor is Lorentzian iff $\beta(\beta - \alpha) < 0$; on the retained $\beta > 0$ component this is $\alpha > \beta > 0$ (A-math conditional). P4/SSB does not select those coefficients or derive the physical clock/universal metric (Open).
3. **The dark matter problem.** No dark-matter particle candidate has been established in direct, indirect or collider searches. The HRC-only algebraic diagnostic models the galactic mass-discrepancy channel without a particle-halo term (C); its source/force map is not derived, and this does not exclude hidden-sector particles.
4. **The cosmological constant problem.** ECT currently supplies an exact fixed-determinant orientation identity, not an unconditional vacuum-decoupling theorem. A separately supplied trace-free scalar–tensor completion is classically shift covariant through $V + C_I$, and a distinct two-slope completion supplies a regular late-de-Sitter proof of class. The derivation of either physical completion from the Φ medium, its radiative protection, universal metric and cosmological state remains Open.
5. **The Hubble tension.** ECT does not yet own a numerical shift: a complete recombination-to-distance-ladder pipeline from one action and state is Open.
6. **JWST early galaxies.** A quantitative abundance and morphology prediction requires the full ECT background, perturbation, halo, baryonic and selection forward model; this remains Open.
7. **Baryon asymmetry.** Only an external observational target and a missing-owner inventory are available; the physical baryon current/violating vertex, nonzero CP source and state, transition, transport, washout and ECT-native derivation remain Open.
8. **Black-hole information paradox.** A thermal marginal is compatible with global purity under a separately supplied Hilbert-space split and isometry. Global ECT unitarity, a physical shell, Page curve and constructive evaporation solution remain Open; the shell/Page toys are Level C/Open.
9. **The measurement problem.** A specified reduced-sector decoherence model can suppress interference, but PES-R supplies no Born or measurement derivation. Exact weights, pointer-basis selection, single outcome, and the update theorem remain Open.
10. **The origin of quantum mechanics.** For supplied Gaussian/free Euclidean kernels satisfying the complete applicable corrected OS-II package, the quotient-and-completion construction gives a conditional Hilbert-space route. Extending it to the interacting Φ theory and identifying the physical observables, time/metric and state remain Open.

Ordinary projector Gleason applies only at complex dimension at least three with the full supplied measure hypotheses; a qubit requires a supplied Busch effect/POVM or named extension. Neither route originates probability, a unique outcome or an update rule.

11. **The arrow of time.** P4 supplies a Euclidean orientation. Retarded orientation exists only in a supplied hyperbolic scalar problem after an initial–boundary choice; it is not a physical or thermodynamic arrow. Neither retardedness nor positive pairwise dephasing fixes the sign of von Neumann entropy. An ECT-specific second-law theorem remains Open pending the reduced map, state or fixed point, entropy functional and detailed-balance/contractivity hypotheses.
12. **The need for inflation.** ECT organises structural or conditional non-inflationary routes to horizon-scale homogeneity and a flat baseline branch. The supplied S^3 orientation target has $\pi_2 = 0$, but its identification with the complete cosmological vacuum manifold and a physical relic conclusion remain Open. A supplied pure-quartic ordering kernel gives an exact flat formal local seed, but the physical band, state, transfer, branch-selection probability and full observed spectrum remain Open; the analysis does not establish that inflation is phenomenologically unnecessary.
13. **The hierarchy problem.** For the displayed matched scales, $M_{\text{Pl}}/M_{\text{EW}} \sim 10^{16}$ and $\mathcal{I}_{\text{EW}} = \ln(\phi_0/v_2) \approx 36.8$. These are external/input arithmetic, not an ECT solution or a scale-generation prediction. Four candidate templates remain possible (asymptotic-freedom-like running, walking/near-conformal dynamics, Miransky/BKT scaling, and topological/Hopf-like routes). This is a reduction, not a solution: the broader programme decomposes into OP-EW-scale, OP-EW-locking, OP-EW-gauge, OP-EW-naturalness, and OP-EW-matter. The first-principles dynamics and the full electroweak completion remain open (*B/Open*).
14. **Why a 3 + 1 split, and what follows for exchange topology?** Given P1 and the supplied P4 vector chart, one coordinate is an aligned temporal seed and three are transverse; its physical realisation and the deeper origin of four-dimensionality remain Open/OP0. For two supplied identical point excitations in the exact defect-free domain $\mathcal{M}_2 = ((\mathbb{R}^3 \times \mathbb{R}^3) \setminus \Delta)/S_2$, $\pi_1(\mathcal{M}_2) = \mathbb{Z}_2$, so the continuous Abelian braid phases characteristic of two-dimensional anyons are unavailable. Bosonic/fermionic characters require an additional one-dimensional representation choice; representation selection, higher-dimensional S_N sectors, CAR and spin–statistics remain Open (*A topology conditional on the supplied domain / Open physical statistics*).
15. **What does the supplied S^3 target say about monopoles?** $\pi_2(S^3) = 0$, so maps into that target have no π_2 charge (A-math conditional). The complete physical vacuum/configuration space, transition owner and relic-production history remain Open, so this is not yet a physical ECT monopole anti-prediction. Later gauge-completion sectors have separate topology and abundance owners.
16. **Amplitude-potential stability.** In the homogeneous P3 amplitude sector, the minima $\Phi = \pm\phi_0$ are locally stable at tree level, and the standard one-loop scalar running closes the Standard-Model-type high-field metastability channel inside that declared amplitude closure: here $\beta_\lambda > 0$, unlike the Higgs sector of the Standard Model where the top-Yukawa contribution drives metastability. The symmetric point $\Phi = 0$ is a hilltop, not a metastable

minimum, so Coleman false-vacuum decay does not apply to the minimal P3 potential. None of this proves stationarity or stability of the P4 gradient chart; the complete ordered-state action/Hessian and any secondary electroweak inheritance remain Open (*B/Open*).

Vacuum decay as a conditional Euclidean benchmark. The Coleman bounce is standard semiclassical mathematics only after a Lorentzian quantum theory, analytic continuation, contour/thimble, measure, state, boundary data, clock and fluctuation determinant are supplied. Bare Euclidean DP alone does not assign a transition amplitude or decay rate. ECT may compare a future completed coherent sector with that benchmark, but currently supplies no distinct decay owner. For the minimal primary condensate potential, the false-vacuum structure required by Coleman decay is absent.

17. **Quantum gravity.** The shared-medium architecture motivates a completion programme, but the physical tensor, source, canonical scale and record channel are not derived (*Open*).
18. **Cosmological coincidence question.** ECT does not yet derive Ω_m , the physical late-time source, or their evolution; the question remains Open.
19. **Physical spin-2 sector.** The fixed map has no linear TT/static-density seed only at fixed α, β , about homogeneous $\bar{n} = w$, and at tangent-director order; quadratic/nonlinear/composite routes remain parametric/Open. A massless Fierz–Pauli action is an Open completion target, and the LIGO mass bound is an external gate for that target.
20. **Are c , G_N , and $\hbar$ already unified?** No. The scalar coefficients fix only $\hat{c}_*$; $c_{\text{char}} = c_u \hat{c}_* / \mathcal{N}_\Phi$ uses the clock map, while $c_{\text{char}} = c$ and $S_0 = \hbar$ are separately adopted Level-B calibrations. $G_N^{\text{nat}} = (8\pi M_G^2)^{-1}$ is a separate tensor matching, and photon/tensor cone equality plus the microscopic origin and universality of the calibrations remain Open.
21. **Can Casimir, Unruh, Hawking, and particle-production effects belong to one family?** ECT poses this as an Open common-response programme. Their displayed calculations currently use different supplied bundles, states, boundaries, detectors and geometries.
22. **Why can a smooth positive kinetic weight not by itself weaken Pauli exclusion in the supplied local CAR model?** In a separately supplied local first-order fermionic/CAR EFT, an invertible canonical normalisation removes a smooth strictly positive temporal kinetic factor while preserving the supplied CAR algebra. This excludes Pauli violation generated solely by that factor. It does not cover singular or sign-changing factors, nonlocal branch mixing, defects/coherence loss, or an altered symplectic/CAR structure (*Level-A conditional mathematics / Open physical ECT*).
23. **Fine-structure constant.** The standard identities and external one-loop running may be used as checks after a photon, charge normalisation, matter representations and matching scale are supplied. ECT currently derives none of those physical owners and therefore predicts no value of α_{fs} (*Open ECT*).
24. **S_8 / lensing–growth sector.** A future covariant $S_n[g, n, \lambda_n]$ could define an orientation contribution to the weak-lensing/growth sector. At present the functions $\mu(k, z)$, $\eta(k, z)$, and $\Sigma(k, z)$ are explored only with external proxies; no ECT orientation stress or S_8 correction is derived (*Level C/Open*).

The technical preprint provides a broad status table comparing about seventy structures—some postulated, some derived within standard frameworks, and some empirical inputs—with ECT’s classifications as conditional mathematics, reconstruction targets, supplied diagnostics or Open problems. A complementary mixed-status table records candidate-completion tests, benchmarks and Open observational handles; only a completed row with its own action, state, observable and uncertainty model can become an ECT-specific prediction.

58. Conclusion

ECT is an attempt to place spacetime, gravity, and quantum behaviour within a single pre-spacetime framework. In its current form, some parts of this programme are structural results, some are closure-dependent effective constructions, and some remain open.

Common pre-geometric architecture of interactions. ECT organises a conditional Abelian route, a candidate weak-sector route and a colour programme in one framework. For colour, $\dim \mathfrak{su}(3) > \dim \mathfrak{so}(4)$ excludes only a direct injective embedding in the displayed minimal algebra; emergent/nonlinear, reducible/singlet, independently supplied bundle and defect-kernel routes remain possible. P7 is an adopted positive-definite Hermitian rank-three bundle postulate with fibrewise $SU(3)$ reduction (Level-B input / Level-A conditional reduction), not a necessary or unique physical gauge completion; its physical-colour map remains Open. Together with the Open gravity completion and generic director coupling this remains an architecture, not a unification theorem. Physical $U(1)_{\text{em}}$, photon, colour connection/action, gauge quotient/reduced Hessian, charges, matter representations, anomaly cancellation and the Weinberg angle remain Open.

Discrete-symmetry architecture. ECT identifies exact Euclidean reflection identities and a conditional route to physical discrete symmetries. Interpreting those reflections as parity or time reversal requires clock, rod and matter maps; OS/CPT reconstruction is controlled only in supplied free/quadratic local sectors and remains Open for the full nonlinear theory. The preferred-direction vector operator has a standard conditional discrete-symmetry classification, but its nonzero coefficient and physical matter owner are Open. It does not change representation-data anomaly coefficients and it does not supply the missing chiral gauge sector.

Flavour architecture. The observed “CKM small, PMNS large” dichotomy can be represented in a supplied overlap-plus-seesaw flavour model: quark and charged-lepton Yukawa matrices and the neutrino seesaw matrix may have different basis misalignments. ECT has not derived those matrices, their complex phases, or the preferred-direction flavour operator, so neither the dichotomy nor leading PMNS/MSW propagation is presently an ECT prediction. Mixing angles, CP and Majorana phases, and the absolute spectrum remain Open.

What is established or supplied at the strongest declared level: the P4 direction is a supplied datum whose $O(3)$ stabiliser is exact group mathematics, while its physical state and selection remain Open; the conditional Lorentzian classification of a scalar principal tensor for independently supplied $\beta(\beta - \alpha) < 0$; one scalar characteristic cone with physical gauge, fermion and tensor cones Open; a status-separated equation inventory: Euclidean scalar boundary problem, supplied KG-form scalar equation, and separately supplied complex Schrödinger sector whose conditional NR relation does not prove common descent; retarded support for declared hyperbolic scalar operators; and generic reduced-state identities with global ECT unitarity Open.

What is conditional or model-internal: the BTFR slope exactly 4 after a separate galactic gravity bridge; selected background diagnostics; and C/Open shell/Page toys that do not derive black-hole entropy or evaporation.

What is obtained at effective, fit, or closure level (mixed B/C/Open): Newton’s constant, Planck’s constant, and c enter through declared matching relations. The primary-amplitude one-loop stability result and the scalar-only metric equation hold inside their stated ansatzes; a covariant orientation stress remains Open. The named two-slope action and state supply a conditional background diagnostic, not a P1–P6 prediction. The 165-galaxy HRC-only SPARC registry has absolute held-out χ^2 /point of order 30 (Level C stress diagnostic); its scalar/ECT source map and any Ξ law are Open. Decoherence, PES retains a B/Open ceiling; physical black-hole area scaling is Open, while Page-curve-like shell behaviour and the shell itself are C/Open. The boundary-sensitive Casimir response remains Open ECT; the exact black-hole coefficient and a constructive Page curve are Open.

What remains an open programme: Three fermion generations; Yukawa hierarchy; $\alpha = 1/137$; renormalisability; Born rule as first-principles theorem; Bell correlators; complete BH evaporation; gradient condensation from bare P3; proton stability as a UV discriminant (three selection-rule patterns conditional on gauge/flavour completion). This also includes the full interacting closure of the conditional CPT route, the exact derivation of the maximally parity-violating weak-sector pattern, and the flavour-architecture derivation targets (CKM and PMNS mixing parameters, CP phases, absolute neutrino spectrum) under OP-Yukawa. The constancy of the dimensionless constants α and μ is recorded only as a conditional selective-coupling protection route (OP-CONST); the unconditional microscopic proof remains open and belongs to OP-matter plus OP-grad completion.

Status-separated architectural summary. The ordered-medium framework supplies candidate reconstructions and common organising language, with explicit status ceilings. The effective-metric route, excitation classification, coherent-sector reconstruction programme, and reduced-record description are not all theorem-level outputs. In particular, coherent interactions can exchange energy/momentum and entangle; decoherence is required only for irreversible record formation in a specified reduced model. The 70.6 ms sphere point is external conditional arithmetic, not a Diósi–Penrose/environment crossover or ECT rate: the physical ECT projector, vertex, state/noise kernel, normalisation and apparatus protocol remain Open, and recoherence remains protocol dependent.

Table 5: Status-separated ECT architecture and foundational questions.

Item	Current ECT statement	Status	Scope
Effective-metric route	Ordered-branch construction inside declared scalar/orientation programmes; complete covariant gravity and physical orientation stress remain Open	B/Open	Architectural
Excitation sectors	Scalar, gauge-candidate, orientation, and topological sectors form a status-separated classification; full Standard-Model matter content is not derived	A/B/Open	Classification

Continued on next page

Table 5 (continued)

Item	Current ECT statement	Status	Scope
Quantum recon- struction	Quadratic RP/OS and wave-kinematic routes are controlled; Born, pointer basis, single outcome, and interacting completion remain conditional/Open	A-math/B/Open	Programme
Coherent coupling and records	Coherent coupling can exchange energy/momentum and entangle; traced uncontrolled channels generate irreversible records in specified reduced models	B/Open	No universal threshold
Fock-space route	Variable excitation number is a reconstruction target, not a foundational ECT theorem	Open	Conceptual
Why a 3 + 1 split?	Given P1 and the supplied P4 vector chart, one aligned coordinate and three transverse coordinates are identified; physical realisation and the origin of four dimensions remain Open/OP0	A-math conditional / Open physical	Structural
Primary monopole topology	$\pi_2(S^3) = 0$ for the supplied orientation target; physical vacuum-manifold identification, transition and relic conclusion remain Open	A-math conditional / Open physical	Target-space limited
Amplitude- potential stability	Tree/local and primary-amplitude one-loop checks hold in their declared P3 closures; P4 gradient-state and global/metastable completion remain Open	B/Open	Closure limited
Quantum gravity	A common-medium architecture reframes the relation; UV, spin-2 RP/OS, and interacting closure remain Open	B/Open	Architectural
Physical spin-2 route	No linear TT/static-density seed only at fixed α, β , homogeneous $\bar{n} = w$ and tangent-director order; quadratic/nonlinear routes, constrained action, source and TT projection Open	A-math conditional negative result / Open physical completion	Completion target
Gravitational record channel	The 51/160, 70.6 ms hard-sphere point and the unit-coefficient guide/crossover relations are distinct external comparisons; none supplies an ECT projector, vertex, state/noise kernel or absolute rate	External A/C; ECT Open	Not ECT-specific
Recoherence	Echo/control recovery is protocol dependent; no universal forward/reverse probability law is derived	Open	Protocol specific
Late-time source	Conditional scalar proof-of-class; physical source and $w(a)$ Open	Open	Owner programme
Director spin/WEP completion	Homogeneous vector term has no spin splitting and no source-generated force; spin-dependent operator and body-force owners are absent	B operator classification / Open observables	External constraints only

ECT places the galactic mass-discrepancy, late-time background and quantum-reconstruction

programmes within one medium architecture. This does not eliminate their independent closure data: the HRC gravity bridge and particle content, the selected cosmological action/state, and the quantum state/operator/probability owners remain B/C/Open. No reduced bare-parameter count is claimed until these owners and their matching constants have been derived.

Whether ECT ultimately becomes a viable theory of nature or remains a productive conceptual laboratory, it does something valuable: it shows how spacetime, gravitational response, and quantum-theoretic structure can be placed within a common condensate architecture, with some parts derived, others reconstructed, and others left as explicit closure targets. It identifies the specific structures — gradient condensation, compact-phase topology, orientation stiffness, and the Principle of Euclidean Stationarity — that make this unification programme possible.

Inside a supplied ordered scalar branch, the programme conditionally models a distinguished direction and scalar propagation structure. Whether this becomes physical time requires the clock, metric, matter and branch-selection maps, which remain Open. Thus the idea that the universe forms time is a conditional programme hypothesis, not a P1–P6 derivation.

Reference scope. The numerical values and observational targets quoted in this companion are external comparison data, not evidence for ECT. Displayed digits are deterministic rounding of the stated arithmetic at declared inputs and conventions, not uncertainty statements or physical precision; inference requires input covariances and an owner-complete action/state-to-observable or likelihood chain. The principal source owners are GW170817 for the tensor-speed gate [3], Planck for the reference cosmology [16], SH0ES for the local Hubble calibration [17], DESI analyses for late-time distance constraints [18, 19], JWST studies for early-galaxy targets [20, 21], SPARC/RAR data for rotation-curve comparisons [22, 23], and MICROSCOPE for the equivalence-principle gate [24]. Every ECT interpretation retains the A/B/C/Open status assigned in the preprint.

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Note. This article is a narrative companion to: “*Euclidean Condensate Theory (ECT): Conditional Routes from a Euclidean Condensate to Lorentzian Kinematics, Quantum Structures, and Gravity*” by Valeriy Blagovidov (2026).

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<https://github.com/chufelo/ECT-preprint-code>

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